

8

Matrices

EXERCISES 8.1

Matrix Algebra

1. 2×4
2. 3×2
3. 3×3
4. 1×3
5. 3×4
6. 8×1
7. Not equal
8. Not equal
9. Not equal
10. Not equal
11. Solving $x = y - 2$, $y = 3x - 2$ we obtain $x = 2$, $y = 4$.
12. Solving $x^2 = 9$, $y = 4x$ we obtain $x = 3$, $y = 12$ and $x = -3$, $t = -12$.
13. $c_{23} = 2(0) - 3(-3) = 9$; $c_{12} = 2(3) - 3(-2) = 12$
14. $c_{23} = 2(1) - 3(0) = 2$; $c_{12} = 2(-1) - 3(0) = -2$
15. (a) $\mathbf{A} + \mathbf{B} = \begin{pmatrix} 4-2 & 5+6 \\ -6+8 & 9-10 \end{pmatrix} = \begin{pmatrix} 2 & 11 \\ 2 & -1 \end{pmatrix}$
 (b) $\mathbf{B} - \mathbf{A} = \begin{pmatrix} -2-4 & 6-5 \\ 8+6 & -10-9 \end{pmatrix} = \begin{pmatrix} -6 & 1 \\ 14 & -19 \end{pmatrix}$
 (c) $2\mathbf{A} + 3\mathbf{B} = \begin{pmatrix} 8 & 10 \\ -12 & 18 \end{pmatrix} + \begin{pmatrix} -6 & 18 \\ 24 & -30 \end{pmatrix} = \begin{pmatrix} 2 & 28 \\ 12 & -12 \end{pmatrix}$
16. (a) $\mathbf{A} - \mathbf{B} = \begin{pmatrix} -2-3 & 0+1 \\ 4-0 & 1-2 \\ 7+4 & 3+2 \end{pmatrix} = \begin{pmatrix} -5 & 1 \\ 4 & -1 \\ 11 & 5 \end{pmatrix}$
 (b) $\mathbf{B} - \mathbf{A} = \begin{pmatrix} 3+2 & -1-0 \\ 0-4 & 2-1 \\ -4-7 & -2-3 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -4 & 1 \\ -11 & -5 \end{pmatrix}$
 (c) $2(\mathbf{A} + \mathbf{B}) = 2 \begin{pmatrix} 1 & -1 \\ 4 & 3 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 8 & 6 \\ 6 & 2 \end{pmatrix}$
17. (a) $\mathbf{AB} = \begin{pmatrix} -2-9 & 12-6 \\ 5+12 & -30+8 \end{pmatrix} = \begin{pmatrix} -11 & 6 \\ 17 & -22 \end{pmatrix}$
 (b) $\mathbf{BA} = \begin{pmatrix} -2-30 & 3+24 \\ 6-10 & -9+8 \end{pmatrix} = \begin{pmatrix} -32 & 27 \\ -4 & -1 \end{pmatrix}$
 (c) $\mathbf{A}^2 = \begin{pmatrix} 4+15 & -6-12 \\ -10-20 & 15+16 \end{pmatrix} = \begin{pmatrix} 19 & -18 \\ -30 & 31 \end{pmatrix}$

8.1 Matrix Algebra

$$(d) \mathbf{B}^2 = \begin{pmatrix} 1+18 & -6+12 \\ -3+6 & 18+4 \end{pmatrix} = \begin{pmatrix} 19 & 6 \\ 3 & 22 \end{pmatrix}$$

$$18. (a) \mathbf{AB} = \begin{pmatrix} -4+4 & 6-12 & -3+8 \\ -20+10 & 30-30 & -15+20 \\ -32+12 & 48-36 & -24+24 \end{pmatrix} = \begin{pmatrix} 0 & -6 & 5 \\ -10 & 0 & 5 \\ -20 & 12 & 0 \end{pmatrix}$$

$$(b) \mathbf{BA} = \begin{pmatrix} -4+30-24 & -16+60-36 \\ 1-15+16 & 4-30+24 \end{pmatrix} = \begin{pmatrix} 2 & 8 \\ 2 & -2 \end{pmatrix}$$

$$19. (a) \mathbf{BC} = \begin{pmatrix} 9 & 24 \\ 3 & 8 \end{pmatrix}$$

$$(b) \mathbf{A}(\mathbf{BC}) = \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 9 & 24 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 3 & 8 \\ -6 & -16 \end{pmatrix}$$

$$(c) \mathbf{C}(\mathbf{BA}) = \begin{pmatrix} 0 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$(d) \mathbf{A}(\mathbf{B} + \mathbf{C}) = \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 6 & 5 \\ 5 & 5 \end{pmatrix} = \begin{pmatrix} -4 & -5 \\ 8 & 10 \end{pmatrix}$$

$$20. (a) \mathbf{AB} = \begin{pmatrix} 5 & -6 & 7 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} = (-16)$$

$$(b) \mathbf{BA} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} \begin{pmatrix} 5 & -6 & 7 \end{pmatrix} = \begin{pmatrix} 15 & -18 & 21 \\ 20 & -24 & 28 \\ -5 & 6 & -7 \end{pmatrix}$$

$$(c) (\mathbf{BA})\mathbf{C} = \begin{pmatrix} 15 & -18 & 21 \\ 20 & -24 & 28 \\ -5 & 6 & -7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 78 & 54 & 99 \\ 104 & 72 & 132 \\ -26 & -18 & -33 \end{pmatrix}$$

(d) Since $\mathbf{AB}$ is 1×1 and $\mathbf{C}$ is 3×3 the product $(\mathbf{AB})\mathbf{C}$ is not defined.

$$21. (a) \mathbf{A}^T \mathbf{A} = \begin{pmatrix} 4 & 8 & -10 \end{pmatrix} \begin{pmatrix} 4 \\ 8 \\ -10 \end{pmatrix} = (180)$$

$$(b) \mathbf{B}^T \mathbf{B} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} \begin{pmatrix} 2 & 4 & 5 \end{pmatrix} = \begin{pmatrix} 4 & 8 & 10 \\ 8 & 16 & 20 \\ 10 & 20 & 25 \end{pmatrix}$$

$$(c) \mathbf{A} + \mathbf{B}^T = \begin{pmatrix} 4 \\ 8 \\ -10 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \\ -5 \end{pmatrix}$$

$$22. (a) \mathbf{A} + \mathbf{B}^T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} + \begin{pmatrix} -2 & 5 \\ 3 & 7 \end{pmatrix} = \begin{pmatrix} -1 & 7 \\ 5 & 11 \end{pmatrix}$$

$$(b) 2\mathbf{A}^T - \mathbf{B}^T = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix} - \begin{pmatrix} -2 & 5 \\ 3 & 7 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ 1 & 1 \end{pmatrix}$$

$$(c) \mathbf{A}^T(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -3 & -3 \end{pmatrix} = \begin{pmatrix} -3 & -7 \\ -6 & -14 \end{pmatrix}$$

$$23. (a) (\mathbf{AB})^T = \begin{pmatrix} 7 & 10 \\ 38 & 75 \end{pmatrix}^T = \begin{pmatrix} 7 & 38 \\ 10 & 75 \end{pmatrix}$$

$$(b) \mathbf{B}^T \mathbf{A}^T = \begin{pmatrix} 5 & -2 \\ 10 & -5 \end{pmatrix} \begin{pmatrix} 3 & 8 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 38 \\ 10 & 75 \end{pmatrix}$$

$$24. (a) \mathbf{A}^T + \mathbf{B} = \begin{pmatrix} 5 & -4 \\ 9 & 6 \end{pmatrix} + \begin{pmatrix} -3 & 11 \\ -7 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 2 & 8 \end{pmatrix}$$

$$(b) 2\mathbf{A} + \mathbf{B}^T = \begin{pmatrix} 10 & 18 \\ -8 & 12 \end{pmatrix} + \begin{pmatrix} -3 & -7 \\ 11 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 11 \\ 3 & 14 \end{pmatrix}$$

$$25. \begin{pmatrix} -4 \\ 8 \end{pmatrix} - \begin{pmatrix} 4 \\ 16 \end{pmatrix} + \begin{pmatrix} -6 \\ 9 \end{pmatrix} = \begin{pmatrix} -14 \\ 1 \end{pmatrix}$$

$$26. \begin{pmatrix} 6 \\ 3 \\ -3 \end{pmatrix} + \begin{pmatrix} -5 \\ -5 \\ 15 \end{pmatrix} + \begin{pmatrix} -6 \\ -8 \\ 10 \end{pmatrix} = \begin{pmatrix} -5 \\ -10 \\ 22 \end{pmatrix}$$

$$27. \begin{pmatrix} -19 \\ 18 \end{pmatrix} - \begin{pmatrix} 19 \\ 20 \end{pmatrix} = \begin{pmatrix} -38 \\ -2 \end{pmatrix}$$

$$28. \begin{pmatrix} -7 \\ 17 \\ -6 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 8 \\ -6 \end{pmatrix} = \begin{pmatrix} -10 \\ 10 \\ 4 \end{pmatrix}$$

$$29. 4 \times 5$$

$$30. 3 \times 2$$

$$31. \mathbf{A}^T = \begin{pmatrix} 2 & -3 \\ 4 & 2 \end{pmatrix}; \quad (\mathbf{A}^T)^T = \begin{pmatrix} 2 & 4 \\ -3 & 2 \end{pmatrix} = \mathbf{A} \quad 32. (\mathbf{A} + \mathbf{B})^T = \begin{pmatrix} 6 & -6 \\ 14 & 10 \end{pmatrix} = \mathbf{A}^T + \mathbf{B}^T$$

$$33. (\mathbf{AB})^T = \begin{pmatrix} 16 & 40 \\ -8 & -20 \end{pmatrix}^T = \begin{pmatrix} 16 & -8 \\ 40 & -20 \end{pmatrix}; \quad \mathbf{B}^T \mathbf{A}^T = \begin{pmatrix} 4 & 2 \\ 10 & 5 \end{pmatrix} \begin{pmatrix} 2 & -3 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 16 & -8 \\ 40 & -20 \end{pmatrix}$$

$$34. (6\mathbf{A})^T = \begin{pmatrix} 12 & -18 \\ 24 & 12 \end{pmatrix} = 6\mathbf{A}^T$$

$$35. \mathbf{B} = \mathbf{AA}^T = \begin{pmatrix} 2 & 1 \\ 6 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 2 & 6 & 2 \\ 1 & 3 & 5 \end{pmatrix} = \begin{pmatrix} 5 & 15 & 9 \\ 15 & 39 & 27 \\ 9 & 27 & 29 \end{pmatrix} = \mathbf{B}^T$$

$$36. \text{ Using Problem 33 we have } (\mathbf{AA}^T)^T = (\mathbf{A}^T)^T \mathbf{A}^T = \mathbf{AA}^T, \text{ so that } \mathbf{AA}^T \text{ is symmetric.}$$

$$37. \text{ Let } \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \text{ Then } \mathbf{AB} = \mathbf{0}.$$

$$38. \text{ We see that } \mathbf{A} \neq \mathbf{B}, \text{ but } \mathbf{AC} = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{pmatrix} = \mathbf{BC}.$$

$$39. \text{ Since } (\mathbf{A} + \mathbf{B})^2 = (\mathbf{A} + \mathbf{B})(\mathbf{A} + \mathbf{B}) \neq \mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2, \text{ and } \mathbf{AB} \neq \mathbf{BA} \text{ in general, } (\mathbf{A} + \mathbf{B})^2 \neq \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2.$$

$$40. \text{ Since } (\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 - \mathbf{AB} + \mathbf{BA} - \mathbf{B}^2, \text{ and } \mathbf{AB} \neq \mathbf{BA} \text{ in general, } (\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) \neq \mathbf{A}^2 - \mathbf{B}^2.$$

$$41. a_{11}x_1 + a_{12}x_2 = b_1; \quad a_{21}x_1 + a_{22}x_2 = b_2$$

$$42. \begin{pmatrix} 2 & 6 & 1 \\ 1 & 2 & -1 \\ 5 & 7 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \\ 9 \end{pmatrix}$$

8.1 Matrix Algebra

$$43. \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by/2 & bx/2 + cy \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax^2 + bxy/2 + bxy/2 + cy^2 \end{pmatrix} = \begin{pmatrix} ax^2 + bxy + cy^2 \end{pmatrix}$$

$$44. \begin{pmatrix} 0 & -\partial/\partial z & \partial/\partial y \\ \partial/\partial z & 0 & -\partial/\partial x \\ -\partial/\partial y & \partial/\partial x & 0 \end{pmatrix} \begin{pmatrix} P \\ Q \\ R \end{pmatrix} = \begin{pmatrix} -\partial Q/\partial z + \partial R/\partial y \\ \partial P/\partial z - \partial R/\partial x \\ -\partial P/\partial y + \partial Q/\partial x \end{pmatrix} = \text{curl } \mathbf{F}$$

$$45. (a) M_Y \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \cos \gamma + y \sin \gamma \\ -x \sin \gamma + y \cos \gamma \\ z \end{pmatrix} = \begin{pmatrix} x_Y \\ y_Y \\ z_Y \end{pmatrix}$$

$$(b) M_R = \begin{pmatrix} \cos \beta & 0 & -\sin \beta \\ 0 & 1 & 0 \\ \sin \beta & 0 & \cos \beta \end{pmatrix}; \quad M_P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{pmatrix}$$

$$(c) M_P \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & \sin 30^\circ \\ 0 & -\sin 30^\circ & \cos 30^\circ \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{2}(\sqrt{3} - 1) \end{pmatrix}$$

$$M_R M_P \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & 0 & -\sin 45^\circ \\ 0 & 1 & 0 \\ \sin 45^\circ & 0 & \cos 45^\circ \end{pmatrix} \begin{pmatrix} 1 \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{2}(\sqrt{3} - 1) \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 1 \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{2}(\sqrt{3} - 1) \end{pmatrix} \\ = \begin{pmatrix} \frac{1}{4}(3\sqrt{2} - \sqrt{6}) \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{pmatrix}$$

$$M_Y M_R M_P \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \cos 60^\circ & \sin 60^\circ & 0 \\ -\sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{4}(3\sqrt{2} - \sqrt{6}) \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{pmatrix} \\ = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{4}(3\sqrt{2} - \sqrt{6}) \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{pmatrix} = \begin{pmatrix} \frac{1}{8}(3\sqrt{2} - \sqrt{6} + 6 + 2\sqrt{3}) \\ \frac{1}{8}(-3\sqrt{6} + 3\sqrt{2} + 2\sqrt{3} + 2) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{pmatrix}$$

$$46. (a) \mathbf{LU} = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 1 & 2 \end{pmatrix} = \mathbf{A}$$

$$(b) \mathbf{LU} = \begin{pmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{pmatrix} \begin{pmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 4 & 1 \end{pmatrix} = \mathbf{A}$$

$$(c) \mathbf{LU} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 10 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 2 & 6 & 1 \end{pmatrix} = \mathbf{A}$$

$$(d) \mathbf{LU} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} = \mathbf{A}$$

$$47. (a) \mathbf{AB} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{pmatrix} = \begin{pmatrix} 17 & 43 \\ 3 & 75 \\ -14 & 51 \end{pmatrix}$$

since

$$A_{11}B_1 + A_{12}B_2 = \begin{pmatrix} 13 & 25 \\ -9 & 49 \end{pmatrix} + \begin{pmatrix} 4 & 18 \\ 12 & 26 \end{pmatrix} = \begin{pmatrix} 17 & 43 \\ 3 & 75 \end{pmatrix}$$

and

$$A_{21}B_1 + A_{22}B_2 = \begin{pmatrix} -24 & 34 \end{pmatrix} + \begin{pmatrix} 10 & 17 \end{pmatrix} = \begin{pmatrix} -14 & 51 \end{pmatrix}.$$

- (b) It is easier to enter smaller strings of numbers and the chance of error is decreased. Also, if the large matrix has submatrices consisting of all zeros or diagonal matrices, these are easily entered without listing all of the entries.

EXERCISES 8.2

Systems of Linear Algebraic Equations

$$1. \left(\begin{array}{cc|c} 1 & -1 & 11 \\ 4 & 3 & -5 \end{array} \right) \xrightarrow{-4R_1+R_2} \left(\begin{array}{cc|c} 1 & -1 & 11 \\ 0 & 7 & -49 \end{array} \right) \xrightarrow{\frac{1}{7}R_2} \left(\begin{array}{cc|c} 1 & -1 & 11 \\ 0 & 1 & -7 \end{array} \right) \xrightarrow{R_3+R_1} \left(\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -7 \end{array} \right)$$

The solution is $x_1 = 4$, $x_2 = -7$.

$$2. \left(\begin{array}{cc|c} 3 & -2 & 4 \\ 1 & -1 & -2 \end{array} \right) \xrightarrow{R_{12}} \left(\begin{array}{cc|c} 1 & -1 & -2 \\ 3 & -2 & 4 \end{array} \right) \xrightarrow{-3R_1+R_2} \left(\begin{array}{cc|c} 1 & -1 & -2 \\ 0 & 1 & 10 \end{array} \right) \xrightarrow{R_2+R_1} \left(\begin{array}{cc|c} 1 & 0 & 8 \\ 0 & 1 & 10 \end{array} \right)$$

The solution is $x_1 = 8$, $x_2 = 10$.

$$3. \left(\begin{array}{cc|c} 9 & 3 & -5 \\ 2 & -1 & -1 \end{array} \right) \xrightarrow{\frac{1}{9}R_1} \left(\begin{array}{cc|c} 1 & \frac{1}{3} & -\frac{5}{9} \\ 2 & -1 & -1 \end{array} \right) \xrightarrow{-2R_1+R_2} \left(\begin{array}{cc|c} 1 & \frac{1}{3} & -\frac{5}{9} \\ 0 & \frac{1}{3} & \frac{1}{9} \end{array} \right) \xrightarrow{3R_2} \left(\begin{array}{cc|c} 1 & \frac{1}{3} & -\frac{5}{9} \\ 0 & 1 & \frac{1}{3} \end{array} \right) \\ \xrightarrow{-\frac{1}{3}R_2+R_1} \left(\begin{array}{cc|c} 1 & 0 & -\frac{2}{3} \\ 0 & 1 & \frac{1}{3} \end{array} \right)$$

The solution is $x_1 = -\frac{2}{3}$, $x_2 = \frac{1}{3}$.

$$4. \left(\begin{array}{cc|c} 10 & 15 & 1 \\ 3 & 2 & -1 \end{array} \right) \xrightarrow{\frac{1}{10}R_1} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & \frac{1}{10} \\ 3 & 2 & -1 \end{array} \right) \xrightarrow{-3R_1+R_2} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & \frac{1}{10} \\ 0 & -\frac{5}{2} & -\frac{13}{10} \end{array} \right) \xrightarrow{-\frac{2}{5}R_2} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & \frac{1}{10} \\ 0 & 1 & \frac{13}{25} \end{array} \right) \\ \xrightarrow{-\frac{3}{2}R_2+R_1} \left(\begin{array}{cc|c} 1 & 0 & -\frac{17}{25} \\ 0 & 1 & \frac{13}{25} \end{array} \right)$$

The solution is $x_1 = -\frac{17}{25}$, $x_2 = \frac{13}{25}$.

$$5. \left(\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 2 & 3 & 5 & 7 \\ 1 & -2 & 3 & -11 \end{array} \right) \xrightarrow[-R_1+R_3]{-2R_1+R_2} \left(\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 0 & 5 & 7 & 13 \\ 0 & -1 & 4 & -8 \end{array} \right) \xrightarrow{\frac{1}{5}R_2} \left(\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 0 & 1 & \frac{7}{5} & \frac{13}{5} \\ 0 & -1 & 4 & -8 \end{array} \right) \\ \xrightarrow[R_2+R_3]{R_2+R_1} \left(\begin{array}{ccc|c} 1 & 0 & \frac{2}{5} & -\frac{2}{5} \\ 0 & 1 & \frac{7}{5} & \frac{13}{5} \\ 0 & 0 & \frac{27}{5} & -\frac{27}{5} \end{array} \right) \xrightarrow{\frac{5}{27}R_3} \left(\begin{array}{ccc|c} 1 & 0 & \frac{2}{5} & -\frac{2}{5} \\ 0 & 1 & \frac{7}{5} & \frac{13}{5} \\ 0 & 0 & 1 & -1 \end{array} \right) \xrightarrow[-\frac{7}{5}R_3+R_2]{-\frac{2}{5}R_3+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -1 \end{array} \right)$$

The solution is $x_1 = 0$, $x_2 = 4$, $x_3 = -1$.

8.2 Systems of Linear Algebraic Equations

$$\begin{aligned}
 6. \quad & \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 1 & 2 & 9 \\ 1 & -1 & 1 & 3 \end{array} \right) \xrightarrow[-R_1+R_3]{-2R_1+R_2} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 4 & 9 \\ 0 & -3 & 2 & 3 \end{array} \right) \xrightarrow{-\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -\frac{4}{3} & -3 \\ 0 & -3 & 2 & 3 \end{array} \right) \\
 & \xrightarrow[3R_2+R_3]{-2R_2+R_1} \left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & 6 \\ 0 & 1 & -\frac{4}{3} & -3 \\ 0 & 0 & -2 & -6 \end{array} \right) \xrightarrow{-\frac{1}{2}R_3} \left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & 6 \\ 0 & 1 & -\frac{4}{3} & -3 \\ 0 & 0 & 1 & 3 \end{array} \right) \xrightarrow[\frac{4}{3}R_3+R_2]{-\frac{5}{3}R_3+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array} \right)
 \end{aligned}$$

The solution is $x_1 = 1$, $x_2 = 1$, $x_3 = 3$.

$$7. \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 \end{array} \right) \xrightarrow{-R_1+R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right)$$

Since $x_3 = 0$, setting $x_2 = t$ we obtain $x_1 = -t$, $x_2 = t$, $x_3 = 0$.

$$8. \quad \left(\begin{array}{ccc|c} 1 & 2 & -4 & 9 \\ 5 & -1 & 2 & 1 \end{array} \right) \xrightarrow{-5R_1+R_2} \left(\begin{array}{ccc|c} 1 & 2 & -4 & 9 \\ 0 & -11 & 22 & -44 \end{array} \right) \xrightarrow{-\frac{1}{11}R_2} \left(\begin{array}{ccc|c} 1 & 2 & -4 & 9 \\ 0 & 1 & -2 & 4 \end{array} \right) \xrightarrow{-2R_2+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -2 & 4 \end{array} \right)$$

If $x_3 = t$, the solution is $x_1 = 1$, $x_2 = 4 + 2t$, $x_3 = t$

$$9. \quad \left(\begin{array}{ccc|c} 1 & -1 & -1 & 8 \\ 1 & -1 & 1 & 3 \\ -1 & 1 & 1 & 4 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -1 & -1 & 8 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 0 & 12 \end{array} \right)$$

Since the bottom row implies $0 = 12$, the system is inconsistent.

$$10. \quad \left(\begin{array}{cc|c} 3 & 1 & 4 \\ 4 & 3 & -3 \\ 2 & -1 & 11 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{4}{3} \\ 0 & 1 & -5 \\ 0 & 0 & 0 \end{array} \right)$$

The solution is $x_1 = 3$, $x_2 = -5$.

$$11. \quad \left(\begin{array}{ccc|c} 2 & 2 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 3 & 0 & 1 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

The solution is $x_1 = x_2 = x_3 = 0$.

$$12. \quad \left(\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 2 & 4 & 5 & 0 \\ 6 & 0 & -3 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 0 & 1 & \frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

The solution is $x_1 = \frac{1}{2}t$, $x_2 = -\frac{3}{2}t$, $x_3 = t$.

$$13. \quad \left(\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 0 \\ 1 & -3 & -1 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right)$$

The solution is $x_1 = -2$, $x_2 = 2$, $x_3 = 4$.

$$14. \quad \left(\begin{array}{ccc|c} 1 & -2 & 1 & 2 \\ 3 & -1 & 2 & 5 \\ 2 & 1 & 1 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 2 \\ 0 & 1 & -\frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 0 & -2 \end{array} \right)$$

Since the bottom row implies $0 = -2$, the system is inconsistent.

$$15. \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & -1 & -1 & -1 \\ 3 & 1 & 1 & 5 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

If $x_3 = t$ the solution is $x_1 = 1$, $x_2 = 2 - t$, $x_3 = t$.

$$16. \left(\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ -3 & -2 & 1 & -7 \\ 2 & 3 & 1 & 8 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

If $x_3 = t$ the solution is $x_1 = 1 + t$, $x_2 = 2 - t$, $x_3 = t$.

$$17. \left(\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 1 \\ 0 & 2 & 1 & 1 & 3 \\ 1 & -1 & 0 & 1 & -1 \\ 1 & 1 & 1 & 1 & 2 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$$

The solution is $x_1 = 0$, $x_2 = 1$, $x_3 = 1$, $x_4 = 0$.

$$18. \left(\begin{array}{cccc|c} 2 & 1 & 1 & 0 & 3 \\ 3 & 1 & 1 & 1 & 4 \\ 1 & 2 & 2 & 3 & 3 \\ 4 & 5 & -2 & 1 & 16 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 & \frac{3}{2} \\ 0 & 1 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$$

The solution is $x_1 = 1$, $x_2 = 2$, $x_3 = -1$, $x_4 = 0$.

$$19. \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 1 \\ 0 & 1 & 1 & -1 & 4 \\ 1 & 2 & 5 & -4 & -2 \\ 1 & 4 & 6 & -2 & 6 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 1 \\ 0 & 1 & 1 & -1 & 4 \\ 0 & 0 & 1 & -4 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

Since the bottom row implies $0 = 1$, the system is inconsistent.

$$20. \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 0 \\ 4 & 9 & 1 & 12 & 0 \\ 3 & 9 & 6 & 21 & 0 \\ 1 & 3 & 1 & 9 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 8 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

If $x_4 = t$ the solution is $x_1 = 19t$, $x_2 = -10t$, $x_3 = 2t$, $x_4 = t$.

$$21. \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4.280 \\ 0.2 & -0.1 & -0.5 & -1.978 \\ 4.1 & 0.3 & 0.12 & 1.686 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4.28 \\ 0 & 1 & 2.333 & 9.447 \\ 0 & 0 & 1 & 4.1 \end{array} \right)$$

The solution is $x_1 = 0.3$, $x_2 = -0.12$, $x_3 = 4.1$.

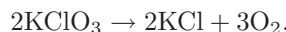
$$22. \left(\begin{array}{ccc|c} 2.5 & 1.4 & 4.5 & 2.6170 \\ 1.35 & 0.95 & 1.2 & 0.7545 \\ 2.7 & 3.05 & -1.44 & -1.4292 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 0.56 & 1.8 & 1.0468 \\ 0 & 1 & -6.3402 & -3.3953 \\ 0 & 0 & 1 & 0.28 \end{array} \right)$$

The solution is $x_1 = 1.45$, $x_2 = -1.62$, $x_3 = 0.28$.

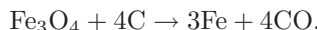
23. From $x_1\text{Na} + x_2\text{H}_2\text{O} \rightarrow x_3\text{NaOH} + x_4\text{H}_2$ we obtain the system $x_1 = x_3$, $2x_2 = x_3 + 2x_4$, $x_2 = x_3$. We see that $x_1 = x_2 = x_3$, so the second equation becomes $2x_1 = x_1 + 2x_4$ or $x_1 = 2x_4$. A solution of the system is $x_1 = x_2 = x_3 = 2t$, $x_4 = t$. Letting $t = 1$ we obtain the balanced equation $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$.

8.2 Systems of Linear Algebraic Equations

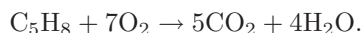
24. From $x_1\text{KClO}_3 \rightarrow x_2\text{KCl} + x_3\text{O}_2$ we obtain the system $x_1 = x_2$, $x_1 = x_2$, $3x_1 = 2x_3$. Letting $x_3 = t$ we see that a solution of the system is $x_1 = x_2 = \frac{2}{3}t$, $x_3 = t$. Taking $t = 3$ we obtain the balanced equation



25. From $x_1\text{Fe}_3\text{O}_4 + x_2\text{C} \rightarrow x_3\text{Fe} + x_4\text{CO}$ we obtain the system $3x_1 = x_3$, $4x_1 = x_4$, $x_2 = x_4$. Letting $x_1 = t$ we see that $x_3 = 3t$ and $x_2 = x_4 = 4t$. Taking $t = 1$ we obtain the balanced equation



26. From $x_1\text{C}_5\text{H}_8 + x_2\text{O}_2 \rightarrow x_3\text{CO}_2 + x_4\text{H}_2\text{O}$ we obtain the system $5x_1 = x_3$, $8x_1 = 2x_4$, $2x_2 = 2x_3 + x_4$. Letting $x_1 = t$ we see that $x_3 = 5t$, $x_4 = 4t$, and $x_2 = 7t$. Taking $t = 1$ we obtain the balanced equation



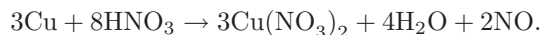
27. From $x_1\text{Cu} + x_2\text{HNO}_3 \rightarrow x_3\text{Cu}(\text{NO}_3)_2 + x_4\text{H}_2\text{O} + x_5\text{NO}$ we obtain the system

$$x_1 = 3, \quad x_2 = 2x_4, \quad x_2 = 2x_3 + x_5, \quad 3x_2 = 6x_3 + x_4 + x_5.$$

Letting $x_4 = t$ we see that $x_2 = 2t$ and

$$\begin{array}{ccc} 2t = 2x_3 + x_5 & & 2x_3 + x_5 = 2t \\ 6t = 6x_3 + t + x_5 & \text{or} & 6x_3 + x_5 = 5t. \end{array}$$

Then $x_3 = \frac{3}{4}t$ and $x_5 = \frac{1}{2}t$. Finally, $x_1 = x_3 = \frac{3}{4}t$. Taking $t = 4$ we obtain the balanced equation



28. From $x_1\text{Ca}_3(\text{PO}_4)_2 + x_2\text{H}_3\text{PO}_4 \rightarrow x_3\text{Ca}(\text{H}_2\text{PO}_4)_2$ we obtain the system

$$3x_1 = x_3, \quad 2x_1 + x_2 = 2x_3, \quad 8x_1 + 4x_2 = 8x_3, \quad 3x_2 = 4x_3.$$

Letting $x_1 = t$ we see from the first equation that $x_3 = 3t$ and from the fourth equation that $x_2 = 4t$. These choices also satisfy the second and third equations. Taking $t = 1$ we obtain the balanced equation



29. The system of equations is

$$\begin{array}{ccc} -i_1 + i_2 - i_3 = 0 & & -i_1 + i_2 - i_3 = 0 \\ 10 - 3i_1 + 5i_3 = 0 & \text{or} & 3i_1 - 5i_3 = 10 \\ 27 - 6i_2 - 5i_3 = 0 & & 6i_2 + 5i_3 = 27 \end{array}$$

Gaussian elimination gives

$$\left(\begin{array}{ccc|c} -1 & 1 & -1 & 0 \\ 3 & 0 & -5 & 10 \\ 0 & 6 & 5 & 27 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -8/3 & 10/3 \\ 0 & 0 & 1 & 1/3 \end{array} \right).$$

The solution is $i_1 = \frac{35}{9}$, $i_2 = \frac{38}{9}$, $i_3 = \frac{1}{3}$.

30. The system of equations is

$$\begin{array}{ccc} i_1 - i_2 - i_3 = 0 & & i_1 - i_2 - i_3 = 0 \\ 52 - i_1 - 5i_2 = 0 & \text{or} & i_1 + 5i_2 = 52 \\ -10i_3 + 5i_2 = 0 & & 5i_2 - 10i_3 = 0 \end{array}$$

Gaussian elimination gives

$$\left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 1 & 5 & 0 & 52 \\ 0 & 5 & -10 & 0 \end{array}\right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 1 & 1/6 & 26/3 \\ 0 & 0 & 1 & 4 \end{array}\right).$$

The solution is $i_1 = 12$, $i_2 = 8$, $i_3 = 4$.

31. Interchange row 1 and row in $\mathbf{I}_3$.

32. Multiply row 3 by c in $\mathbf{I}_3$.

33. Add c times row 2 to row 3 in $\mathbf{I}_3$.

34. Add row 4 to row 1 in $\mathbf{I}_4$.

$$\mathbf{35. EA} = \begin{pmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\mathbf{36. EA} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ca_{31} & ca_{32} & ca_{33} \end{pmatrix}$$

$$\mathbf{37. EA} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ca_{21} + a_{31} & ca_{22} + a_{32} & ca_{23} + a_{33} \end{pmatrix}$$

$$\mathbf{38. E_1 E_2 A = E_1} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ca_{21} + a_{31} & ca_{22} + a_{32} & ca_{23} + a_{33} \end{pmatrix} = \begin{pmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ ca_{21} + a_{31} & ca_{22} + a_{32} & ca_{23} + a_{33} \end{pmatrix}$$

39. The system is equivalent to

$$\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 0 & 3 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

Letting

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 0 & 3 \end{pmatrix} \mathbf{X}$$

we have

$$\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

This implies $y_1 = 2$ and $\frac{1}{2}y_1 + y_2 = 1 + y_2 = 6$ or $y_2 = 5$. Then

$$\begin{pmatrix} 2 & -2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix},$$

which implies $3x_2 = 5$ or $x_2 = \frac{5}{3}$ and $2x_1 - 2x_2 = 2x_1 - \frac{10}{3} = 2$ or $x_1 = \frac{8}{3}$. The solution is $\mathbf{X} = (\frac{8}{3}, \frac{5}{3})$.

40. The system is equivalent to

$$\begin{pmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{pmatrix} \begin{pmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{pmatrix} \mathbf{X} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Letting

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{pmatrix} \mathbf{X}$$

we have

$$\begin{pmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

8.2 Systems of Linear Algebraic Equations

This implies $y_1 = 1$ and $\frac{2}{3}y_1 + y_2 = \frac{2}{3} + y_2 = -1$ or $y_2 = -\frac{5}{3}$. Then

$$\begin{pmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{5}{3} \end{pmatrix},$$

which implies $-\frac{1}{3}x_2 = -\frac{5}{3}$ or $x_2 = 5$ and $6x_1 + 2x_2 = 6x_1 + 10 = 1$ or $x_1 = -\frac{3}{2}$. The solution is $\mathbf{X} = (-\frac{3}{2}, 5)$.

41. The system is equivalent to

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 10 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

Letting

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{pmatrix} \mathbf{X}$$

we have

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 10 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

This implies $y_1 = 2$, $y_2 = -1$, and $2y_1 + 10y_2 + y_3 = 4 - 10 + y_3 = 1$ or $y_3 = 7$. Then

$$\begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 7 \end{pmatrix},$$

which implies $-21x_3 = 7$ or $x_3 = -\frac{1}{3}$, $x_2 + 2x_3 = x_2 - \frac{2}{3} = -1$ or $x_2 = -\frac{1}{3}$, and $x_1 - 2x_2 + x_3 = x_1 + \frac{2}{3} - \frac{1}{3} = 2$ or $x_1 = \frac{5}{3}$. The solution is $\mathbf{X} = (\frac{5}{3}, -\frac{1}{3}, -\frac{1}{3})$.

42. The system is equivalent to

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}.$$

Letting

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{X}$$

we have

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}.$$

This implies $y_1 = 0$, $3y_1 + y_2 = y_2 = 1$, and $y_1 + y_2 + y_3 = 0 + 1 + y_3 = 4$ or $y_3 = 3$. Then

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix},$$

which implies $x_3 = 3$, $-2x_2 - x_3 = -2x_2 - 3 = 1$ or $x_2 = -2$, and $x_1 + x_2 + x_3 = x_1 - 2 + 3 = 0$ or $x_1 = -1$. The solution is $\mathbf{X} = (-1, -2, 3)$.

43. Using the **Solve** function in *Mathematica* we find $x_1 = -0.0717393 - 1.43084c$, $x_2 = -0.332591 + 0.855709c$, $x_3 = c$, where c is any real number
44. Using the **Solve** function in *Mathematica* we find $x_1 = c/3$, $x_2 = 5c/6$, $x_3 = c$, where c is any real number
45. Using the **Solve** function in *Mathematica* we find $x_1 = -3.76993$, $x_2 = -1.09071$, $x_3 = -4.50461$, $x_4 = -3.12221$
46. Using the **Solve** function in *Mathematica* we find $x_1 = \frac{8}{3} - \frac{7}{3}b + \frac{2}{3}c$, $x_2 = \frac{2}{3} - \frac{1}{3}b - \frac{1}{3}c$, $x_3 = -3$, $x_4 = b$, $x_5 = c$, where b and c are any real numbers.

EXERCISES 8.3

Rank of a Matrix

1. $\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$; The rank is 2.
2. $\begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$; The rank is 1.
3. $\begin{pmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ -1 & -\frac{1}{2} & -\frac{3}{2} \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; The rank is 1.
4. $\begin{pmatrix} 1 & 1 & 2 \\ -1 & 2 & 4 \\ -1 & 0 & 3 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}$; The rank is 3.
5. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 4 \\ 1 & 4 & 1 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}$; The rank is 3.
6. $\begin{pmatrix} 3 & -1 & 2 & 0 \\ 6 & 2 & 4 & 5 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 1 & 0 & \frac{5}{4} \end{pmatrix}$; The rank is 2.
7. $\begin{pmatrix} 1 & -2 \\ 3 & -6 \\ 7 & -1 \\ 4 & 5 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$; The rank is 2.
8. $\begin{pmatrix} 1 & -2 & 3 & 4 \\ 1 & 4 & 6 & 8 \\ 0 & 1 & 0 & 0 \\ 2 & 5 & 6 & 8 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}$; The rank is 3.

8.3 Rank of a Matrix

$$9. \begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 1 & 0 & 5 & 1 \\ 2 & 1 & \frac{2}{3} & 3 & \frac{1}{3} \\ 6 & 6 & 6 & 12 & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{3}{2} & \frac{1}{6} \\ 0 & 1 & \frac{4}{3} & 1 & -\frac{1}{3} \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}; \quad \text{The rank is 3.}$$

$$10. \begin{pmatrix} 1 & -2 & 1 & 8 & -1 & 1 & 1 & 6 \\ 0 & 0 & 1 & 3 & -1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 & -1 & 2 & 10 & 8 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 3 \\ 1 & -2 & 1 & 8 & -1 & 1 & 2 & 6 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -2 & 1 & 8 & -1 & 1 & 1 & 6 \\ 0 & 0 & 1 & 3 & -1 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 1 & 9 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}; \quad \text{The rank is 4.}$$

$$11. \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & -1 & 5 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix};$$

Since the rank of the matrix is 3 and there are 3 vectors, the vectors are linearly independent.

$$12. \begin{pmatrix} 2 & 6 & 3 \\ 1 & -1 & 4 \\ 3 & 2 & 1 \\ 2 & 5 & 4 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -1 & 4 \\ 0 & 1 & -\frac{5}{8} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Since the rank of the matrix is 3 and there are 4 vectors, the vectors are linearly dependent.

$$13. \begin{pmatrix} 1 & -1 & 3 & -1 \\ 1 & -1 & 4 & 2 \\ 1 & -1 & 5 & 7 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -1 & 3 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since the rank of the matrix is 3 and there are 3 vectors, the vectors are linearly independent.

$$14. \begin{pmatrix} 2 & 1 & 1 & 5 \\ 2 & 2 & 1 & 1 \\ 3 & -1 & 6 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & -7 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since the rank of the matrix is 4 and there are 4 vectors, the vectors are linearly independent.

15. Since the number of unknowns is $n = 8$ and the rank of the coefficient matrix is $r = 3$, the solution of the system has $n - r = 5$ parameters.
16. (a) The maximum possible rank of $\mathbf{A}$ is the number of rows in $\mathbf{A}$, which is 4.
 (b) The system is inconsistent if $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}/\mathbf{B}) = 2$ and consistent if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}/\mathbf{B}) = 2$.
 (c) The system has $n = 6$ unknowns and the rank of $\mathbf{A}$ is $r = 3$, so the solution of the system has $n - r = 3$ parameters.
17. Since $2\mathbf{v}_1 + 3\mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$ we conclude that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are linearly dependent. Thus, the rank of $\mathbf{A}$ is at most 2.
18. Since the rank of $\mathbf{A}$ is $r = 3$ and the number of equations is $n = 6$, the solution of the system has $n - r = 3$ parameters. Thus, the solution of the system is not unique.
19. The system consists of 4 equations, so the rank of the coefficient matrix is at most 4, and the maximum number of linearly independent rows is 4. However, the maximum number of linearly independent columns is the same

as the maximum number of linearly independent rows. Thus, the coefficient matrix has at most 4 linearly independent columns. Since there are 5 column vectors, they must be linearly dependent.

20. Using the **RowReduce** in *Mathematica* we find that the reduced row-echelon form of the augmented matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \frac{834}{2215} & -\frac{261}{443} \\ 0 & 1 & 0 & 0 & 0 & \frac{1818}{2215} & \frac{282}{443} \\ 0 & 0 & 1 & 0 & 0 & \frac{13}{443} & -\frac{6}{443} \\ 0 & 0 & 0 & 1 & 0 & \frac{4214}{2215} & -\frac{130}{443} \\ 0 & 0 & 0 & 0 & 1 & -\frac{6079}{2215} & \frac{677}{443} \end{pmatrix}.$$

We conclude that the system is consistent and the solution is $x_1 = -\frac{226}{443} - \frac{834}{2215}c$, $x_2 = \frac{282}{443} - \frac{1818}{2215}c$, $x_3 = -\frac{6}{443} - \frac{13}{443}c$, $x_4 = -\frac{130}{443} - \frac{4214}{2215}c$, $x_5 = \frac{677}{443} + \frac{6079}{2215}c$, $x_6 = c$.

EXERCISES 8.4

Determinants

1. $M_{12} = \begin{vmatrix} 1 & 2 \\ -2 & 5 \end{vmatrix} = 9$

2. $M_{32} = \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix}$

3. $C_{13} = (-1)^{1+3} \begin{vmatrix} 1 & -1 \\ -2 & 3 \end{vmatrix} = 1$

4. $C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 4 \\ -2 & 5 \end{vmatrix} = 18$

5. $M_{33} = \begin{vmatrix} 0 & 2 & 0 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{vmatrix} = 2$

6. $M_{41} = \begin{vmatrix} 2 & 4 & 0 \\ 2 & -2 & 3 \\ 1 & 0 & -1 \end{vmatrix} = 24$

7. $C_{34} = (-1)^{3+4} \begin{vmatrix} 0 & 2 & 4 \\ 1 & 2 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 10$

8. $C_{23} = (-1)^{2+3} \begin{vmatrix} 0 & 2 & 0 \\ 5 & 1 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 22$

9. -7

10. 2

11. 17

12. $-1/2$

13. $(1 - \lambda)(2 - \lambda) - 6 = \lambda^2 - 3\lambda - 4$

14. $(-3 - \lambda)(5 - \lambda) - 8 = \lambda^2 - 2\lambda - 23$

15. $\begin{vmatrix} 0 & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 5 & 8 \end{vmatrix} = -3 \begin{vmatrix} 2 & 0 \\ 5 & 8 \end{vmatrix} = -48$

16. $\begin{vmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 5 \begin{vmatrix} -3 & 0 \\ 0 & 2 \end{vmatrix} = 5(-3)(2) = -30$

17. $\begin{vmatrix} 3 & 0 & 2 \\ 2 & 7 & 1 \\ 2 & 6 & 4 \end{vmatrix} = 3 \begin{vmatrix} 7 & 1 \\ 6 & 4 \end{vmatrix} + 2 \begin{vmatrix} 2 & 7 \\ 2 & 6 \end{vmatrix} = 3(22) + 2(-2) = 62$

8.4 Determinants

$$18. \begin{vmatrix} 1 & -1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & 9 \end{vmatrix} = \begin{vmatrix} 2 & -2 \\ 1 & 9 \end{vmatrix} - 2 \begin{vmatrix} -1 & -1 \\ 1 & 9 \end{vmatrix} + \begin{vmatrix} -1 & -1 \\ 2 & -2 \end{vmatrix} = 20 - 2(-8) + 4 = 40$$

$$19. \begin{vmatrix} 4 & 5 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{vmatrix} = 4 \begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix} - 5 \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 0$$

$$20. \begin{vmatrix} \frac{1}{4} & 6 & 0 \\ \frac{1}{3} & 8 & 0 \\ \frac{1}{2} & 9 & 0 \end{vmatrix} = 0, \text{ expanding along the third column.}$$

$$21. \begin{vmatrix} -2 & -1 & 4 \\ -3 & 6 & 1 \\ -3 & 4 & 8 \end{vmatrix} = -2 \begin{vmatrix} 6 & 1 \\ 4 & 8 \end{vmatrix} + 3 \begin{vmatrix} -1 & 4 \\ 4 & 8 \end{vmatrix} - 3 \begin{vmatrix} -1 & 4 \\ 6 & 1 \end{vmatrix} = -2(44) + 3(-24) - 3(-25) = -85$$

$$22. \begin{vmatrix} 3 & 5 & 1 \\ -1 & 2 & 5 \\ 7 & -4 & 10 \end{vmatrix} = 3 \begin{vmatrix} 2 & 5 \\ -4 & 10 \end{vmatrix} - 5 \begin{vmatrix} -1 & 5 \\ 7 & 10 \end{vmatrix} + \begin{vmatrix} -1 & 2 \\ 7 & -4 \end{vmatrix} = 3(40) - 5(-45) + (-10) = 335$$

$$23. \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 2 & 3 & 4 \end{vmatrix} = \begin{vmatrix} y & z \\ 3 & 4 \end{vmatrix} - \begin{vmatrix} x & z \\ 2 & 4 \end{vmatrix} + \begin{vmatrix} x & y \\ 2 & 3 \end{vmatrix} = (4y - 3z) - (4x - 2z) + (3x - 2y) = -x + 2y - z$$

$$24. \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 2+x & 3+y & 4+z \end{vmatrix} = \begin{vmatrix} y & z \\ 3+y & 4+z \end{vmatrix} - \begin{vmatrix} x & z \\ 2+x & 4+z \end{vmatrix} + \begin{vmatrix} x & y \\ 2+x & 3+y \end{vmatrix} \\ = (4y + yz - 3z - yz) - (4x + xz - 2z - xz) + (3x + xy - 2y - xy) = -x + 2y - z$$

$$25. \begin{vmatrix} 1 & 1 & -3 & 0 \\ 1 & 5 & 3 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 8 & 0 & 0 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & -3 \\ 1 & -2 & 1 \\ 4 & 8 & 0 \end{vmatrix} = 2(4) \begin{vmatrix} 1 & -3 \\ -2 & 1 \end{vmatrix} - 2(8) \begin{vmatrix} 1 & -3 \\ 1 & 1 \end{vmatrix} = 8(-5) - 16(4) = -104$$

$$26. \begin{vmatrix} 2 & 1 & -2 & 1 \\ 0 & 5 & 0 & 4 \\ 1 & 6 & 1 & 0 \\ 5 & -1 & 1 & 1 \end{vmatrix} = 5 \begin{vmatrix} 2 & -2 & 1 \\ 1 & 1 & 0 \\ 5 & 1 & 1 \end{vmatrix} + 4 \begin{vmatrix} 2 & 1 & -2 \\ 1 & 6 & 1 \\ 5 & -1 & 1 \end{vmatrix} = 5(0) + 4(80) = 320$$

27. Expanding along the first column in the original matrix and each succeeding minor, we obtain $3(1)(2)(4)(2) = 48$.

28. Expanding along the bottom row we obtain

$$-1 \begin{vmatrix} 2 & 0 & 0 & -2 \\ 1 & 6 & 0 & 5 \\ 1 & 2 & -1 & 1 \\ 2 & 1 & -2 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 2 & 0 & 0 \\ 1 & 1 & 6 & 0 \\ 1 & 0 & 2 & -1 \\ 2 & 0 & 1 & -2 \end{vmatrix} = -1(-48) + 0 = 48.$$

29. Solving $\lambda^2 - 2\lambda - 15 - 20 = \lambda^2 - 2\lambda - 35 = (\lambda - 7)(\lambda + 5) = 0$ we obtain $\lambda = 7$ and -5 .

30. Solving $-\lambda^3 + 3\lambda^2 - 2\lambda = -\lambda(\lambda - 2)(\lambda - 1) = 0$ we obtain $\lambda = 0, 1$, and 2 .

EXERCISES 8.5

Properties of Determinants

1. Theorem 8.11
2. Theorem 8.14
3. Theorem 8.14
4. Theorem 8.12 and 8.11
5. Theorem 8.12 (twice)
6. Theorem 8.11 (twice)
7. Theorem 8.10
8. Theorem 8.12 and 8.9
9. Theorem 8.8
10. Theorem 8.11 (twice)
11. $\det \mathbf{A} = -5$
12. $\det \mathbf{B} = 2(3)(5) = 30$
13. $\det \mathbf{C} = -5$
14. $\det \mathbf{D} = 5$
15. $\det \mathbf{A} = 6(\frac{2}{3})(-4)(-5) = 80$
16. $\det \mathbf{B} = -a_{13}a_{22}a_{31}$
17. $\det \mathbf{C} = (-5)(7)(3) = -105$
18. $\det \mathbf{D} = 4(7)(-2) = -56$
19. $\det \mathbf{A} = 14 = \det \mathbf{A}^T$
20. $\det \mathbf{A} = 96 = \det^T$

$$21. \det \mathbf{AB} = \begin{vmatrix} 0 & -2 & 2 \\ 10 & 7 & 23 \\ 8 & 4 & 16 \end{vmatrix} = -80 = 20(-4) = \det \mathbf{A} \det \mathbf{B}$$

22. From Problem 21, $(\det \mathbf{A})^2 = \det \mathbf{A}^2 = \det \mathbf{I} = 1$, so $\det \mathbf{A} = \pm 1$.

$$23. \text{ Using Theorems 8.14, 8.12, and 8.9, } \det \mathbf{A} = \begin{vmatrix} a & 1 & 2 \\ b & 1 & 2 \\ c & 1 & 2 \end{vmatrix} = 2 \begin{vmatrix} a & 1 & 1 \\ b & 1 & 1 \\ c & 1 & 1 \end{vmatrix} = 0.$$

24. Using Theorems 8.14 and 8.9,

$$\det \mathbf{A} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x+y+z & x+y+z & x+y+z \end{vmatrix} = (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

$$25. \begin{vmatrix} 1 & 1 & 5 \\ 4 & 3 & 6 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 5 \\ 0 & -1 & -14 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 5 \\ 0 & -1 & -14 \\ 0 & 0 & 15 \end{vmatrix} = 1(-1)(15) = -15$$

$$26. \begin{vmatrix} 2 & 4 & 5 \\ 4 & 2 & 0 \\ 8 & 7 & -2 \end{vmatrix} = \begin{vmatrix} 2 & 4 & 5 \\ 0 & -6 & -10 \\ 0 & -9 & -22 \end{vmatrix} = -2 \begin{vmatrix} 2 & 4 & 5 \\ 0 & 3 & 5 \\ 0 & -9 & -22 \end{vmatrix} = -2 \begin{vmatrix} 2 & 4 & 5 \\ 0 & 3 & 5 \\ 0 & 0 & -7 \end{vmatrix} = -2(2)(3)(-7) = 84$$

8.5 Properties of Determinants

$$27. \begin{vmatrix} -1 & 2 & 3 \\ 4 & -5 & -2 \\ 9 & -9 & 6 \end{vmatrix} = \begin{vmatrix} -1 & 2 & 3 \\ 0 & 3 & 10 \\ 0 & 9 & 33 \end{vmatrix} = \begin{vmatrix} -1 & 2 & 3 \\ 0 & 3 & 10 \\ 0 & 0 & 3 \end{vmatrix} = -1(3)(3) = -9$$

$$28. \begin{vmatrix} -2 & 2 & -6 \\ 5 & 0 & 1 \\ 1 & -2 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 2 \\ 5 & 0 & 1 \\ -2 & 2 & -6 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 2 \\ 0 & 10 & -9 \\ 0 & -2 & -2 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 2 \\ 0 & 10 & -9 \\ 0 & 0 & -\frac{19}{5} \end{vmatrix} = -1(10)(-\frac{19}{5}) = 38$$

$$29. \begin{vmatrix} 1 & -2 & 2 & 1 \\ 2 & 1 & -2 & 3 \\ 3 & 4 & -8 & 1 \\ 3 & -11 & 12 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 2 & 1 \\ 0 & 5 & -6 & 1 \\ 0 & 10 & -14 & -2 \\ 0 & -5 & 6 & -1 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 2 & 1 \\ 0 & 5 & -6 & 1 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{vmatrix} = 1(5)(-2)(0) = 0$$

$$30. \begin{vmatrix} 0 & 1 & 4 & 5 \\ 2 & 5 & 0 & 1 \\ 1 & 2 & 2 & 0 \\ 3 & 1 & 3 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 2 & 5 & 0 & 1 \\ 0 & 1 & 4 & 5 \\ 3 & 1 & 3 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 1 & 4 & 5 \\ 0 & -5 & -3 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 8 & 4 \\ 0 & 0 & -23 & 7 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 8 & 4 \\ 0 & 0 & -23 & \frac{37}{2} \end{vmatrix} \\ = -(1)(1)(8)(\frac{37}{2}) = -148$$

$$31. \begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 2 & 3 & 6 & 7 \\ 1 & 5 & 8 & 20 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & 3 & 5 & 16 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -1 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 8 \end{vmatrix} = 1(1)(2)(8) = 16$$

$$32. \begin{vmatrix} 2 & 9 & 1 & 8 \\ 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 3 & 1 & 4 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 3 & 7 & 4 \\ 2 & 9 & 1 & 8 \\ 0 & 1 & 6 & 5 \\ 3 & 1 & 4 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 3 & -13 & 0 \\ 0 & 1 & 6 & 5 \\ 0 & -8 & -17 & -10 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 0 & 3 & -13 & 0 \\ 0 & -8 & -17 & -10 \end{vmatrix} \\ = \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & -31 & -15 \\ 0 & 0 & 31 & 30 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & -31 & -15 \\ 0 & 0 & 0 & 15 \end{vmatrix} = 1(1)(-31)(15) = -465$$

33. We first use the second row to reduce the third row. Then we use the first row to reduce the second row.

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 0 & b^2 - ab & c^2 - ac \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & b - a & c - a \\ 0 & b(b - a) & c(c - a) \end{vmatrix} = (b - a)(c - a) \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & b & c \end{vmatrix}.$$

Expanding along the first row gives $(b - a)(c - a)(c - b)$.

34. In order, we use the third row to reduce the fourth row, the second row to reduce the third row, and the first row to reduce the second row. We then pull out a common factor from each column.

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b - a & c - a & d - a \\ 0 & b^2 - ab & c^2 - ac & d^2 - ad \\ 0 & b^3 - ab^2 & c^3 - ac^2 & d^3 - ad^2 \end{vmatrix} = (b - a)(c - a)(d - a) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & b & c & d \\ 0 & b^2 & c^2 & d^2 \end{vmatrix}.$$

Expanding along the first column and using Problem 33 we obtain $(b - a)(c - a)(d - a)(c - b)(d - b)(d - c)$.

35. Since $C_{11} = 4$, $C_{12} = 5$, and $C_{13} = -6$, we have $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = (-1)(4) + 2(5) + 1(-6) = 0$. Since $C_{12} = 5$, $C_{22} = -7$, and $C_{23} = -3$, we have $a_{13}C_{12} + a_{23}C_{22} + a_{33}C_{32} = 2(5) + 1(-7) + 1(-3) = 0$.
36. Since $C_{11} = -7$, $C_{12} = -8$, and $C_{13} = -10$ we have $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = -2(-7) + 3(-8) - 1(-10) = 0$. Since $C_{12} = -8$, $C_{22} = -19$, and $C_{32} = -7$ we have $a_{13}C_{12} + a_{23}C_{22} + a_{33}C_{32} = 5(-8) - 1(-19) - 3(-7) = 0$.
37. $\det(\mathbf{A} + \mathbf{B}) = \begin{vmatrix} 10 & 0 \\ 0 & -3 \end{vmatrix} = -30$; $\det \mathbf{A} + \det \mathbf{B} = 10 - 31 = -21$
38. $\det(2\mathbf{A}) = 2^5 \det \mathbf{A} = 32(-7) = -224$
39. Factoring -1 out of each row we see that $\det(-\mathbf{A}) = (-1)^5 \det \mathbf{A} = -\det \mathbf{A}$. Then $-\det \mathbf{A} = \det(-\mathbf{A}) = \det \mathbf{A}^T = \det \mathbf{A}$ and $\det \mathbf{A} = 0$.
40. (a) Cofactors: $25! \approx 1.55(10^{25})$; Row reduction: $25^3/3 \approx 5.2(10^3)$
 (b) Cofactors: about 90 billion centuries; Row reduction: about $\frac{1}{10}$ second

EXERCISES 8.6

Inverse of a Matrix

1. $\mathbf{AB} = \begin{pmatrix} 3-2 & -1+1 \\ 6-6 & -2+3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
2. $\mathbf{AB} = \begin{pmatrix} 2-1 & -1+1 & -2+2 \\ 6-6 & -3+4 & 6-6 \\ 2+1-3 & -1-1+2 & 2+2-3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
3. $\det \mathbf{A} = 9$. $\mathbf{A}$ is nonsingular. $\mathbf{A}^{-1} = \frac{1}{9} \begin{pmatrix} 1 & 1 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} \frac{1}{9} & \frac{1}{9} \\ -\frac{4}{9} & \frac{5}{9} \end{pmatrix}$
4. $\det \mathbf{A} = 5$. $\mathbf{A}$ is nonsingular. $\mathbf{A}^{-1} = \frac{1}{5} \begin{pmatrix} 3 & 1 \\ -4 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{4}{5} & \frac{1}{15} \end{pmatrix}$
5. $\det \mathbf{A} = 12$. $\mathbf{A}$ is nonsingular. $\mathbf{A}^{-1} = \frac{1}{12} \begin{pmatrix} 2 & 0 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & 0 \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix}$
6. $\det \mathbf{A} = -3\pi^2$. $\mathbf{A}$ is nonsingular. $\mathbf{A}^{-1} = -\frac{1}{3\pi^2} \begin{pmatrix} \pi & \pi \\ \pi & -2\pi \end{pmatrix} = \begin{pmatrix} -\frac{1}{3\pi} & -\frac{1}{3\pi} \\ -\frac{1}{3\pi} & \frac{2}{3\pi} \end{pmatrix}$
7. $\det \mathbf{A} = -16$. $\mathbf{A}$ is nonsingular. $\mathbf{A}^{-1} = -\frac{1}{16} \begin{pmatrix} 8 & -8 & -8 \\ 2 & -4 & 6 \\ -6 & 4 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{8} & \frac{1}{4} & -\frac{3}{8} \\ \frac{3}{8} & -\frac{1}{4} & \frac{1}{8} \end{pmatrix}$
8. $\det \mathbf{A} = 0$. $\mathbf{A}$ is singular.
9. $\det \mathbf{A} = -30$. $\mathbf{A}$ is nonsingular. $\mathbf{A}^{-1} = -\frac{1}{30} \begin{pmatrix} -14 & 13 & 16 \\ -2 & 4 & -2 \\ -4 & -7 & -4 \end{pmatrix} = \begin{pmatrix} \frac{7}{15} & -\frac{13}{30} & -\frac{8}{15} \\ \frac{1}{15} & -\frac{2}{15} & \frac{1}{15} \\ \frac{2}{15} & \frac{7}{30} & \frac{2}{15} \end{pmatrix}$

8.6 Inverse of a Matrix

$$10. \det \mathbf{A} = 78. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{78} \begin{pmatrix} 8 & 20 & 2 \\ -2 & -5 & 19 \\ 12 & -9 & 3 \end{pmatrix} \begin{pmatrix} \frac{4}{39} & \frac{10}{39} & \frac{1}{39} \\ -\frac{1}{39} & -\frac{5}{78} & \frac{19}{78} \\ \frac{2}{13} & -\frac{3}{26} & \frac{1}{26} \end{pmatrix}$$

$$11. \det \mathbf{A} = -36. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{36} \begin{pmatrix} -12 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 18 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{6} & 0 \\ 0 & 0 & -\frac{1}{2} \end{pmatrix}$$

$$12. \det \mathbf{A} = 16. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{16} \begin{pmatrix} 0 & 0 & 2 \\ 8 & 0 & 0 \\ 0 & 16 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & \frac{1}{8} \\ \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$13. \det \mathbf{A} = 27. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{27} \begin{pmatrix} 6 & 21 & -9 & -36 \\ -1 & 1 & 6 & -3 \\ 10 & 17 & -6 & -51 \\ 4 & -4 & 3 & 12 \end{pmatrix} = \begin{pmatrix} \frac{2}{9} & \frac{7}{9} & -\frac{1}{3} & -\frac{4}{3} \\ -\frac{1}{27} & \frac{1}{27} & \frac{2}{9} & -\frac{1}{9} \\ \frac{10}{27} & \frac{17}{27} & -\frac{2}{9} & -\frac{17}{9} \\ \frac{4}{27} & -\frac{4}{27} & \frac{1}{9} & \frac{4}{9} \end{pmatrix}$$

$$14. \det \mathbf{A} = -6. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{6} \begin{pmatrix} 0 & 1 & -3 & 3 \\ 0 & 1 & 3 & -9 \\ 0 & -2 & 0 & 0 \\ -6 & -1 & -3 & 15 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{6} & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{6} & -\frac{1}{2} & \frac{3}{2} \\ 0 & \frac{1}{3} & 0 & 0 \\ 1 & \frac{1}{6} & \frac{1}{2} & -\frac{5}{2} \end{pmatrix}$$

$$15. \left(\begin{array}{cc|cc} 6 & -2 & 1 & 0 \\ 0 & 4 & 0 & 1 \end{array} \right) \xrightarrow[\frac{1}{4}R_2]{\frac{1}{6}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & \frac{1}{6} & 0 \\ 0 & 1 & 0 & \frac{1}{4} \end{array} \right) \xrightarrow{\frac{1}{3}R_2+R_1} \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{6} & \frac{1}{12} \\ 0 & 1 & 0 & \frac{1}{4} \end{array} \right); \mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{6} & \frac{1}{12} \\ 0 & \frac{1}{4} \end{pmatrix}$$

$$16. \left(\begin{array}{cc|cc} 8 & 0 & 1 & 0 \\ 0 & \frac{1}{2} & 0 & 1 \end{array} \right) \xrightarrow[2R_2]{\frac{1}{8}R_1} \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{8} & 0 \\ 0 & 1 & 0 & 2 \end{array} \right); \mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{8} & 0 \\ 0 & 2 \end{pmatrix}$$

$$17. \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array} \right) \xrightarrow{-5R_1+R_2} \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & -12 & -5 & 1 \end{array} \right) \xrightarrow{-\frac{1}{12}R_2} \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & \frac{5}{12} & -\frac{1}{12} \end{array} \right) \xrightarrow{-3R_2+R_1} \left(\begin{array}{cc|cc} 1 & 0 & -\frac{1}{4} & \frac{1}{4} \\ 0 & 1 & \frac{5}{12} & -\frac{1}{12} \end{array} \right);$$

$$\mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{4} & \frac{1}{4} \\ \frac{5}{12} & -\frac{1}{12} \end{pmatrix}$$

$$18. \left(\begin{array}{cc|cc} 2 & -3 & 1 & 0 \\ -2 & 4 & 0 & 1 \end{array} \right) \xrightarrow{\frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{3}{2} & \frac{1}{2} & 0 \\ -2 & 4 & 0 & 1 \end{array} \right) \xrightarrow{2R_1+R_2} \left(\begin{array}{cc|cc} 1 & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 1 & 1 \end{array} \right) \xrightarrow{\frac{3}{2}R_2+R_1} \left(\begin{array}{cc|cc} 1 & 0 & 2 & \frac{3}{2} \\ 0 & 1 & 1 & 1 \end{array} \right);$$

$$\mathbf{A}^{-1} = \begin{pmatrix} 2 & \frac{3}{2} \\ 1 & 1 \end{pmatrix}$$

$$19. \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & \frac{4}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right); \mathbf{A} \text{ is singular.}$$

$$20. \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 & 1 & 0 \\ 2 & -1 & 3 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{9} & -\frac{1}{9} & \frac{2}{9} \\ 0 & 1 & 0 & -\frac{2}{9} & -\frac{5}{9} & \frac{1}{9} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{9} & \frac{2}{9} \end{array} \right); \mathbf{A}^{-1} = \begin{pmatrix} \frac{5}{9} & -\frac{1}{9} & \frac{2}{9} \\ -\frac{2}{9} & -\frac{5}{9} & \frac{1}{9} \\ -\frac{4}{9} & -\frac{1}{9} & \frac{2}{9} \end{pmatrix}$$

$$21. \left(\begin{array}{ccc|ccc} 4 & 2 & 3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ -1 & -2 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_{13}} \left(\begin{array}{ccc|ccc} -1 & -2 & 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & 2 & 3 & 1 & 0 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 1 & 0 & 0 & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{2}{3} & 0 \end{array} \right);$$

$$\mathbf{A}^{-1} = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & 0 \end{pmatrix}$$

$$22. \left(\begin{array}{ccc|ccc} 2 & 4 & -2 & 1 & 0 & 0 \\ 4 & 2 & -2 & 0 & 1 & 0 \\ 8 & 10 & -6 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & \frac{1}{3} & -\frac{1}{6} & 0 \\ 0 & 0 & 0 & -2 & -1 & 1 \end{array} \right); \quad \mathbf{A} \text{ is singular.}$$

$$23. \left(\begin{array}{ccc|ccc} -1 & 3 & 0 & 1 & 0 & 0 \\ 3 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & 6 & -3 \\ 0 & 1 & 0 & 2 & 2 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right);$$

$$\mathbf{A}^{-1} = \begin{pmatrix} 5 & 6 & -3 \\ 2 & 2 & -1 \\ -1 & -1 & 1 \end{pmatrix}$$

$$24. \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 8 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & \frac{5}{8} \\ 0 & 1 & 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{8} \end{array} \right); \quad \mathbf{A}^{-1} = \begin{pmatrix} 1 & -2 & \frac{5}{8} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{8} \end{pmatrix}$$

$$25. \left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 2 & 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & -3 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & 1 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & -\frac{2}{3} & \frac{1}{3} & -1 & -\frac{2}{3} & 0 \\ 0 & 0 & 0 & 1 & -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \end{array} \right)$$

$$\xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{1}{2} & -\frac{2}{3} & -\frac{1}{6} & \frac{7}{6} \\ 0 & 1 & 0 & 0 & 1 & \frac{1}{3} & \frac{1}{3} & -\frac{4}{3} \\ 0 & 0 & 1 & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \end{array} \right); \quad \mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{2} & -\frac{2}{3} & -\frac{1}{6} & \frac{7}{6} \\ 1 & \frac{1}{3} & \frac{1}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$26. \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{interchange}]{\text{row}} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right); \quad \mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$27. (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ -1 & \frac{10}{3} \end{pmatrix}$$

$$28. (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} -1 & -4 & 20 \\ 2 & 6 & -30 \\ 3 & 6 & -32 \end{pmatrix}$$

$$29. \mathbf{A} = (\mathbf{A}^{-1})^{-1} = \begin{pmatrix} -2 & 3 \\ 3 & -4 \end{pmatrix}$$

$$30. \mathbf{A}^T = \begin{pmatrix} 1 & 2 \\ 4 & 10 \end{pmatrix}; \quad (\mathbf{A}^T)^{-1} = \begin{pmatrix} 5 & -1 \\ -2 & \frac{1}{2} \end{pmatrix}; \quad \mathbf{A}^{-1} = \begin{pmatrix} 5 & -2 \\ -1 & \frac{1}{2} \end{pmatrix}; \quad (\mathbf{A}^{-1})^T = \begin{pmatrix} 5 & -1 \\ -2 & \frac{1}{2} \end{pmatrix}$$

$$31. \text{ Multiplying } \begin{pmatrix} 4 & -3 \\ x & -4 \end{pmatrix} \begin{pmatrix} 4 & -3 \\ x & -4 \end{pmatrix} = \begin{pmatrix} 16-3x & 0 \\ 0 & 16-3x \end{pmatrix} \text{ we see that } x = 5.$$

8.6 Inverse of a Matrix

$$32. \mathbf{A}^{-1} = \begin{pmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{pmatrix}$$

$$33. \text{ (a) } \mathbf{A}^T = \begin{pmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{pmatrix} = \mathbf{A}^{-1} \qquad \text{ (b) } \mathbf{A}^T = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix} = \mathbf{A}^{-1}$$

34. Since $\det \mathbf{A} \cdot \det \mathbf{A}^{-1} = \det \mathbf{A}\mathbf{A}^{-1} = \det \mathbf{I} = 1$, we see that $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$. If $\mathbf{A}$ is orthogonal, $\det \mathbf{A} = \det \mathbf{A}^T = \det \mathbf{A}^{-1} = 1/\det \mathbf{A}$ and $(\det \mathbf{A})^2 = 1$, so $\det \mathbf{A} = \pm 1$.

35. Since $\mathbf{A}$ and $\mathbf{B}$ are nonsingular, $\det \mathbf{AB} = \det \mathbf{A} \cdot \det \mathbf{B} \neq 0$, and $\mathbf{AB}$ is nonsingular.

36. Suppose $\mathbf{A}$ is singular. Then $\det \mathbf{A} = 0$, $\det \mathbf{AB} = \det \mathbf{A} \cdot \det \mathbf{B} = 0$, and $\mathbf{AB}$ is singular.

37. Since $\det \mathbf{A} \cdot \det \mathbf{A}^{-1} = \det \mathbf{A}\mathbf{A}^{-1} = \det \mathbf{I} = 1$, $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.

38. Suppose $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{A}$ is nonsingular. Then $\mathbf{A}^2\mathbf{A}^{-1} = \mathbf{A}\mathbf{A}^{-1}$, and $\mathbf{A} = \mathbf{I}$. Thus, if $\mathbf{A}^2 = \mathbf{A}$, either $\mathbf{A}$ is singular or $\mathbf{A} = \mathbf{I}$.

39. If $\mathbf{A}$ is nonsingular, then $\mathbf{A}^{-1}$ exists, and $\mathbf{AB} = \mathbf{0}$ implies $\mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{0}$, so $\mathbf{B} = \mathbf{0}$.

40. If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1}$ exists, and $\mathbf{AB} = \mathbf{AC}$ implies $\mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{AC}$, so $\mathbf{B} = \mathbf{C}$.

41. No, consider $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

42. $\mathbf{A}$ is nonsingular if $a_{11}a_{22}a_{33} \neq 0$ or a_{11} , a_{22} , and a_{33} are all nonzero.

$$\mathbf{A}^{-1} = \begin{pmatrix} 1/a_{11} & 0 & 0 \\ 0 & 1/a_{22} & 0 \\ 0 & 0 & 1/a_{33} \end{pmatrix}$$

For any diagonal matrix, the inverse matrix is obtained by taking the reciprocals of the diagonal entries and leaving all other entries 0.

$$43. \mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 4 \\ 14 \end{pmatrix} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}; \quad x_1 = 6, \quad x_2 = -2$$

$$44. \mathbf{A}^{-1} = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{6} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix}; \quad x_1 = \frac{1}{2}, \quad x_2 = -\frac{3}{2}$$

$$45. \mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{16} & \frac{3}{8} \\ -\frac{1}{8} & \frac{1}{4} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 6 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} \\ -\frac{1}{2} \end{pmatrix}; \quad x_1 = \frac{3}{4}, \quad x_2 = -\frac{1}{2}$$

$$46. \mathbf{A}^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} -11 \\ \frac{15}{2} \end{pmatrix}; \quad x_1 = -11, \quad x_2 = \frac{15}{2}$$

$$47. \mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ -1 & 1 & 0 \\ \frac{6}{5} & -\frac{1}{5} & -\frac{1}{5} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} -4 \\ 0 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ -6 \end{pmatrix}; \quad x_1 = 2, \quad x_2 = 4, \quad x_3 = -6$$

$$48. \mathbf{A}^{-1} = \begin{pmatrix} \frac{5}{12} & -\frac{1}{12} & \frac{1}{4} \\ -\frac{2}{3} & \frac{1}{3} & 0 \\ -\frac{1}{12} & \frac{5}{12} & -\frac{1}{4} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ \frac{3}{2} \end{pmatrix}; \quad x_1 = -\frac{1}{2}, \quad x_2 = 0, \quad x_3 = \frac{3}{2}$$

$$49. \mathbf{A}^{-1} = \begin{pmatrix} -2 & -3 & 2 \\ \frac{1}{4} & -\frac{1}{4} & 0 \\ \frac{5}{4} & \frac{7}{4} & -1 \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 1 \\ -3 \\ 7 \end{pmatrix} = \begin{pmatrix} 21 \\ 1 \\ -11 \end{pmatrix}; \quad x_1 = 21, \quad x_2 = 1, \quad x_3 = -11$$

$$50. \mathbf{A}^{-1} = \begin{pmatrix} 2 & -1 & 1 & 1 \\ -1 & 2 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} 2 \\ 1 \\ -5 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \\ -4 \end{pmatrix}; \quad x_1 = 1, \quad x_2 = 2, \quad x_3 = -1, \quad x_4 = -4$$

$$51. \begin{pmatrix} 7 & -2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}; \quad \mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{10} & \frac{1}{10} \\ -\frac{3}{20} & \frac{7}{20} \end{pmatrix}; \quad \mathbf{X} = \mathbf{A}^{-1} \begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} \frac{9}{10} \\ \frac{13}{20} \end{pmatrix}; \quad \mathbf{X} = \mathbf{A}^{-1} \begin{pmatrix} 10 \\ 50 \end{pmatrix} = \begin{pmatrix} 6 \\ 16 \end{pmatrix};$$

$$\mathbf{X} = \mathbf{A}^{-1} \begin{pmatrix} 0 \\ -20 \end{pmatrix} = \begin{pmatrix} -2 \\ -7 \end{pmatrix}$$

$$52. \begin{pmatrix} 1 & 2 & 5 \\ 2 & 3 & 8 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}; \quad \mathbf{A}^{-1} = \begin{pmatrix} 2 & -1 & -1 \\ 12 & -7 & -2 \\ -5 & 3 & 1 \end{pmatrix}; \quad \mathbf{X} = \mathbf{A}^{-1} \begin{pmatrix} -1 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} -12 \\ -52 \\ 23 \end{pmatrix};$$

$$\mathbf{X} = \mathbf{A}^{-1} \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 9 \\ -3 \end{pmatrix}; \quad \mathbf{X} = \mathbf{A}^{-1} \begin{pmatrix} 0 \\ -5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 27 \\ -11 \end{pmatrix}$$

53. $\det \mathbf{A} = 18 \neq 0$, so the system has only the trivial solution.

54. $\det \mathbf{A} = 0$, so the system has a nontrivial solution.

55. $\det \mathbf{A} = 0$, so the system has a nontrivial solution.

56. $\det \mathbf{A} = 12 \neq 0$, so the system has only the trivial solution.

$$57. (a) \begin{pmatrix} 1 & 1 & 1 \\ -R_1 & R_2 & 0 \\ 0 & -R_2 & R_3 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 0 \\ E_2 - E_1 \\ E_3 - E_2 \end{pmatrix}$$

(b) $\det \mathbf{A} = R_1 R_2 + R_1 R_3 + R_2 R_3 > 0$, so $\mathbf{A}$ is nonsingular.

$$(c) \mathbf{A}^{-1} = \frac{1}{R_1 R_2 + R_1 R_3 + R_2 R_3} \begin{pmatrix} R_2 R_3 & -R_2 - R_3 & -R_2 \\ R_1 R_3 & R_3 & -R_1 \\ R_1 R_2 & R_2 & R_1 + R_2 \end{pmatrix};$$

$$\mathbf{A}^{-1} \begin{pmatrix} 0 \\ E_2 - E_1 \\ E_3 - E_2 \end{pmatrix} = \frac{1}{R_1 R_2 + R_1 R_3 + R_2 R_3} \begin{pmatrix} R_2 E_1 - R_2 E_3 + R_3 E_1 - R_3 E_2 \\ R_1 E_2 - R_1 E_3 - R_3 E_1 + R_3 E_2 \\ -R_1 E_2 + R_1 E_3 - R_2 E_1 + R_2 E_3 \end{pmatrix}$$

58. (a) We write the equations in the form

$$-4u_1 + u_2 + u_4 = -200$$

$$u_1 - 4u_2 + u_3 = -300$$

$$u_2 - 4u_3 + u_4 = -300$$

$$u_1 + u_3 - 4u_4 = -200.$$

In matrix form this becomes

$$\begin{pmatrix} -4 & 1 & 0 & 1 \\ 1 & -4 & 1 & 0 \\ 0 & 1 & -4 & 1 \\ 1 & 0 & 1 & -4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} -200 \\ -300 \\ -300 \\ -200 \end{pmatrix}.$$

8.6 Inverse of a Matrix

$$(b) \mathbf{A}^{-1} = \begin{pmatrix} -\frac{7}{24} & -\frac{1}{12} & -\frac{1}{24} & -\frac{1}{12} \\ -\frac{1}{12} & -\frac{7}{24} & -\frac{1}{12} & -\frac{1}{24} \\ -\frac{1}{24} & -\frac{1}{12} & -\frac{7}{24} & -\frac{1}{12} \\ -\frac{1}{12} & -\frac{1}{24} & -\frac{1}{12} & -\frac{7}{24} \end{pmatrix}; \quad \mathbf{A}^{-1} \begin{pmatrix} -200 \\ -300 \\ -300 \\ -200 \end{pmatrix} = \begin{pmatrix} \frac{225}{2} \\ \frac{275}{2} \\ \frac{275}{2} \\ \frac{225}{2} \end{pmatrix}; \quad u_1 = u_4 = \frac{225}{2}, \quad u_2 = u_3 = \frac{275}{2}$$

EXERCISES 8.7

Cramer's Rule

- $\det \mathbf{A} = 10$, $\det \mathbf{A}_1 = -6$, $\det \mathbf{A}_2 = 12$; $x_1 = \frac{-6}{10} = -\frac{3}{5}$, $x_2 = \frac{12}{10} = \frac{6}{5}$
- $\det \mathbf{A} = -3$, $\det \mathbf{A}_1 = -6$, $\det \mathbf{A}_2 = -6$; $x_1 = \frac{-6}{-3} = 2$, $x_2 = \frac{-6}{-3} = 2$
- $\det \mathbf{A} = 0.3$, $\det \mathbf{A}_1 = 0.03$, $\det \mathbf{A}_2 = -0.09$; $x_1 = \frac{0.03}{0.3} = 0.1$, $x_2 = \frac{-0.09}{0.3} = -0.3$
- $\det \mathbf{A} = -0.015$, $\det \mathbf{A}_1 = -0.00315$, $\det \mathbf{A}_2 = -0.00855$; $x_1 = \frac{-0.00315}{-0.015} = 0.21$, $x_2 = \frac{-0.00855}{-0.015} = 0.57$
- $\det \mathbf{A} = 1$, $\det \mathbf{A}_1 = 4$, $\det \mathbf{A}_2 = -7$; $x = 4$, $y = -7$
- $\det \mathbf{A} = -70$, $\det \mathbf{A}_1 = -14$, $\det \mathbf{A}_2 = 35$; $r = \frac{-14}{-70} = \frac{1}{5}$, $s = \frac{35}{-70} = -\frac{1}{2}$
- $\det \mathbf{A} = 11$, $\det \mathbf{A}_1 = -44$, $\det \mathbf{A}_2 = 44$, $\det \mathbf{A}_3 = -55$; $x_1 = \frac{-44}{11} = -4$, $x_2 = \frac{44}{11} = 4$, $x_3 = \frac{-55}{11} = -5$
- $\det \mathbf{A} = -63$, $\det \mathbf{A}_1 = 173$, $\det \mathbf{A}_2 = -136$, $\det \mathbf{A}_3 = -\frac{61}{2}$; $x_1 = -\frac{173}{63}$, $x_2 = \frac{136}{63}$, $x_3 = \frac{61}{126}$
- $\det \mathbf{A} = -12$, $\det \mathbf{A}_1 = -48$, $\det \mathbf{A}_2 = -18$, $\det \mathbf{A}_3 = -12$; $u = \frac{48}{12} = 4$, $v = \frac{18}{12} = \frac{3}{2}$, $w = 1$
- $\det \mathbf{A} = 1$, $\det \mathbf{A}_1 = -2$, $\det \mathbf{A}_2 = 2$, $\det \mathbf{A}_3 = 5$; $x = -2$, $y = 2$, $z = 5$
- $\det \mathbf{A} = 6 - 5k$, $\det \mathbf{A}_1 = 12 - 7k$, $\det \mathbf{A}_2 = 6 - 7k$; $x_1 = \frac{12 - 7k}{6 - 5k}$, $x_2 = \frac{6 - 7k}{6 - 5k}$. The system is inconsistent for $k = 6/5$.
- (a) $\det \mathbf{A} = \epsilon - 1$, $\det \mathbf{A}_1 = \epsilon - 2$, $\det \mathbf{A}_2 = 1$; $x_1 = \frac{\epsilon - 2}{\epsilon - 1} = \frac{\epsilon - 1 - 1}{\epsilon - 1} = 1 - \frac{1}{\epsilon - 1}$, $x_2 = \frac{1}{\epsilon - 1}$
(b) When $\epsilon = 1.01$, $x_1 = -99$ and $x_2 = 100$. When $\epsilon = 0.99$, $x_1 = 101$ and $x_2 = -100$.
- $\det \mathbf{A} \approx 0.6428$, $\det \mathbf{A}_1 \approx 289.8$, $\det \mathbf{A}_2 \approx 271.9$; $x_1 \approx \frac{289.8}{0.6428} \approx 450.8$, $x_2 \approx \frac{271.9}{0.6428} \approx 423$
- We have $(\sin 30^\circ)F + (\sin 30^\circ)(0.5N) + N \sin 60^\circ = 400$ and $(\cos 30^\circ)F + (\cos 30^\circ)(0.5N) - N \cos 60^\circ = 0$. The system is
$$(\sin 30^\circ)F + (0.5 \sin 30^\circ + \sin 60^\circ)N = 400$$

$$(\cos 30^\circ)F + (0.5 \cos 30^\circ - \cos 60^\circ)N = 0.$$

$$\det \mathbf{A} \approx -1, \det \mathbf{A}_1 \approx -26.795, \det \mathbf{A}_2 \approx -346.41; F \approx 26.795, N \approx 346.41$$
- The system is
$$i_1 + i_2 - i_3 = 0$$

$$r_1 i_1 - r_2 i_2 = E_1 - E_2$$

$$r_2 i_2 + R i_3 = E_2$$

$$\det \mathbf{A} = -r_1 R - r_2 R - r_1 r_2, \det \mathbf{A}_3 = -r_1 E_2, -r_2 E_1; i_3 = \frac{r_1 E_2 + r_2 E_1}{r_1 R + r_2 R + r_1 r_2}$$

EXERCISES 8.8

The Eigenvalue Problem

1. $\mathbf{K}_3$ since $\begin{pmatrix} 4 & 2 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \end{pmatrix} = (-1) \begin{pmatrix} -2 \\ 5 \end{pmatrix}; \quad \lambda = -1$
2. $\mathbf{K}_1$ and $\mathbf{K}_2$ since $\begin{pmatrix} 2 & -1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 - \sqrt{2} \end{pmatrix} = \begin{pmatrix} \sqrt{2} \\ -2 + 2\sqrt{2} \end{pmatrix} = \sqrt{2} \begin{pmatrix} 1 \\ 2 - \sqrt{2} \end{pmatrix}, \quad \lambda = \sqrt{2}$
 $\begin{pmatrix} 2 & -1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 2 + \sqrt{2} \\ 2 \end{pmatrix} = \begin{pmatrix} 2 + 2\sqrt{2} \\ 2\sqrt{2} \end{pmatrix} = \sqrt{2} \begin{pmatrix} 2 + \sqrt{2} \\ 2 \end{pmatrix}; \quad \lambda = \sqrt{2}$
3. $\mathbf{K}_3$ since $\begin{pmatrix} 6 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -5 \\ 10 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} -5 \\ 10 \end{pmatrix}; \quad \lambda = 0$
4. $\mathbf{K}_2$ since $\begin{pmatrix} 2 & 8 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 2 + 2i \\ -1 \end{pmatrix} = \begin{pmatrix} -4 + 4i \\ -2i \end{pmatrix} = 2i \begin{pmatrix} 2 + 2i \\ -1 \end{pmatrix}; \quad \lambda = 2i$
5. $\mathbf{K}_2$ and $\mathbf{K}_3$ since $\begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 12 \\ -12 \\ 0 \end{pmatrix} = 3 \begin{pmatrix} 4 \\ -4 \\ 0 \end{pmatrix}; \quad \lambda = 3$
 $\begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}; \quad \lambda = 1$
6. $\mathbf{K}_2$ since $\begin{pmatrix} -1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 12 \\ 9 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}; \quad \lambda = 3$
7. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -1 - \lambda & 2 \\ -7 & 8 - \lambda \end{vmatrix} = (\lambda - 6)(\lambda - 1) = 0.$

For $\lambda_1 = 6$ we have

$$\left(\begin{array}{cc|c} -7 & 2 & 0 \\ -7 & 2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -2/7 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = \frac{2}{7}k_2$. If $k_2 = 7$ then $\mathbf{K}_1 = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$. For $\lambda_2 = 1$ we have

$$\left(\begin{array}{cc|c} -2 & 2 & 0 \\ -7 & 7 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = k_2$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

8. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 2 - \lambda & 1 \\ 2 & 1 - \lambda \end{vmatrix} = \lambda(\lambda - 3) = 0.$

For $\lambda_1 = 0$ we have

$$\left(\begin{array}{cc|c} 2 & 1 & 0 \\ 2 & 1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & 1/2 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

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so that $k_1 = -\frac{1}{2}k_2$. If $k_2 = 2$ then $\mathbf{K}_1 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. For $\lambda_2 = 3$ we have

$$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 2 & -2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = k_2$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

9. We solve
$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -8 - \lambda & -1 \\ 16 & -\lambda \end{vmatrix} = (\lambda + 4)^2 = 0.$$

For $\lambda_1 = \lambda_2 = -4$ we have
$$\left(\begin{array}{cc|c} -4 & -1 & 0 \\ 16 & 4 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & 1/4 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = -\frac{1}{4}k_2$. If $k_2 = 4$ then $\mathbf{K}_1 = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

10. We solve
$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & 1 \\ 1/4 & 1 - \lambda \end{vmatrix} = (\lambda - 3/2)(\lambda - 1/2) = 0.$$

For $\lambda_1 = 3/2$ we have
$$\left(\begin{array}{cc|c} -1/2 & 1 & 0 \\ 1/4 & -1/2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = 2k_2$. If $k_2 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. If $\lambda_2 = 1/2$ then

$$\left(\begin{array}{cc|c} 1/2 & 1 & 0 \\ 1/4 & 1/2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = -2k_2$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

11. We solve
$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -1 - \lambda & 2 \\ -5 & 1 - \lambda \end{vmatrix} = \lambda^2 + 9 = (\lambda - 3i)(\lambda + 3i) = 0.$$

For $\lambda_1 = 3i$ we have
$$\left(\begin{array}{cc|c} -1 - 3i & 2 & 0 \\ -5 & 1 - 3i & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -(1/5) + (3/5)i & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = (\frac{1}{5} - \frac{3}{5}i)k_2$. If $k_2 = 5$ then $\mathbf{K}_1 = \begin{pmatrix} 1 - 3i \\ 5 \end{pmatrix}$. For $\lambda_2 = -3i$ we have

$$\left(\begin{array}{cc|c} -1 + 3i & 2 & 0 \\ -5 & 1 + 3i & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -\frac{1}{5} - \frac{3}{5}i & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = (\frac{1}{5} + \frac{3}{5}i)k_2$. If $k_2 = 5$ then $\mathbf{K}_2 = \begin{pmatrix} 1 + 3i \\ 5 \end{pmatrix}$.

12. We solve
$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & -1 \\ 1 & 1 - \lambda \end{vmatrix} = \lambda^2 - 2\lambda + 2 = 0.$$

For $\lambda_1 = 1 - i$ we have
$$\left(\begin{array}{cc|c} i & -1 & 0 \\ 1 & i & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} i & -1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = -ik_2$. If $k_2 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$ and $\mathbf{K}_2 = \overline{\mathbf{K}}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$.

13. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 4 - \lambda & 8 \\ 0 & -5 - \lambda \end{vmatrix} = (\lambda - 4)(\lambda + 5) = 0.$

For $\lambda_1 = 4$ we have $\begin{pmatrix} 0 & 8 & | & 0 \\ 0 & -9 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix}$

so that $k_2 = 0$. If $k_1 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. For $\lambda_2 = -5$ we have

$$\begin{pmatrix} 9 & 8 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & \frac{8}{9} & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix}$$

so that $k_1 = -\frac{8}{9}k_2$. If $k_2 = 9$ then $\mathbf{K}_2 = \begin{pmatrix} -8 \\ 9 \end{pmatrix}$.

14. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 7 - \lambda & 0 \\ 0 & 13 - \lambda \end{vmatrix} = (\lambda - 7)(\lambda - 13) = 0.$

For $\lambda_1 = 7$ we have $\begin{pmatrix} 0 & 0 & | & 0 \\ 0 & 6 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix}$

so that $k_2 = 0$. If $k_1 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. For $\lambda_2 = 13$ we have

$$\begin{pmatrix} -6 & 0 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix}$$

so that $k_1 = 0$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

15. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 5 - \lambda & -1 & 0 \\ 0 & -5 - \lambda & 9 \\ 5 & -1 & -\lambda \end{vmatrix} = \begin{vmatrix} 4 - \lambda & -1 & 0 \\ 4 - \lambda & -5 - \lambda & 9 \\ 4 - \lambda & -1 & -\lambda \end{vmatrix} = \lambda(4 - \lambda)(\lambda + 4) = 0.$

For $\lambda_1 = 0$ we have $\begin{pmatrix} 5 & -1 & 0 & | & 0 \\ 0 & -5 & 9 & | & 0 \\ 5 & -1 & 0 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & -9/25 & | & 0 \\ 0 & 1 & -9/5 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$

so that $k_1 = \frac{9}{25}k_3$ and $k_2 = \frac{9}{5}k_3$. If $k_3 = 25$ then $\mathbf{K}_1 = \begin{pmatrix} 9 \\ 45 \\ 25 \end{pmatrix}$. If $\lambda_2 = 4$ then

$$\begin{pmatrix} 1 & -1 & 0 & | & 0 \\ 0 & -9 & 9 & | & 0 \\ 5 & -1 & -4 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

so that $k_1 = k_3$ and $k_2 = k_3$. If $k_3 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. If $\lambda_3 = -4$ then

$$\begin{pmatrix} 9 & -1 & 0 & | & 0 \\ 0 & -1 & 9 & | & 0 \\ 5 & -1 & 4 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & -1 & | & 0 \\ 0 & 1 & -9 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

so that $k_1 = k_3$ and $k_2 = 9k_3$. If $k_3 = 1$ then $\mathbf{K}_3 = \begin{pmatrix} 1 \\ 9 \\ 1 \end{pmatrix}$.

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16. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 4 & 0 & 1-\lambda \end{vmatrix} = (3-\lambda)(2-\lambda)(1-\lambda) = 0.$

For $\lambda_1 = 1$ we have $\left(\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_1 = 0$ and $k_2 = 0$. If $k_3 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. If $\lambda_2 = 2$ then

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & -1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = 0$ and $k_3 = 0$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. If $\lambda_3 = 3$ then

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 4 & 0 & -2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = \frac{1}{2}k_3$ and $k_2 = 0$. If $k_3 = 2$ then $\mathbf{K}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$.

17. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & 4 & 0 \\ -1 & -4-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix} = -(\lambda+2)^3 = 0.$

For $\lambda_1 = \lambda_2 = \lambda_3 = -2$ we have $\left(\begin{array}{ccc|c} 2 & 4 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_1 = -2k_2$. If $k_2 = 1$ and $k_3 = 1$ then

$$\mathbf{K}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

18. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1-\lambda & 6 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 1 & 2-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & 6 & 0 \\ 0 & 3-\lambda & 3-\lambda \\ 0 & 1 & 2-\lambda \end{vmatrix} = (3-\lambda)(1-\lambda)^2 = 0.$

For $\lambda_1 = 3$ we have $\left(\begin{array}{ccc|c} -2 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_1 = 3k_3$ and $k_2 = k_3$. If $k_3 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$. For $\lambda_2 = \lambda_3 = 1$ we have

$$\left(\begin{array}{ccc|c} 0 & 6 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_2 = 0$ and $k_3 = 0$. If $k_1 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

19. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} -\lambda & 0 & -1 \\ 1 & -\lambda & 0 \\ 1 & 1 & -1-\lambda \end{vmatrix} = -(\lambda+1)(\lambda^2+1) = 0.$

For $\lambda_1 = -1$ we have $\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_1 = k_3$ and $k_2 = -k_3$. If $k_3 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$. For $\lambda_2 = i$ we have

$$\left(\begin{array}{ccc|c} -i & 0 & -1 & 0 \\ 1 & -i & 0 & 0 \\ 1 & 1 & -1-i & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -i & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = ik_3$ and $k_2 = k_3$. If $k_3 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} i \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{K}_3 = \overline{\mathbf{K}_2} = \begin{pmatrix} -i \\ 1 \\ 1 \end{pmatrix}$.

20. We solve

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 2-\lambda & -1 & 0 \\ 5 & 2-\lambda & 4 \\ 0 & 1 & 2-\lambda \end{vmatrix} = -\lambda^3 + 6\lambda^2 - 13\lambda + 10 = (\lambda-2)(-\lambda^2 + 4\lambda - 5) \\ = (\lambda-2)(\lambda-(2+i))(\lambda-(2-i)) = 0.$$

For $\lambda_1 = 2$ we have $\left(\begin{array}{ccc|c} 0 & -1 & 0 & 0 \\ 5 & 0 & 4 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 4/5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_1 = -\frac{4}{5}k_3$ and $k_2 = 0$. If $k_3 = 5$ then $\mathbf{K}_1 = \begin{pmatrix} -4 \\ 0 \\ 5 \end{pmatrix}$. For $\lambda_2 = 2+i$ we have

$$\left(\begin{array}{ccc|c} -i & -1 & 0 & 0 \\ 5 & -i & 4 & 0 \\ 0 & 1 & -i & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & -i & 0 & 0 \\ 0 & 1 & -i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

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so that $k_1 = ik_2$ and $k_2 = ik_3$. If $k_3 = i$ then $\mathbf{K}_2 = \begin{pmatrix} -i \\ -1 \\ i \end{pmatrix}$. For $\lambda_3 = 2 - i$ we have

$$\left(\begin{array}{ccc|c} i & -1 & 0 & 0 \\ 5 & i & 4 & 0 \\ 0 & 1 & i & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & i & 0 & 0 \\ 0 & 1 & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = -ik_2$ and $k_2 = -ik_3$. If $k_3 = i$ then $\mathbf{K}_3 = \begin{pmatrix} -1 \\ 1 \\ i \end{pmatrix}$.

21. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & 5-\lambda & 6 \\ 0 & 0 & -7-\lambda \end{vmatrix} = -(\lambda-1)(\lambda-5)(\lambda+7) = 0.$

For $\lambda_1 = 1$ we have $\left(\begin{array}{ccc|c} 0 & 2 & 3 & 0 \\ 0 & 4 & 6 & 0 \\ 0 & 0 & -6 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_2 = k_3 = 0$. If $k_1 = 1$ then $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. For $\lambda_2 = 5$ we have

$$\left(\begin{array}{ccc|c} -4 & 2 & 3 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & -12 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_3 = 0$ and $k_2 = 2k_1$. If $k_1 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$. For $\lambda_3 = -7$ we have

$$\left(\begin{array}{ccc|c} 8 & 2 & 3 & 0 \\ 0 & 12 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so that $k_1 = -\frac{1}{4}k_3$ and $k_2 = -\frac{1}{2}k_3$. If $k_3 = 4$ then $\mathbf{K}_3 = \begin{pmatrix} -1 \\ -2 \\ 4 \end{pmatrix}$.

22. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} -\lambda & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = -\lambda^2(\lambda-1) = 0.$

For $\lambda_1 = \lambda_2 = 0$ we have $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$

so that $k_3 = 0$. If $k_1 = 1$ and $k_2 = 0$ then $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and if $k_1 = 0$ and $k_2 = 1$ then $\mathbf{K}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. For $\lambda_3 = 1$

we have

$$\left(\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

so that $k_1 = k_2 = 0$. If $k_3 = 1$ then $\mathbf{K}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

23. The eigenvalues and eigenvectors of $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$ are

$$\lambda_1 = 4, \quad \lambda_2 = 6, \quad \mathbf{K}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and the eigenvalues and eigenvectors of $\mathbf{A}^{-1} = \frac{1}{24} \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix}$ are

$$\lambda_1 = \frac{1}{4}, \quad \lambda_2 = \frac{1}{6}, \quad \mathbf{K}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

24. The eigenvalues and eigenvectors of $\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{pmatrix}$ are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3, \quad \mathbf{K}_1 = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}, \quad \mathbf{K}_3 = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}.$$

and the eigenvalues and eigenvectors of $\mathbf{A}^{-1} = \frac{1}{6} \begin{pmatrix} 4 & -6 & 2 \\ -1 & 9 & -2 \\ -4 & 12 & -2 \end{pmatrix}$ are

$$\lambda_1 = 1, \quad \lambda_2 = \frac{1}{2}, \quad \lambda_3 = \frac{1}{3}, \quad \mathbf{K}_1 = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}, \quad \mathbf{K}_3 = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}.$$

25. Since $\det \mathbf{A} = \begin{vmatrix} 6 & 0 \\ 3 & 0 \end{vmatrix} = 0$ the matrix is singular. Now from

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 6 - \lambda & 0 \\ 3 & -\lambda \end{vmatrix} = \lambda(\lambda - 6)$$

we see $\lambda = 0$ is an eigenvalue.

26. Since $\det \mathbf{A} = \begin{vmatrix} 1 & 0 & 1 \\ 4 & -4 & 5 \\ 7 & -4 & 8 \end{vmatrix} = 0$ the matrix is singular. Now from

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & 0 & 1 \\ 4 & -4 - \lambda & 5 \\ 7 & -4 & 8 - \lambda \end{vmatrix} = -\lambda(\lambda^2 - 5\lambda - 15)$$

we see $\lambda = 0$ is an eigenvalue.

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27. (a) Since $p + 1 - p = 1$ and $q + 1 - q = 1$, the first matrix $\mathbf{A}$ is stochastic. Since $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$, $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, and $\frac{1}{6} + \frac{1}{3} + \frac{1}{2} = 1$, the second matrix $\mathbf{A}$ is stochastic.

(b) The matrix from part (a) is shown with its eigenvalues and corresponding eigenvectors.

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}; \quad \text{eigenvalues: } 1, \frac{1}{6} - \frac{1}{12}\sqrt{2}, \frac{1}{6} + \frac{1}{12}\sqrt{2};$$

$$\text{eigenvectors: } (1, 1, 1), \left(-\frac{3(-1+\sqrt{2})}{-6+\sqrt{2}}, \frac{2(2+\sqrt{2})}{-6+\sqrt{2}}, 1\right), \left(-\frac{3(1+\sqrt{2})}{6+\sqrt{2}}, \frac{2(-2+\sqrt{2})}{6+\sqrt{2}}, 1\right)$$

Further examples indicate that 1 is always an eigenvalue with corresponding eigenvector $(1, 1, 1)$. To prove this, let $\mathbf{A}$ be a stochastic matrix and $\mathbf{K} = (1, 1, 1)$. Then

$$\mathbf{AK} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} + \cdots + a_{1n} \\ \vdots \\ a_{n1} + \cdots + a_{nn} \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = 1\mathbf{K},$$

and 1 is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $(1, 1, 1)$.

(c) For the 3×3 matrix in part (a) we have

$$\mathbf{A}^2 = \begin{pmatrix} \frac{3}{8} & \frac{7}{24} & \frac{1}{3} \\ \frac{1}{3} & \frac{11}{36} & \frac{13}{36} \\ \frac{5}{18} & \frac{23}{72} & \frac{29}{72} \end{pmatrix}, \quad \mathbf{A}^3 = \begin{pmatrix} \frac{49}{144} & \frac{29}{96} & \frac{103}{288} \\ \frac{71}{216} & \frac{11}{36} & \frac{79}{216} \\ \frac{5}{16} & \frac{67}{216} & \frac{163}{432} \end{pmatrix}.$$

These powers of $\mathbf{A}$ are also stochastic matrices. To prove that this is true in general for 2×2 matrices, we prove the more general theorem that any product of 2×2 stochastic matrices is stochastic. Let

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

be stochastic matrices. Then

$$\mathbf{AB} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}.$$

The sums of the rows are

$$\begin{aligned} a_{11}b_{11} + a_{12}b_{21} + a_{11}b_{12} + a_{12}b_{22} &= a_{11}(b_{11} + b_{12}) + a_{12}(b_{21} + b_{22}) \\ &= a_{11}(1) + a_{12}(1) = a_{11} + a_{12} = 1 \\ a_{21}b_{11} + a_{22}b_{21} + a_{21}b_{12} + a_{22}b_{22} &= a_{21}(b_{11} + b_{12}) + a_{22}(b_{21} + b_{22}) \\ &= a_{21}(1) + a_{22}(1) = a_{21} + a_{22} = 1. \end{aligned}$$

Thus, the product matrix $\mathbf{AB}$ is stochastic. It follows that any power of a 2×2 matrix is stochastic. The proof in the case of an $n \times n$ matrix is very similar.

EXERCISES 8.9

Powers of Matrices

1. The characteristic equation is $\lambda^2 - 6\lambda + 13 = 0$. Then

$$\mathbf{A}^2 - 6\mathbf{A} + 13\mathbf{I} = \begin{pmatrix} -7 & -12 \\ 24 & 17 \end{pmatrix} - \begin{pmatrix} 6 & -12 \\ 24 & 30 \end{pmatrix} + \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

2. The characteristic equation is $-\lambda^3 + \lambda^2 + 4\lambda - 1 = 0$. Then

$$-\mathbf{A}^3 + \mathbf{A}^2 + 4\mathbf{A} - \mathbf{I} = -\begin{pmatrix} 2 & 6 & 13 \\ 4 & 5 & 17 \\ 1 & 5 & 9 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 5 \\ 0 & 4 & 5 \\ 1 & 1 & 4 \end{pmatrix} + 4\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

3. The characteristic equation is $\lambda^2 - 3\lambda - 10 = 0$, with eigenvalues -2 and 5 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$\begin{aligned} (-2)^m &= c_0 - 2c_1 \\ 5^m &= c_0 + 5c_1. \end{aligned}$$

Solving the system gives

$$c_0 = \frac{1}{7}[5(-2)^m + 2(5)^m], \quad c_1 = \frac{1}{7}[(-2)^m + 5^m].$$

Thus

$$\mathbf{A}^m = c_0\mathbf{I} + c_1\mathbf{A} = \begin{pmatrix} \frac{1}{7}[3(-1)^m 2^{m+1} + 5^m] & \frac{3}{7}[(-2)^m + 5^m] \\ \frac{2}{7}[(-2)^m + 5^m] & \frac{1}{7}[(-2)^m + 6(5)^m] \end{pmatrix}$$

and

$$\mathbf{A}^3 = \begin{pmatrix} 11 & 57 \\ 38 & 106 \end{pmatrix}.$$

4. The characteristic equation is $\lambda^2 - 10\lambda + 16 = 0$, with eigenvalues 2 and 8 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$\begin{aligned} 2^m &= c_0 + 2c_1 \\ 8^m &= c_0 + 8c_1. \end{aligned}$$

Solving the system gives

$$c_0 = \frac{1}{3}(2^{m+2} - 8^m), \quad c_1 = \frac{1}{6}(-2^m + 8^m).$$

Thus

$$\mathbf{A}^m = c_0\mathbf{I} + c_1\mathbf{A} = \begin{pmatrix} \frac{1}{2}(2^m + 8^m) & \frac{1}{2}(2^m - 8^m) \\ \frac{1}{2}(2^m - 8^m) & \frac{1}{2}(2^m + 8^m) \end{pmatrix}$$

and

$$\mathbf{A}^4 = \begin{pmatrix} 2056 & -2040 \\ -2040 & 2056 \end{pmatrix}.$$

8.9 Powers of Matrices

5. The characteristic equation is $\lambda^2 - 8\lambda - 20 = 0$, with eigenvalues -2 and 10 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$\begin{aligned}(-2)^m &= c_0 - 2c_1 \\ 10^m &= c_0 + 10c_1.\end{aligned}$$

Solving the system gives

$$c_0 = \frac{1}{6}[5(-2)^m + 10^m], \quad c_1 = \frac{1}{12}[-(-2)^m + 10^m].$$

Thus

$$\mathbf{A}^m = c_0\mathbf{I} + c_1\mathbf{A} = \begin{pmatrix} \frac{1}{6}[(-2)^m + 2^m 5^{m+1}] & \frac{5}{12}[-(-2)^m + 10^m] \\ \frac{1}{3}[-(-2)^m + 10^m] & \frac{1}{6}[5(-2)^m + 10^m] \end{pmatrix}$$

and

$$\mathbf{A}^5 = \begin{pmatrix} 83328 & 41680 \\ 33344 & 16640 \end{pmatrix}.$$

6. The characteristic equation is $\lambda^2 + 4\lambda + 3 = 0$, with eigenvalues -3 and -1 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$\begin{aligned}(-3)^m &= c_0 - 3c_1 \\ (-1)^m &= c_0 - c_1.\end{aligned}$$

Solving the system gives

$$c_0 = \frac{1}{2}[-(-3)^m + 3(-1)^m], \quad c_1 = \frac{1}{2}[-(-3)^m + (-1)^m].$$

Thus

$$\mathbf{A}^m = c_0\mathbf{I} + c_1\mathbf{A} = \begin{pmatrix} (-1)^m & -(-3)^m + (-1)^m \\ 0 & (-3)^m \end{pmatrix}$$

and

$$\mathbf{A}^6 = \begin{pmatrix} 1 & -728 \\ 0 & 729 \end{pmatrix}.$$

7. The characteristic equation is $-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$, with eigenvalues -1 , 1 , and 2 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned}(-1)^m &= c_0 - c_1 + c_2 \\ 1 &= c_0 + c_1 + c_2 \\ 2^m &= c_0 + 2c_1 + 4c_2.\end{aligned}$$

Solving the system gives

$$\begin{aligned}c_0 &= \frac{1}{3}[3 + (-1)^m - 2^m], \\ c_1 &= \frac{1}{2}[1 - (-1)^m], \\ c_2 &= \frac{1}{6}[-3 + (-1)^m + 2^{m+1}].\end{aligned}$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2 = \begin{pmatrix} 1 & -1 + 2^m & -1 + 2^m \\ 0 & \frac{1}{3}[(-1)^m + 2^{m+1}] & -\frac{2}{3}[(-1)^m - 2^m] \\ 0 & \frac{1}{3}[(-1)^m + 2^m] & \frac{1}{3}[2(-1)^m + 2^m] \end{pmatrix}$$

and

$$\mathbf{A}^{10} = \begin{pmatrix} 1 & 1023 & 1023 \\ 0 & 683 & 682 \\ 0 & 341 & 342 \end{pmatrix}.$$

8. The characteristic equation is $-\lambda^3 - \lambda^2 + 2\lambda + 2 = 0$, with eigenvalues -1 , $-\sqrt{2}$, and $\sqrt{2}$. Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned} (-1)^m &= c_0 - c_1 + c_2 \\ (-\sqrt{2})^m &= c_0 - \sqrt{2}c_1 + 2c_2 \\ (\sqrt{2})^m &= c_0 + \sqrt{2}c_1 + 2c_2. \end{aligned}$$

Solving the system gives

$$\begin{aligned} c_0 &= [2 - (\sqrt{2})^{m-1} - (\sqrt{2})^{m-2}](-1)^m + (\sqrt{2} - 1)(\sqrt{2})^{m-2}, \\ c_1 &= \frac{1}{2}[1 - (-1)^m](\sqrt{2})^{m-1}, \\ c_2 &= (-1)^{m+1} + \frac{1}{2}(1 + \sqrt{2})(-1)^m(\sqrt{2})^{m-1} + \frac{1}{2}(\sqrt{2} - 1)(\sqrt{2})^{m-1}. \end{aligned}$$

Thus $\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2$ and

$$\mathbf{A}^6 = \begin{pmatrix} 1 & 0 & 7 \\ 7 & 8 & -7 \\ 0 & 0 & 8 \end{pmatrix}.$$

9. The characteristic equation is $-\lambda^3 + 3\lambda^2 + 6\lambda - 8 = 0$, with eigenvalues -2 , 1 , and 4 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned} (-2)^m &= c_0 - 2c_1 + 4c_2 \\ 1 &= c_0 + c_1 + c_2 \\ 4^m &= c_0 + 4c_1 + 16c_2. \end{aligned}$$

Solving the system gives

$$\begin{aligned} c_0 &= \frac{1}{9}[8 + (-1)^m 2^{m+1} - 4^m], \\ c_1 &= \frac{1}{18}[4 - 5(-2)^m + 4^m], \\ c_2 &= \frac{1}{18}[-2 + (-2)^m + 4^m]. \end{aligned}$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2 = \begin{pmatrix} \frac{1}{9}[(-2)^m + (-1)^m 2^{m+1} + 3 \cdot 2^{2m+1}] & \frac{1}{3}[(-2)^m + 4^m] & 0 \\ -\frac{2}{3}[(-2)^m - 4^m] & \frac{1}{3}[(-1)^m 2^{2m+1} + 4^m] & 0 \\ \frac{1}{3}[-3 + (-2)^m + 2^{2m+1}] & \frac{1}{3}[(-2)^m + 4^m] & 1 \end{pmatrix}$$

8.9 Powers of Matrices

and

$$\mathbf{A}^{10} = \begin{pmatrix} 699392 & 349184 & 0 \\ 698368 & 350208 & 0 \\ 699391 & 349184 & 1 \end{pmatrix}.$$

10. The characteristic equation is $-\lambda^3 - \frac{3}{2}\lambda^2 + \frac{3}{2}\lambda + 1 = 0$, with eigenvalues -2 , $-\frac{1}{2}$, and 1 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned} (-2)^m &= c_0 - 2c_1 + 4c_2 \\ \left(-\frac{1}{2}\right)^m &= c_0 - \frac{1}{2}c_1 + \frac{1}{4}c_2 \\ 1 &= c_0 + c_1 + c_2. \end{aligned}$$

Solving the system gives

$$\begin{aligned} c_0 &= \frac{1}{9}[2^{-m}[(-4)^m + 8(-1)^m + 2^{m+1} - (-1)^m 2^{2m+1}], \\ c_1 &= -\frac{1}{9}2^{-m}[(-4)^m + 4(-1)^m - 5 \cdot 2^m], \\ c_2 &= \frac{2}{9}[1 + (-2)^m - (-1)^m 2^{m-1}]. \end{aligned}$$

Thus

$$\begin{aligned} \mathbf{A}^m &= c_0\mathbf{I} + c_1\mathbf{A} + c_2\mathbf{A}^2 \\ &= \begin{pmatrix} \frac{1}{3}2^{-m}[2(-1)^m + 2^m] & \frac{1}{3}[-1 + (-\frac{1}{2})^m] & 0 \\ \frac{2}{3}[-1 + (-\frac{1}{2})^m] & \frac{1}{3}[2 + (-\frac{1}{2})^m] & 0 \\ -\frac{1}{9}2^{-m}[7(-4)^m - 6(-1)^m - 3 \cdot 2^m + (-1)^m 2^{2m+1}] & \frac{1}{3}[-1 + (-\frac{1}{2})^m] & \frac{1}{3}[(-2)^m + (-1)^m 2^{m+1}] \end{pmatrix} \end{aligned}$$

and

$$\mathbf{A}^8 = \begin{pmatrix} \frac{43}{128} & -\frac{85}{256} & 0 \\ -\frac{85}{128} & \frac{171}{256} & 0 \\ -\frac{32725}{128} & -\frac{85}{256} & 256 \end{pmatrix}.$$

11. The characteristic equation is $\lambda^2 - 8\lambda + 16 = 0$, with eigenvalues 4 and 4 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$\begin{aligned} 4^m &= c_0 + 4c_1 \\ 4^{m-1}m &= c_1. \end{aligned}$$

Solving the system gives

$$c_0 = -4^m(m-1), \quad c_1 = 4^{m-1}m.$$

Thus

$$\mathbf{A}^m = c_0\mathbf{I} + c_1\mathbf{A} = \begin{pmatrix} 4^{m-1}(3m+4) & 3 \cdot 4^{m-1}m \\ -3 \cdot 4^{m-1}m & 4^{m-1}(-3m+4) \end{pmatrix}$$

and

$$\mathbf{A}^6 = \begin{pmatrix} 22528 & 18432 \\ -18432 & -14336 \end{pmatrix}.$$

12. The characteristic equation is $-\lambda^3 - \lambda^2 + 21\lambda + 45 = 0$, with eigenvalues -3 , -3 , and 5 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned}(-3)^m &= c_0 - 3c_1 + 9c_2 \\ (-3)^{m-1}m &= c_1 - 6c_2 \\ 5^m &= c_0 + 5c_1 + 25c_2.\end{aligned}$$

Solving the system gives

$$\begin{aligned}c_0 &= \frac{1}{64}[73(-3)^m - 2(-1)^m 3^{m+2} + 9 \cdot 5^m - 40(-3)^m m], \\ c_1 &= \frac{1}{96}[-(-1)^m 3^{m+2} + 9 \cdot 5^m - 8(-3)^m m], \\ c_2 &= \frac{1}{64}[-(-3)^m + 5^m - 8(-3)^{m-1}m].\end{aligned}$$

Thus

$$\begin{aligned}\mathbf{A}^m &= c_0\mathbf{I} + c_1\mathbf{A} + c_2\mathbf{A}^2 \\ &= \begin{pmatrix} \frac{1}{32}[31(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{16}[-(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{32}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] \\ \frac{1}{16}[-(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{8}[7(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{16}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] \\ \frac{3}{32}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] & \frac{3}{16}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] & \frac{1}{32}[29(-3)^m - (-1)^m 3^{m+2} + 12 \cdot 5^m] \end{pmatrix}\end{aligned}$$

and

$$\mathbf{A}^5 = \begin{pmatrix} 178 & 842 & -421 \\ 842 & 1441 & -842 \\ -1263 & -2526 & 1020 \end{pmatrix}.$$

13. (a) The characteristic equation is $\lambda^2 - 4\lambda = \lambda(\lambda - 4) = 0$, so 0 is an eigenvalue. Since the matrix satisfies the characteristic equation, $\mathbf{A}^2 = 4\mathbf{A}$, $\mathbf{A}^3 = 4\mathbf{A}^2 = 4^2\mathbf{A}$, $\mathbf{A}^4 = 4^2\mathbf{A}^2 = 4^3\mathbf{A}$, and, in general,

$$\mathbf{A}^m = 4^m \mathbf{A} = \begin{pmatrix} 4^m & 4^m \\ 3(4)^m & 3(4)^m \end{pmatrix}.$$

- (b) The characteristic equation is $\lambda^2 = 0$, so 0 is an eigenvalue. Since the matrix satisfies the characteristic equation, $\mathbf{A}^2 = \mathbf{0}$, $\mathbf{A}^3 = \mathbf{A}\mathbf{A}^2 = \mathbf{0}$, and, in general, $\mathbf{A}^m = \mathbf{0}$.
- (c) The characteristic equation is $-\lambda^3 + 5\lambda^2 - 6\lambda = 0$, with eigenvalues 0 , 2 , and 3 . Substituting $\lambda = 0$ into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ we find that $c_0 = 0$. Using the nonzero eigenvalues, we find

$$\begin{aligned}2^m &= 2c_1 + 4c_2 \\ 3^m &= 3c_1 + 9c_2.\end{aligned}$$

Solving the system gives

$$c_1 = \frac{1}{6}[9(2)^m - 4(3)^m], \quad c_2 = \frac{1}{6}[-3(2)^m + 2(3)^m].$$

Thus $\mathbf{A}^m = c_1\mathbf{A} + c_2\mathbf{A}^2$ and

$$\mathbf{A}^m = \begin{pmatrix} 2(3)^{m-1} & 3^{m-1} & 3^{m-1} \\ \frac{1}{6}[9(2)^m - 4(3)^m] & \frac{1}{6}[3(2)^m - 2(3)^m] & \frac{1}{6}[-3(2)^m - 2(3)^m] \\ \frac{1}{6}[-9(2)^m + 8(3)^m] & \frac{1}{6}[-3(2)^m + 4(3)^m] & \frac{1}{6}[3(2)^m + 4(3)^m] \end{pmatrix}.$$

8.9 Powers of Matrices

14. (a) Let

$$\mathbf{X}_{n-1} = \begin{pmatrix} x_{n-1} \\ y_{n-1} \end{pmatrix} \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$\mathbf{X}_n = \mathbf{A}\mathbf{X}_{n-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ y_{n-1} \end{pmatrix} = \begin{pmatrix} x_{n-1} + y_{n-1} \\ x_{n-1} \end{pmatrix}.$$

(b) The characteristic equation of $\mathbf{A}$ is $\lambda^2 - \lambda - 1 = 0$, with eigenvalues $\lambda_1 = \frac{1}{2}(1 - \sqrt{5})$ and $\lambda_2 = \frac{1}{2}(1 + \sqrt{5})$. From $\lambda^m = c_0 + c_1\lambda$ we get $\lambda_1^m = c_0 + c_1\lambda_1$ and $\lambda_2^m = c_0 + c_1\lambda_2$. Solving this system gives

$$c_0 = (\lambda_2\lambda_1^m - \lambda_1\lambda_2^m)/(\lambda_2 - \lambda_1) \quad \text{and} \quad c_1 = (\lambda_2^m - \lambda_1^m)/(\lambda_2 - \lambda_1).$$

Thus

$$\begin{aligned} \mathbf{A}^m &= c_0\mathbf{I} + c_1\mathbf{A} \\ &= \frac{1}{2^{m+1}\sqrt{5}} \begin{pmatrix} (1 + \sqrt{5})^{m+1} - (1 - \sqrt{5})^{m+1} & 2(1 + \sqrt{5})^m - 2(1 - \sqrt{5})^m \\ 2(1 + \sqrt{5})^m - 2(1 - \sqrt{5})^m & (1 + \sqrt{5})(1 - \sqrt{5})^m - (1 - \sqrt{5})(1 + \sqrt{5})^m \end{pmatrix}. \end{aligned}$$

(c) From part (a), $\mathbf{X}_2 = \mathbf{A}\mathbf{X}_1$, $\mathbf{X}_3 = \mathbf{A}\mathbf{X}_2 = \mathbf{A}^2\mathbf{X}_1$, $\mathbf{X}_4 = \mathbf{A}\mathbf{X}_3 = \mathbf{A}^3\mathbf{X}_1$, and, in general, $\mathbf{X}_n = \mathbf{A}^{n-1}\mathbf{X}_1$. With

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{we have} \quad \mathbf{X}_{12} = \mathbf{A}^{11}\mathbf{X}_1 = \begin{pmatrix} 144 & 89 \\ 89 & 55 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 233 \\ 144 \end{pmatrix},$$

so the number of adult pairs is 233. With

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{we have} \quad \mathbf{A}^{11}\mathbf{X}_1 = \begin{pmatrix} 144 & 89 \\ 89 & 55 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 144 \\ 89 \end{pmatrix},$$

so the number of baby pairs is 144. With

$$\mathbf{X}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \text{we have} \quad \mathbf{A}^{11}\mathbf{X}_1 = \begin{pmatrix} 144 & 89 \\ 89 & 55 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 377 \\ 233 \end{pmatrix},$$

so the total number of pairs is 377.

15. The characteristic equation of $\mathbf{A}$ is $\lambda^2 - 5\lambda + 10 = 0$, so $\mathbf{A}^2 - 5\mathbf{A} + 10\mathbf{I} = \mathbf{0}$ and $\mathbf{I} = -\frac{1}{10}\mathbf{A}^2 + \frac{1}{2}\mathbf{A}$. Multiplying by $\mathbf{A}^{-1}$ we find

$$\mathbf{A}^{-1} = -\frac{1}{10}\mathbf{A} + \frac{1}{2}\mathbf{I} = -\frac{1}{10} \begin{pmatrix} 2 & -4 \\ 1 & 3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{10} & \frac{2}{5} \\ -\frac{1}{10} & \frac{1}{5} \end{pmatrix}.$$

16. The characteristic equation of $\mathbf{A}$ is $-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$, so $-\mathbf{A}^3 + 2\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I} = \mathbf{0}$ and $\mathbf{I} = -\frac{1}{2}\mathbf{A}^3 + \mathbf{A}^2 + \frac{1}{2}\mathbf{A}$. Multiplying by $\mathbf{A}^{-1}$ we find

$$\mathbf{A}^{-1} = -\frac{1}{2}\mathbf{A}^2 + \mathbf{A} + \frac{1}{2}\mathbf{I} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{3}{2} \end{pmatrix}.$$

17. (a) Since

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix} \quad \text{we see that} \quad \mathbf{A}^m = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}$$

for all integers $m \geq 2$. Thus $\mathbf{A}$ is not nilpotent.

(b) Since $\mathbf{A}^2 = \mathbf{0}$, the matrix is nilpotent with index 2.

(c) Since $\mathbf{A}^3 = \mathbf{0}$, the matrix is nilpotent with index 3.

- (d) Since $\mathbf{A}^2 = \mathbf{0}$, the matrix is nilpotent with index 2.
- (e) Since $\mathbf{A}^4 = \mathbf{0}$, the matrix is nilpotent with index 4.
- (f) Since $\mathbf{A}^4 = \mathbf{0}$, the matrix is nilpotent with index 4.
18. (a) If $\mathbf{A}^m = \mathbf{0}$ for some m , then $(\det \mathbf{A})^m = \det \mathbf{A}^m = \det \mathbf{0} = 0$, and $\mathbf{A}$ is a singular matrix.
- (b) By (1) of Section 8.8 we have $\mathbf{A}\mathbf{K} = \lambda\mathbf{K}$, $\mathbf{A}^2\mathbf{K} = \lambda\mathbf{A}\mathbf{K} = \lambda^2\mathbf{K}$, $\mathbf{A}^3\mathbf{K} = \lambda^2\mathbf{A}\mathbf{K} = \lambda^3\mathbf{K}$, and, in general, $\mathbf{A}^m\mathbf{K} = \lambda^m\mathbf{K}$. If $\mathbf{A}$ is nilpotent with index m , then $\mathbf{A}^m = \mathbf{0}$ and $\lambda^m = 0$.

EXERCISES 8.10

Orthogonal Matrices

1. (a)–(b) $\begin{pmatrix} 0 & 0 & -4 \\ 0 & -4 & 0 \\ -4 & 0 & 15 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \\ 0 \end{pmatrix} = -4 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \quad \lambda_1 = -4$
- $\begin{pmatrix} 0 & 0 & -4 \\ 0 & -4 & 0 \\ -4 & 0 & 15 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \\ 1 \end{pmatrix} = (-1) \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}; \quad \lambda_2 = -1$
- $\begin{pmatrix} 0 & 0 & -4 \\ 0 & -4 & 0 \\ -4 & 0 & 15 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = \begin{pmatrix} 16 \\ 0 \\ -64 \end{pmatrix} = 16 \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}; \quad \lambda_3 = 16$
- (c) $\mathbf{K}_1^T \mathbf{K}_2 = (0 \ 1 \ 0) \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = (0 \ 1 \ 0) \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = 0; \quad \mathbf{K}_2^T \mathbf{K}_3 = (4 \ 0 \ 1) \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = 0$
2. (a)–(b) $\begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}; \quad \lambda_1 = 2$
- $\begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}; \quad \lambda_2 = 2$
- $\begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix} = (-1) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}; \quad \lambda_3 = -1$
- (c) $\mathbf{K}_1^T \mathbf{K}_2 = (-2 \ 1 \ 1) \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 1 - 1 = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = (-2 \ 1 \ 1) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -2 + 1 + 1 = 0$

8.10 Orthogonal Matrices

$$\mathbf{K}_2^T \mathbf{K}_3 = \begin{pmatrix} 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 - 1 = 0$$

$$3. \text{ (a)-(b)} \quad \begin{pmatrix} 5 & 13 & 0 \\ 13 & 5 & 0 \\ 0 & 0 & -8 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} 9\sqrt{2} \\ 9\sqrt{2} \\ 0 \end{pmatrix} = 18 \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix}; \quad \lambda_1 = 18$$

$$\begin{pmatrix} 5 & 13 & 0 \\ 13 & 5 & 0 \\ 0 & 0 & -8 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix} = \begin{pmatrix} -\frac{8\sqrt{2}}{3} \\ \frac{8\sqrt{3}}{3} \\ -\frac{8\sqrt{3}}{3} \end{pmatrix} = (-8) \begin{pmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix}; \quad \lambda_2 = -8$$

$$\begin{pmatrix} 5 & 13 & 0 \\ 13 & 5 & 0 \\ 0 & 0 & -8 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{pmatrix} = \begin{pmatrix} -\frac{8\sqrt{6}}{6} \\ \frac{8\sqrt{6}}{6} \\ \frac{8\sqrt{6}}{3} \end{pmatrix} = (-8) \begin{pmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{pmatrix}; \quad \lambda_3 = -8$$

$$(c) \quad \mathbf{K}_1^T \mathbf{K}_2 = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix} = \frac{\sqrt{6}}{6} - \frac{\sqrt{6}}{6} = 0;$$

$$\mathbf{K}_1^T \mathbf{K}_3 = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{pmatrix} = \frac{\sqrt{12}}{12} - \frac{\sqrt{12}}{12} = 0$$

$$\mathbf{K}_2^T \mathbf{K}_3 = \begin{pmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{pmatrix} = \frac{\sqrt{18}}{18} + \frac{\sqrt{18}}{18} - \frac{\sqrt{18}}{9} = 0$$

$$4. \text{ (a)-(b)} \quad \begin{pmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{pmatrix} \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}; \quad \lambda_1 = 0$$

$$\begin{pmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}; \quad \lambda_2 = 3$$

$$\begin{pmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \\ 12 \end{pmatrix} = 6 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}; \quad \lambda_3 = 6$$

$$(c) \quad \mathbf{K}_1^T \mathbf{K}_2 = \begin{pmatrix} -2 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} = -2 + 4 - 2 = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = \begin{pmatrix} -2 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = -4 + 2 + 2 = 0$$

$$\mathbf{K}_2^T \mathbf{K}_3 = \begin{pmatrix} 1 & 2 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = 2 + 2 - 4 = 0$$

5. Orthogonal. Columns form an orthonormal set.

6. Not orthogonal. Columns one and three are not unit vectors.
7. Orthogonal. Columns form an orthonormal set.
8. Not orthogonal. The matrix is singular.
9. Not orthogonal. Columns are not unit vectors.
10. Orthogonal. Columns form an orthogonal set.
11. $\lambda_1 = -8$, $\lambda_2 = 10$, $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$
12. $\lambda_1 = 7$, $\lambda_2 = 4$, $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
13. $\lambda_1 = 0$, $\lambda_2 = 10$, $\mathbf{K}_1 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ -\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{pmatrix}$
14. $\lambda_1 = \frac{1}{2} + \frac{\sqrt{5}}{2}$, $\lambda_2 = \frac{1}{2} - \frac{\sqrt{5}}{2}$, $\mathbf{K}_1 = \begin{pmatrix} 1 + \sqrt{5} \\ 2 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 - \sqrt{5} \\ 2 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} \frac{1+\sqrt{5}}{\sqrt{10+2\sqrt{5}}} & \frac{1-\sqrt{5}}{\sqrt{10-2\sqrt{5}}} \\ \frac{2}{\sqrt{10+2\sqrt{5}}} & \frac{2}{\sqrt{10-2\sqrt{5}}} \end{pmatrix}$
15. $\lambda_1 = 0$, $\lambda_2 = 2$, $\lambda_3 = 1$, $\mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{K}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$
16. $\lambda_1 = -1$, $\lambda_2 = 1 - \sqrt{2}$, $\lambda_3 = 1 + \sqrt{2}$, $\mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$, $\mathbf{K}_3 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$
17. $\lambda_1 = -11$, $\lambda_2 = 0$, $\lambda_3 = 6$, $\mathbf{K}_1 = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 \\ -4 \\ 7 \end{pmatrix}$, $\mathbf{K}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} -\frac{3}{\sqrt{11}} & \frac{1}{\sqrt{66}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{11}} & -\frac{4}{\sqrt{66}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{11}} & \frac{7}{\sqrt{66}} & \frac{1}{\sqrt{6}} \end{pmatrix}$
18. $\lambda_1 = -18$, $\lambda_2 = 0$, $\lambda_3 = 9$, $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{K}_3 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{pmatrix}$
19. $\begin{pmatrix} \frac{3}{5} & a \\ \frac{4}{5} & b \end{pmatrix} \begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ a & b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ implies $\frac{9}{25} + a^2 = 1$ and $\frac{16}{25} + b^2 = 1$. These equations give $a = \pm \frac{4}{5}$, $b = \pm \frac{3}{5}$.

But $\frac{12}{25} + ab = 0$ indicates a and b must have opposite signs. Therefore choose $a = -\frac{4}{5}$, $b = \frac{3}{5}$.

The matrix $\begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}$ is orthogonal.

20. $\begin{pmatrix} \frac{1}{\sqrt{5}} & b \\ a & \frac{1}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{5}} & a \\ b & \frac{1}{\sqrt{5}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ implies $\frac{1}{5} + b^2 = 1$ and $a^2 + \frac{1}{5} = 1$. These give $a = \pm \frac{2}{\sqrt{5}}$, $b = \pm \frac{2}{\sqrt{5}}$.

But $\frac{a}{\sqrt{5}} + \frac{b}{\sqrt{5}} = 0$ indicates a and b must have opposite signs. Therefore choose $a = -\frac{2}{\sqrt{5}}$, $b = \frac{2}{\sqrt{5}}$.

The matrix $\begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$ is orthogonal.

8.10 Orthogonal Matrices

21. (a)–(b) We compute

$$\begin{aligned}\mathbf{AK}_1 &= \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = -2\mathbf{K}_1 \\ \mathbf{AK}_2 &= \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = -2\mathbf{K}_2 \\ \mathbf{AK}_3 &= \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 4\mathbf{K}_3,\end{aligned}$$

and observe that $\mathbf{K}_1$ is an eigenvector with corresponding eigenvalue -2 , $\mathbf{K}_2$ is an eigenvector with corresponding eigenvalue -2 , and $\mathbf{K}_3$ is an eigenvector with corresponding eigenvalue 4 .

- (c) Since $\mathbf{K}_1 \cdot \mathbf{K}_2 = 1 \neq 0$, $\mathbf{K}_1$ and $\mathbf{K}_2$ are not orthogonal, while $\mathbf{K}_1 \cdot \mathbf{K}_3 = 0$ and $\mathbf{K}_2 \cdot \mathbf{K}_3 = 0$ so $\mathbf{K}_3$ is orthogonal to both $\mathbf{K}_1$ and $\mathbf{K}_2$. To transform $\{\mathbf{K}_1, \mathbf{K}_2\}$ into an orthogonal set we let $\mathbf{V}_1 = \mathbf{K}_1$ and compute $\mathbf{K}_2 \cdot \mathbf{V}_1 = 1$ and $\mathbf{V}_1 \cdot \mathbf{V}_1 = 2$. Then

$$\mathbf{V}_2 = \mathbf{K}_2 - \frac{\mathbf{K}_2 \cdot \mathbf{V}_1}{\mathbf{V}_1 \cdot \mathbf{V}_1} \mathbf{V}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \end{pmatrix}.$$

Now, $\{\mathbf{V}_1, \mathbf{V}_2, \mathbf{K}_3\}$ is an orthogonal set of eigenvectors with

$$\|\mathbf{V}_1\| = \sqrt{2}, \quad \|\mathbf{V}_2\| = \frac{3}{\sqrt{6}}, \quad \text{and} \quad \|\mathbf{K}_3\| = \sqrt{3}.$$

An orthonormal set of vectors is

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix},$$

and so the matrix

$$\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

is orthogonal.

22. (a)–(b) We compute

$$\begin{aligned}\mathbf{AK}_1 &= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = 0\mathbf{K}_1 \\ \mathbf{AK}_2 &= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 0\mathbf{K}_2 \\ \mathbf{AK}_3 &= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 0\mathbf{K}_3 \\ \mathbf{AK}_4 &= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 4\mathbf{K}_4,\end{aligned}$$

and observe that $\mathbf{K}_1$ is an eigenvector with corresponding eigenvalue 0, $\mathbf{K}_2$ is an eigenvector with corresponding eigenvalue 0, $\mathbf{K}_3$ is an eigenvector with corresponding eigenvalue 0, and $\mathbf{K}_4$ is an eigenvector with corresponding eigenvalue 4.

- (c) Since $\mathbf{K}_1 \cdot \mathbf{K}_2 = 1 \neq 0$, $\mathbf{K}_1$ and $\mathbf{K}_2$ are not orthogonal. Similarly, $\mathbf{K}_1 \cdot \mathbf{K}_3 = 1 \neq 0$ and $\mathbf{K}_2 \cdot \mathbf{K}_3 = 1 \neq 0$ so $\mathbf{K}_1$ and $\mathbf{K}_3$ and $\mathbf{K}_2$ and $\mathbf{K}_3$ are not orthogonal. However, $\mathbf{K}_1 \cdot \mathbf{K}_4 = 0$, $\mathbf{K}_2 \cdot \mathbf{K}_4 = 0$, and $\mathbf{K}_3 \cdot \mathbf{K}_4 = 0$, so each of $\mathbf{K}_1, \mathbf{K}_2$, and $\mathbf{K}_3$ is orthogonal to $\mathbf{K}_4$. To transform $\{\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3\}$ into an orthogonal set we let $\mathbf{V}_1 = \mathbf{K}_1$ and compute $\mathbf{K}_2 \cdot \mathbf{V}_1 = 1$ and $\mathbf{V}_1 \cdot \mathbf{V}_1 = 2$. Then

$$\mathbf{V}_2 = \mathbf{K}_2 - \frac{\mathbf{K}_2 \cdot \mathbf{V}_1}{\mathbf{V}_1 \cdot \mathbf{V}_1} \mathbf{V}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ -\frac{1}{2} \end{pmatrix}.$$

Next, using $\mathbf{K}_3 \cdot \mathbf{V}_1 = 1$, $\mathbf{K}_3 \cdot \mathbf{V}_2 = \frac{1}{2}$, and $\mathbf{V}_2 \cdot \mathbf{V}_2 = \frac{3}{2}$, we obtain

$$\mathbf{V}_3 = \mathbf{K}_3 - \frac{\mathbf{K}_3 \cdot \mathbf{V}_1}{\mathbf{V}_1 \cdot \mathbf{V}_1} \mathbf{V}_1 - \frac{\mathbf{K}_3 \cdot \mathbf{V}_2}{\mathbf{V}_2 \cdot \mathbf{V}_2} \mathbf{V}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} - \frac{1/2}{3/2} \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} \\ 1 \\ -\frac{1}{3} \\ -\frac{1}{3} \end{pmatrix}.$$

Now, $\{\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{K}_4\}$ is an orthogonal set of eigenvectors with

$$\|\mathbf{V}_1\| = \sqrt{2}, \quad \|\mathbf{V}_2\| = \frac{3}{\sqrt{6}}, \quad \|\mathbf{K}_3\| = \frac{2}{\sqrt{3}} \quad \text{and} \quad \|\mathbf{K}_4\| = 2.$$

An orthonormal set of vectors is

$$\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ 0 \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{2\sqrt{3}} \\ \frac{3}{2\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix},$$

8.10 Orthogonal Matrices

and so the matrix

$$\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{3}} & \frac{1}{2} \\ 0 & 0 & \frac{3}{2\sqrt{3}} & \frac{1}{2} \\ 0 & \frac{2}{\sqrt{6}} & -\frac{1}{2\sqrt{3}} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{3}} & \frac{1}{2} \end{pmatrix}$$

is orthogonal.

23. If we take $\mathbf{K}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ as in Example 4 in the text then we look for a vector $\mathbf{K}_2 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ such that $1(a) + \frac{1}{4}b - \frac{1}{4}c = 0$ and $\mathbf{K}_1 \cdot \mathbf{K}_2 = 0$ or $b + c = 0$. The last equation implies $c = -b$ so $a + \frac{1}{4}b - \frac{1}{4}(-b) = a + \frac{1}{2}b = 0$. If we let $b = -2$, then $a = 1$ and $c = 2$, so a second eigenvector with eigenvalue -9 and orthogonal to $\mathbf{K}_1$ is $\mathbf{K}_2 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.

24. The eigenvalues and corresponding eigenvectors of $\mathbf{A}$ are

$$\lambda_1 = \lambda_2 = -1, \quad \lambda_3 = \lambda_4 = 3, \quad \text{and} \quad \mathbf{K}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{K}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{K}_4 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Since $\mathbf{K}_1 \cdot \mathbf{K}_2 = \mathbf{K}_1 \cdot \mathbf{K}_3 = \mathbf{K}_1 \cdot \mathbf{K}_4 = \mathbf{K}_2 \cdot \mathbf{K}_3 = \mathbf{K}_2 \cdot \mathbf{K}_4 = \mathbf{K}_3 \cdot \mathbf{K}_4 = 0$, the vectors are orthogonal. Using $\|\mathbf{K}_1\| = \|\mathbf{K}_2\| = \|\mathbf{K}_3\| = \|\mathbf{K}_4\| = \sqrt{2}$, we construct the orthogonal matrix

$$\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}.$$

25. Suppose $\mathbf{A}$ and $\mathbf{B}$ are orthogonal matrices. Then $\mathbf{A}^{-1} = \mathbf{A}^T$ and $\mathbf{B}^{-1} = \mathbf{B}^T$ and

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \mathbf{B}^T\mathbf{A}^T = (\mathbf{AB})^T.$$

Thus $\mathbf{AB}$ is an orthogonal matrix.

EXERCISES 8.11

Approximation of Eigenvalues

1. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and computing $\mathbf{X}_i = \mathbf{A}\mathbf{X}_{i-1}$ for $i = 1, 2, 3, 4$ we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 8 \\ 8 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 16 \\ 16 \end{pmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ with corresponding eigenvalue $\lambda = \frac{\mathbf{AK} \cdot \mathbf{K}}{\mathbf{K} \cdot \mathbf{K}} = \frac{4}{2} = 2$.

2. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and computing $\mathbf{X}_i = \mathbf{A}\mathbf{X}_{i-1}$ for $i = 1, 2, 3, 4, 5$ we obtain

$$\mathbf{X}_1 = \begin{pmatrix} -5 \\ 7 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 49 \\ -47 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} -437 \\ 439 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 3937 \\ -3935 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} -35429 \\ 35431 \end{pmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \frac{1}{35439} \begin{pmatrix} -35429 \\ 35431 \end{pmatrix} \approx \begin{pmatrix} -0.99994 \\ 1 \end{pmatrix}$ with corresponding eigenvalue $\lambda = \frac{\mathbf{A}\mathbf{K} \cdot \mathbf{K}}{\mathbf{K} \cdot \mathbf{K}} = -8.9998$.

3. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{pmatrix} 6 \\ 16 \end{pmatrix}$, we define $\mathbf{X}_1 = \frac{1}{16} \begin{pmatrix} 6 \\ 16 \end{pmatrix} = \begin{pmatrix} 0.375 \\ 1 \end{pmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{pmatrix} 0.3363 \\ 1 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 0.3335 \\ 1 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 0.3333 \\ 1 \end{pmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{pmatrix} 0.3333 \\ 1 \end{pmatrix}$ with corresponding eigenvalue $\lambda = 14$.

4. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, we define $\mathbf{X}_1 = \frac{1}{5} \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 0.2 \\ 1 \end{pmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{pmatrix} 0.2727 \\ 1 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 0.2676 \\ 1 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 0.2680 \\ 1 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 0.2679 \\ 1 \end{pmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{pmatrix} 0.2679 \\ 1 \end{pmatrix}$ with corresponding eigenvalue $\lambda = 6.4641$.

5. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{pmatrix} 11 \\ 11 \\ 6 \end{pmatrix}$, we define $\mathbf{X}_1 = \frac{1}{11} \begin{pmatrix} 11 \\ 11 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0.5455 \end{pmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{pmatrix} 1 \\ 1 \\ 0.5045 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 1 \\ 1 \\ 0.5005 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 1 \\ 1 \\ 0.5 \end{pmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{pmatrix} 1 \\ 1 \\ 0.5 \end{pmatrix}$ with corresponding eigenvalue $\lambda = 10$.

6. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix}$, we define $\mathbf{X}_1 = \frac{1}{5} \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0.4 \\ 0.4 \end{pmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{pmatrix} 1 \\ 0.2105 \\ 0.2105 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 1 \\ 0.1231 \\ 0.1231 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 1 \\ 0.0758 \\ 0.0758 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 1 \\ 0.0481 \\ 0.0481 \end{pmatrix}.$$

At this point if we restart with $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ we see that $\mathbf{K} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is a dominant eigenvector with corresponding eigenvalue $\lambda = 3$.

8.11 Approximation of Eigenvalues

7. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 0.625 \\ 1 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 0.5345 \\ 1 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 0.5098 \\ 1 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 0.5028 \\ 1 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 0.5008 \\ 1 \end{pmatrix}.$$

Taking $\mathbf{K} = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}$ as the dominant eigenvector we find $\lambda_1 = 7$. Now the normalized eigenvector is

$$\mathbf{K}_1 = \begin{pmatrix} 0.4472 \\ 0.8944 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1.6 & -0.8 \\ -0.8 & 0.4 \end{pmatrix}. \text{ Taking } \mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and using scaling again we obtain } \mathbf{X}_1 = \begin{pmatrix} 1 \\ -0.5 \end{pmatrix}, \mathbf{X}_2 = \begin{pmatrix} 1 \\ -0.5 \end{pmatrix}. \text{ Taking } \mathbf{K} = \begin{pmatrix} 1 \\ -0.5 \end{pmatrix} \text{ we find } \lambda_2 = 2. \text{ The eigenvalues are 7 and 2.}$$

8. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and using scaling we obtain $\mathbf{X}_1 = \begin{pmatrix} 0.3333 \\ 1 \end{pmatrix}, \mathbf{X}_2 = \begin{pmatrix} 0.3333 \\ 1 \end{pmatrix}$. Taking $\mathbf{K} = \begin{pmatrix} 1/3 \\ 1 \end{pmatrix}$ as the

dominant eigenvector we find $\lambda_1 = 10$. Now the normalized eigenvector is $\mathbf{K}_1 = \begin{pmatrix} 0.3162 \\ 0.9486 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

An eigenvector for the zero matrix is $\lambda_2 = 0$. The eigenvalues are 10 and 0.

9. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 1 \\ -0.6667 \\ 1 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 1 \\ -0.9091 \\ 1 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 1 \\ -0.9767 \\ 1 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 1 \\ -0.9942 \\ 1 \end{pmatrix}.$$

Taking $\mathbf{K} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ as the dominant eigenvector we find $\lambda_1 = 4$. Now the normalized eigenvector is

$$\mathbf{K}_1 = \begin{pmatrix} 0.5774 \\ -0.5774 \\ 0.5774 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1.6667 & 0.3333 & -1.3333 \\ 0.3333 & 0.6667 & 0.3333 \\ -1.3333 & 0.3333 & 1.6667 \end{pmatrix}. \text{ If } \mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ is now chosen only one more}$$

eigenvalue is found. Thus, try $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ 0.5 \\ -0.5 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 1 \\ 0.2 \\ -0.8 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 1 \\ 0.0714 \\ -0.9286 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 1 \\ 0.0244 \\ -0.9756 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 1 \\ 0.0082 \\ -0.9918 \end{pmatrix}.$$

Taking $\mathbf{K} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ as the eigenvector we find $\lambda_2 = 3$. The normalized eigenvector in this case is

$$\mathbf{K}_2 = \begin{pmatrix} 0.7071 \\ 0 \\ -0.7071 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 0.1667 & 0.3333 & 0.1667 \\ 0.3333 & 0.6667 & 0.3333 \\ 0.1667 & 0.3333 & 0.1667 \end{pmatrix}. \text{ If } \mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ is chosen, and scaling is used we}$$

obtain $\mathbf{X}_1 = \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix}$, $\mathbf{X}_2 = \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix}$. Taking $\mathbf{K} = \begin{pmatrix} 0.5 \\ 1 \\ 0.5 \end{pmatrix}$ we find $\lambda_3 = 1$. The eigenvalues are 4, 3, and 1.

The difficulty in choosing $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ to find the second eigenvector results from the fact that this vector is a

linear combination of the eigenvectors corresponding to the other two eigenvalues, with 0 contribution from the second eigenvector. When this occurs the development of the power method, shown in the text, breaks down.

10. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} -0.3636 \\ -0.3636 \\ 1 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} -0.2431 \\ 0.0884 \\ 1 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} -0.2504 \\ -0.0221 \\ 1 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} -0.2499 \\ -0.0055 \\ 1 \end{pmatrix}.$$

Taking $\mathbf{K} = \begin{pmatrix} -0.25 \\ 0 \\ 1 \end{pmatrix}$ as the dominant eigenvector we find $\lambda_1 = 16$. The normalized eigenvector is

$$\mathbf{K}_1 = \begin{pmatrix} -0.2425 \\ 0 \\ 0.9701 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} -0.9412 & 0 & -0.2353 \\ 0 & -4 & 0 \\ -0.2353 & 0 & -0.0588 \end{pmatrix}. \text{ Taking } \mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ and using scaling we obtain}$$

$$\mathbf{X}_1 = \begin{pmatrix} -0.2941 \\ -1 \\ -0.0735 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 0.0735 \\ 1 \\ 0.0184 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} -0.0184 \\ -1 \\ -0.0046 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 0.0046 \\ 1 \\ 0.0011 \end{pmatrix}.$$

Taking $\mathbf{K} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ as the eigenvector we find $\lambda_2 = -4$. The normalized eigenvector in this case is $\mathbf{K}_2 = \mathbf{K} =$

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} -0.9412 & 0 & -0.2353 \\ 0 & 0 & 0 \\ -0.2353 & 0 & -0.0588 \end{pmatrix}. \text{ Taking } \mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ and using scaling we obtain } \mathbf{X}_1 = \begin{pmatrix} -1 \\ 0 \\ -0.25 \end{pmatrix},$$

$$\mathbf{X}_2 = \begin{pmatrix} 1 \\ 0 \\ 0.25 \end{pmatrix}. \text{ Using } \mathbf{K} = \begin{pmatrix} 1 \\ 0 \\ 0.25 \end{pmatrix} \text{ we find } \lambda_3 = -1. \text{ The eigenvalues are } 16, -4, \text{ and } -1.$$

11. The inverse matrix is $\begin{pmatrix} 4 & -1 \\ -3 & 1 \end{pmatrix}$. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ -0.6667 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 1 \\ -0.7857 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 1 \\ -0.7910 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 1 \\ -0.7913 \end{pmatrix}.$$

Using $\mathbf{K} = \begin{pmatrix} 1 \\ -0.7913 \end{pmatrix}$ we find $\lambda = 4.7913$. The minimum eigenvalue of $\begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix}$ is $1/4.7913 \approx 0.2087$.

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12. The inverse matrix is $\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$. Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 0.6667 \\ 1 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 0.7857 \\ 1 \end{pmatrix}, \dots, \quad \mathbf{X}_{10} = \begin{pmatrix} 0.75 \\ 1 \end{pmatrix}.$$

Using $\mathbf{K} = \begin{pmatrix} 0.75 \\ 1 \end{pmatrix}$ we find $\lambda = 5$. The minimum eigenvalue of $\begin{pmatrix} -0.2 & 0.3 \\ 0.4 & -0.1 \end{pmatrix}$ is $1/5 = 0.2$

13. (a) Replacing the second derivative with the difference expression we obtain

$$EI \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + Py_i = 0 \quad \text{or} \quad EI(y_{i+1} - 2y_i + y_{i-1}) + Ph^2 y_i = 0.$$

(b) Expanding the difference equation for $i = 1, 2, 3$ and using $h = L/4$, $y_0 = 0$, and $y_4 = 0$ we obtain

$$\begin{aligned} EI(y_2 - 2y_1 + y_0) + \frac{PL^2}{16} y_1 &= 0 & 2y_1 - y_2 &= \frac{PL^2}{16EI} y_1 \\ EI(y_3 - 2y_2 + y_1) + \frac{PL^2}{16} y_2 &= 0 & \text{or} & & -y_1 + 2y_2 - y_3 &= \frac{PL^2}{16EI} y_2 \\ EI(y_4 - 2y_3 + y_2) + \frac{PL^2}{16} y_3 &= 0 & -y_2 + 2y_3 &= \frac{PL^2}{16EI} y_3. \end{aligned}$$

In matrix form this becomes

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \frac{PL^2}{16EI} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.$$

(c) $\mathbf{A}^{-1} = \begin{pmatrix} 0.75 & 0.5 & 0.25 \\ 0.5 & 1 & 0.5 \\ 0.25 & 0.5 & 0.75 \end{pmatrix}$

(d) Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 0.75 \\ 1 \\ 0.75 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 0.7143 \\ 1 \\ 0.7143 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 0.7083 \\ 1 \\ 0.7083 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 0.7073 \\ 1 \\ 0.7073 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 0.7071 \\ 1 \\ 0.7071 \end{pmatrix}.$$

Using $\mathbf{K} = \begin{pmatrix} 0.7071 \\ 1 \\ 0.7071 \end{pmatrix}$ we find $\lambda = 1.7071$. Then $1/\lambda = 0.5859$ is the minimum eigenvalue of $\mathbf{A}$.

(e) Solving $\frac{PL^2}{16EI} = 0.5859$ for P we obtain $P = 9.3726 \frac{EI}{L^2}$. In Example 3 of Section 3.9 we saw

$$P = \pi^2 \frac{EI}{L^2} \approx 9.8696 \frac{EI}{L^2}.$$

14. (a) The difference equation is

$$EI_i(y_{i+1} - 2y_i + y_{i-1}) + Ph^2 y_i = 0, \quad i = 1, 2, 3,$$

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where $I_0 = 0.00200$, $I_1 = 0.00175$, $I_2 = 0.00150$, $I_3 = 0.00125$, and $I_4 = 0.00100$. The system of equations is

$$\begin{aligned} 0.00175E(y_2 - 2y_1 + y_0) + \frac{PL^2}{16} y_1 &= 0 & 0.0035y_1 - 0.00175y_2 &= \frac{PL^2}{16E} y_1 \\ 0.00150E(y_3 - 2y_2 + y_1) + \frac{PL^2}{16} y_2 &= 0 & \text{or} & -0.0015y_1 + 0.003y_2 - 0.0015y_3 &= \frac{PL^2}{16E} y_2 \\ 0.00125E(y_4 - 2y_3 + y_2) + \frac{PL^2}{16} y_3 &= 0 & -0.00125y_2 + 0.0025y_3 &= \frac{PL^2}{16E} y_3. \end{aligned}$$

In matrix form this becomes

$$\begin{pmatrix} 0.0035 & -0.00175 & 0 \\ -0.0015 & 0.003 & -0.0015 \\ 0 & -0.00125 & 0.0025 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \frac{PL^2}{16E} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.$$

(b) The inverse of $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \begin{pmatrix} 428.571 & 333.333 & 200 \\ 285.714 & 666.667 & 400 \\ 142.857 & 333.333 & 600 \end{pmatrix}.$$

Taking $\mathbf{X}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{pmatrix} 0.7113 \\ 1 \\ 0.7958 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} 0.6710 \\ 1 \\ 0.7679 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 0.6645 \\ 1 \\ 0.7635 \end{pmatrix}, \quad \mathbf{X}_4 = \begin{pmatrix} 0.6634 \\ 1 \\ 0.7628 \end{pmatrix}, \quad \mathbf{X}_5 = \begin{pmatrix} 0.6632 \\ 1 \\ 0.7627 \end{pmatrix}.$$

This yields the eigenvalue $\lambda = 1161.23$. The smallest eigenvalue of $\mathbf{A}$ is then $1/\lambda = 0.0008612$. The lowest critical load is

$$P = \frac{16E}{L^2}(0.0008612) - 0.01378 \frac{E}{L^2}.$$

$$15. \text{ (a) } \mathbf{A}^{10} = \begin{pmatrix} 67,745,349 & -43,691,832 & 8,258,598 \\ -43,691,832 & 28,182,816 & -5,328,720 \\ 8,258,598 & -5,328,720 & 1,008,180 \end{pmatrix}$$

$$\text{(b) } \mathbf{X}_{10} = \mathbf{A}^{10} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 67,745,349 \\ -43,691,832 \\ 8,258,598 \end{pmatrix} \approx 67,745,349 \begin{pmatrix} 1 \\ -0.644942 \\ 0.121906 \end{pmatrix}$$

$$\mathbf{X}_{12} = \mathbf{A}^{12} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2,680,201,629 \\ -1,728,645,624 \\ 326,775,222 \end{pmatrix} \approx 2,680,201,629 \begin{pmatrix} 1 \\ -0.644968 \\ 0.121922 \end{pmatrix}.$$

The vectors appear to be approaching scalar multiples of $\mathbf{K} = (1, -0.644968, 0.121922)$, which approximates the dominant eigenvector.

(c) The dominant eigenvalue is $\lambda_1 = (\mathbf{A}\mathbf{K} \cdot \mathbf{K})/(\mathbf{K} \cdot \mathbf{K}) = 6.28995$.

EXERCISES 8.12**Diagonalization**

1. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = 5$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} -3 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$$

2. Distinct eigenvalues $\lambda_1 = 0$, $\lambda_2 = 6$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} -5 & -1 \\ 4 & 2 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 6 \end{pmatrix}$$

3. For $\lambda_1 = \lambda_2 = 1$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Hence $\mathbf{A}$ is not diagonalizable.

4. Distinct eigenvalues $\lambda_1 = \sqrt{5}$, $\lambda_2 = -\sqrt{5}$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} \sqrt{5} & -\sqrt{5} \\ 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \sqrt{5} & 0 \\ 0 & -\sqrt{5} \end{pmatrix}$$

5. Distinct eigenvalues $\lambda_1 = -7$, $\lambda_2 = 4$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 13 & 1 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -7 & 0 \\ 0 & 4 \end{pmatrix}$$

6. Distinct eigenvalues $\lambda_1 = -4$, $\lambda_2 = 10$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} -3 & 1 \\ 1 & -5 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -4 & 0 \\ 0 & 10 \end{pmatrix}$$

7. Distinct eigenvalues $\lambda_1 = \frac{1}{3}$, $\lambda_2 = \frac{2}{3}$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{2}{3} \end{pmatrix}$$

8. For $\lambda_1 = \lambda_2 = -3$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Hence $\mathbf{A}$ is not diagonalizable.

9. Distinct eigenvalues $\lambda_1 = -i$, $\lambda_2 = i$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

10. Distinct eigenvalues $\lambda_1 = 1 + i$, $\lambda_2 = 1 - i$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 2 & 2 \\ i & -i \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}$$

11. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = -1$, $\lambda_3 = 2$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

12. Distinct eigenvalues $\lambda_1 = 3$, $\lambda_2 = 4$, $\lambda_3 = 5$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

13. Distinct eigenvalues $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = 2$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

14. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = -3i$, $\lambda_3 = 3i$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 0 & -3i & 3i \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3i & 0 \\ 0 & 0 & 3i \end{pmatrix}$$

15. The eigenvalues are $\lambda_1 = \lambda_2 = 1$, $\lambda_3 = 2$. For $\lambda_1 = \lambda_2 = 1$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

Hence $\mathbf{A}$ is not diagonalizable.

16. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

17. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = \sqrt{5}$, $\lambda_3 = -\sqrt{5}$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{pmatrix} 0 & 1 + \sqrt{5} & 1 - \sqrt{5} \\ 0 & 2 & 2 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & -\sqrt{5} \end{pmatrix}$$

18. For $\lambda_1 = \lambda_2 = \lambda_3 = 1$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$. Hence $\mathbf{A}$ is not diagonalizable.

19. For the eigenvalues $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = 1$, $\lambda_4 = -1$ we obtain four linearly independent eigenvectors. Hence $\mathbf{A}$ is diagonalizable and

$$\mathbf{P} = \begin{pmatrix} -3 & -1 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

20. The eigenvalues are $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = \lambda_4 = 3$. For $\lambda_3 = \lambda_4 = 3$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$.

Hence $\mathbf{A}$ is not diagonalizable.

21. $\lambda_1 = 0$, $\lambda_2 = 2$, $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$

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$$22. \lambda_1 = -1, \lambda_2 = 4, \mathbf{K}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix}$$

$$23. \lambda_1 = 3, \lambda_2 = 10, \mathbf{K}_1 = \begin{pmatrix} -\sqrt{10} \\ 2 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} \sqrt{10} \\ 5 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} -\frac{\sqrt{10}}{\sqrt{14}} & \frac{\sqrt{10}}{\sqrt{35}} \\ \frac{2}{\sqrt{14}} & \frac{15}{\sqrt{35}} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 3 & 0 \\ 0 & 10 \end{pmatrix}$$

$$24. \lambda_1 = -1, \lambda_2 = 3, \mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$$

$$25. \lambda_1 = -1, \lambda_2 = \lambda_3 = 1, \mathbf{K}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{K}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$26. \lambda_1 = \lambda_2 = -1, \lambda_3 = 5, \mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{K}_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix},$$

$$\mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$$27. \lambda_1 = 3, \lambda_2 = 6, \lambda_3 = 9, \mathbf{K}_1 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}, \mathbf{K}_3 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

$$28. \lambda_1 = 1, \lambda_2 = 2 - \sqrt{2}, \lambda_3 = 2 + \sqrt{2}, \mathbf{K}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 1 - \sqrt{2} \\ 0 \\ 1 \end{pmatrix}, \mathbf{K}_3 = \begin{pmatrix} 1 + \sqrt{2} \\ 0 \\ 1 \end{pmatrix},$$

$$\mathbf{P} = \begin{pmatrix} 0 & -\frac{\sqrt{2-\sqrt{2}}}{2} & \frac{\sqrt{2+\sqrt{2}}}{2} \\ 1 & 0 & 0 \\ 0 & \frac{\sqrt{2+\sqrt{2}}}{2} & \frac{\sqrt{2-\sqrt{2}}}{2} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 - 2\sqrt{2} & 0 \\ 0 & 0 & 2 + \sqrt{2} \end{pmatrix}$$

$$29. \lambda_1 = 1, \lambda_2 = -6, \lambda_3 = 8, \mathbf{K}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \mathbf{K}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

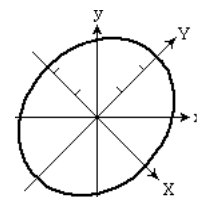
$$30. \lambda_1 = \lambda_2 = 0, \lambda_3 = -2, \lambda_4 = 2, \mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{K}_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \mathbf{K}_4 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

31. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 24$: $(x \ y) \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 24$. Using

$$\lambda_1 = 6, \lambda_2 = 4, \mathbf{K}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \text{ and } \mathbf{X} = \mathbf{P} \mathbf{X}' \text{ we find}$$

$$(X \ Y) \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = 24 \quad \text{or} \quad 6X^2 + 4Y^2 = 24.$$

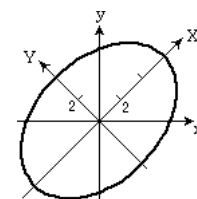


The conic section is an ellipse. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(1, -1)$ and $(1, 1)$ are $(\sqrt{2}, 0)$ and $(0, \sqrt{2})$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.

32. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 288$: $(x \ y) \begin{pmatrix} 13 & -5 \\ -5 & 13 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 288$.

$$\text{Using } \lambda_1 = 8, \lambda_2 = 18, \mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \text{ and } \mathbf{X} = \mathbf{P} \mathbf{X}' \text{ we find}$$

$$(X \ Y) \begin{pmatrix} 8 & 0 \\ 0 & 18 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = 288 \quad \text{or} \quad 8X^2 + 18Y^2 = 288.$$

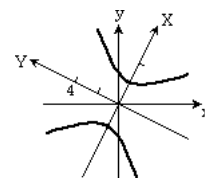


The conic section is an ellipse. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(1, 1)$ and $(1, -1)$ are $(\sqrt{2}, 0)$ and $(0, -\sqrt{2})$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.

33. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 20$: $(x \ y) \begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 20$. Using

$$\lambda_1 = 5, \lambda_2 = -5, \mathbf{K}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix} \text{ and } \mathbf{X} = \mathbf{P} \mathbf{X}' \text{ we find}$$

$$(X \ Y) \begin{pmatrix} 5 & 0 \\ 0 & -5 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = 20 \quad \text{or} \quad 5X^2 - 5Y^2 = 20.$$



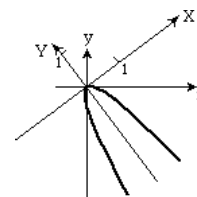
The conic section is a hyperbola. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(1, 2)$ and $(-2, 1)$ are $(\sqrt{5}, 0)$ and $(0, \sqrt{5})$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.

34. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 288$:

$$(x \ y) \begin{pmatrix} 16 & 12 \\ 12 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + (-3 \ 4) \begin{pmatrix} x \\ y \end{pmatrix} = 0.$$

$$\text{Using } \lambda_1 = 25, \lambda_2 = 0, \mathbf{K}_1 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} -3 \\ 4 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{4}{5} \end{pmatrix} \text{ and } \mathbf{X} = \mathbf{P} \mathbf{X}' \text{ we find}$$

$$(X \ Y) \begin{pmatrix} 25 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} + (0 \ 5) \begin{pmatrix} X \\ Y \end{pmatrix} = 0 \quad \text{or} \quad 25X^2 + 5Y = 0.$$



The conic section is a parabola. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(4, 3)$ and $(3, -4)$ are $(5, 0)$ and $(0, -5)$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.

35. Since $\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ we have $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$. Hence

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix}.$$

8.12 Diagonalization

36. Since eigenvectors are mutually orthogonal we use an orthogonal matrix $\mathbf{P}$ and $\mathbf{A} = \mathbf{PDP}^T$.

$$\mathbf{A} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix} = \begin{pmatrix} \frac{8}{3} & \frac{4}{3} & -\frac{1}{3} \\ \frac{4}{3} & \frac{11}{3} & \frac{4}{3} \\ -\frac{1}{3} & \frac{4}{3} & \frac{8}{3} \end{pmatrix}$$

37. Since $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ we have

$$\mathbf{A} = \mathbf{PDP}^{-1}$$

$$\mathbf{A}^2 = \mathbf{PDP}^{-1}\mathbf{PDP}^{-1} = \mathbf{PDDP}^{-1} = \mathbf{PD}^2\mathbf{P}^{-1}$$

$$\mathbf{A}^3 = \mathbf{A}^2\mathbf{A} = \mathbf{PD}^2\mathbf{P}^{-1}\mathbf{PDP}^{-1} = \mathbf{PD}^2\mathbf{DP}^{-1} = \mathbf{PD}^3\mathbf{P}^{-1}$$

and so on.

$$38. \begin{pmatrix} 2^4 & 0 & 0 & 0 \\ 0 & 3^4 & 0 & 0 \\ 0 & 0 & (-1)^4 & 0 \\ 0 & 0 & 0 & (5)^4 \end{pmatrix} = \begin{pmatrix} 16 & 0 & 0 & 0 \\ 0 & 81 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 625 \end{pmatrix}$$

$$39. \lambda_1 = 2, \lambda_2 = -1, \mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}, \mathbf{P}^{-1} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

$$\mathbf{A}^5 = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 32 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 21 & 11 \\ 22 & 10 \end{pmatrix}$$

$$40. \lambda_1 = 0, \lambda_2 = 1, \mathbf{K}_1 = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \mathbf{K}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \mathbf{P}^{-1} = \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$$

$$\mathbf{A}^{10} = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix} = \begin{pmatrix} 6 & -10 \\ 3 & -5 \end{pmatrix}$$

EXERCISES 8.13

Cryptography

1. (a) The message is $\mathbf{M} = \begin{pmatrix} 19 & 5 & 14 & 4 & 0 \\ 8 & 5 & 12 & 16 & 0 \end{pmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{AM} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 19 & 5 & 14 & 4 & 0 \\ 8 & 5 & 12 & 16 & 0 \end{pmatrix} = \begin{pmatrix} 35 & 15 & 38 & 36 & 0 \\ 27 & 10 & 26 & 20 & 0 \end{pmatrix}.$$

(b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 35 & 15 & 38 & 36 & 0 \\ 27 & 10 & 26 & 20 & 0 \end{pmatrix} = \begin{pmatrix} 19 & 5 & 14 & 4 & 0 \\ 8 & 5 & 12 & 16 & 0 \end{pmatrix}.$$

2. (a) The message is $\mathbf{M} = \begin{pmatrix} 20 & 8 & 5 & 0 & 13 & 15 & 14 & 5 & 25 \\ 0 & 9 & 19 & 0 & 8 & 5 & 18 & 5 & 0 \end{pmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{AM} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 20 & 8 & 5 & 0 & 13 & 15 & 14 & 5 & 25 \\ 0 & 9 & 19 & 0 & 8 & 5 & 18 & 5 & 0 \end{pmatrix} = \begin{pmatrix} 60 & 69 & 110 & 0 & 79 & 70 & 132 & 40 & 75 \\ 20 & 26 & 43 & 0 & 29 & 25 & 50 & 15 & 25 \end{pmatrix}.$$

(b) The decoded message is

$$\begin{aligned}\mathbf{M} &= \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 60 & 69 & 110 & 0 & 79 & 70 & 132 & 40 & 75 \\ 20 & 26 & 43 & 0 & 29 & 25 & 50 & 15 & 25 \end{pmatrix} \\ &= \begin{pmatrix} 20 & 8 & 5 & 0 & 13 & 15 & 14 & 5 & 25 \\ 0 & 9 & 19 & 0 & 8 & 5 & 18 & 5 & 0 \end{pmatrix}.\end{aligned}$$

3. (a) The message is $\mathbf{M} = \begin{pmatrix} 16 & 8 & 15 & 14 & 5 \\ 0 & 8 & 15 & 13 & 5 \end{pmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{AM} = \begin{pmatrix} 3 & 5 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 16 & 8 & 15 & 14 & 5 \\ 0 & 8 & 15 & 13 & 5 \end{pmatrix} = \begin{pmatrix} 48 & 64 & 120 & 107 & 40 \\ 32 & 40 & 75 & 67 & 25 \end{pmatrix}.$$

(b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} -3 & 5 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 48 & 64 & 120 & 107 & 40 \\ 32 & 40 & 75 & 67 & 25 \end{pmatrix} = \begin{pmatrix} 16 & 8 & 15 & 14 & 5 \\ 0 & 8 & 15 & 13 & 5 \end{pmatrix}.$$

4. (a) The message is $\mathbf{M} = \begin{pmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{pmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{AM} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 50 & 57 & 30 & 99 & 75 & 44 & 22 \\ 33 & 43 & 15 & 66 & 55 & 31 & 21 \\ 26 & 28 & 15 & 52 & 40 & 13 & 1 \end{pmatrix}.$$

(b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 0 & 1 & -1 \\ 2 & -2 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 50 & 57 & 30 & 99 & 75 & 44 & 22 \\ 33 & 43 & 15 & 66 & 55 & 31 & 21 \\ 26 & 28 & 15 & 52 & 40 & 13 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{pmatrix}.$$

5. (a) The message is $\mathbf{M} = \begin{pmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{pmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{AM} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 31 & 44 & 15 & 61 & 50 & 49 & 41 \\ 24 & 29 & 15 & 47 & 35 & 31 & 21 \\ 1 & -15 & 15 & 0 & -15 & -5 & -19 \end{pmatrix}.$$

(b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 1 \\ -2 & 3 & -1 \end{pmatrix} \begin{pmatrix} 31 & 44 & 15 & 61 & 50 & 49 & 41 \\ 24 & 29 & 15 & 47 & 35 & 31 & 21 \\ 1 & -15 & 15 & 0 & -15 & -5 & -19 \end{pmatrix} = \begin{pmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{pmatrix}.$$

6. (a) The message is $\mathbf{M} = \begin{pmatrix} 4 & 18 & 0 & 10 & 15 & 8 \\ 14 & 0 & 9 & 19 & 0 & 20 \\ 8 & 5 & 0 & 19 & 16 & 25 \end{pmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{AM} = \begin{pmatrix} 5 & 3 & 0 \\ 4 & 3 & -1 \\ 5 & 2 & 2 \end{pmatrix} \begin{pmatrix} 4 & 18 & 0 & 10 & 15 & 8 \\ 14 & 0 & 9 & 19 & 0 & 20 \\ 8 & 5 & 0 & 19 & 16 & 25 \end{pmatrix} = \begin{pmatrix} 62 & 90 & 27 & 107 & 75 & 100 \\ 50 & 67 & 27 & 78 & 44 & 67 \\ 64 & 100 & 18 & 126 & 107 & 130 \end{pmatrix}.$$

8.13 Cryptography

(b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 8 & -6 & -3 \\ -13 & 10 & 5 \\ -7 & 5 & 3 \end{pmatrix} \begin{pmatrix} 62 & 90 & 27 & 107 & 75 & 100 \\ 50 & 67 & 27 & 78 & 44 & 67 \\ 64 & 100 & 18 & 126 & 107 & 130 \end{pmatrix} = \begin{pmatrix} 4 & 18 & 0 & 10 & 15 & 8 \\ 14 & 0 & 9 & 19 & 0 & 20 \\ 8 & 5 & 0 & 19 & 16 & 25 \end{pmatrix}.$$

7. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 2 & -3 \\ -5 & 8 \end{pmatrix} \begin{pmatrix} 152 & 184 & 171 & 86 & 212 \\ 95 & 116 & 107 & 56 & 133 \end{pmatrix} = \begin{pmatrix} 19 & 20 & 21 & 4 & 25 \\ 0 & 8 & 1 & 18 & 4 \end{pmatrix}.$$

From correspondence (1) we obtain: STUDY_HARD.

8. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 46 & -7 & -13 & 22 & -18 & 1 & 10 \\ 23 & -15 & -14 & 2 & -18 & -12 & 5 \end{pmatrix} = \begin{pmatrix} 23 & 8 & 1 & 20 & 0 & 13 & 5 \\ 0 & 23 & 15 & 18 & 18 & 25 & 0 \end{pmatrix}.$$

From correspondence (1) we obtain: WHAT_ME_WORRY_.

9. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 31 & 21 & 21 & 22 & 20 & 9 \\ 19 & 0 & 9 & 13 & 16 & 15 \\ 13 & 1 & 20 & 8 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 13 & 1 & 20 & 8 & 0 & 9 \\ 19 & 0 & 9 & 13 & 16 & 15 \\ 18 & 20 & 1 & 14 & 20 & 0 \end{pmatrix}.$$

From correspondence (1) we obtain: MATH_IS_IMPORTANT.

10. The decoded message is

$$\begin{aligned} \mathbf{M} = \mathbf{A}^{-1}\mathbf{B} &= \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 36 & 32 & 28 & 61 & 26 & 56 & 10 & 12 \\ -9 & -2 & -18 & -1 & -18 & -25 & 0 & 0 \\ 23 & 27 & 23 & 41 & 26 & 43 & 5 & 12 \end{pmatrix} \\ &= \begin{pmatrix} 13 & 5 & 5 & 20 & 0 & 13 & 5 & 0 \\ 1 & 20 & 0 & 20 & 8 & 5 & 0 & 12 \\ 9 & 2 & 18 & 1 & 18 & 25 & 0 & 0 \end{pmatrix}. \end{aligned}$$

From correspondence (1) we obtain: MEET_ME_AT_THE_LIBRARY_..

11. Let $\mathbf{A}^{-1} = \begin{pmatrix} u & v \\ x & y \end{pmatrix}$. Then

$$\mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} u & v \\ x & y \end{pmatrix} \begin{pmatrix} 17 & 16 & 18 & 5 & 34 & 0 & 34 & 20 & 9 & 5 & 25 \\ -30 & -31 & -32 & -10 & -59 & 0 & -54 & -35 & -13 & -6 & -50 \end{pmatrix},$$

so $17u - 30v = 4$, $16u - 31v = 1$ and $5x - 6y = 1$, $25x - 50y = 25$. Then $\mathbf{A}^{-1} = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}$ and

$$\mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 4 & 1 & 4 & 0 & 9 & 0 & 14 & 5 & 5 & 4 & 0 \\ 13 & 15 & 14 & 5 & 25 & 0 & 20 & 15 & 4 & 1 & 25 \end{pmatrix}.$$

From correspondence (1) we obtain: DAD_I_NEED_MONEY_TODAY.

$$12. \text{ (a) } \mathbf{M}^T = \begin{pmatrix} 22 & 8 & 19 & 27 & 21 & 3 & 3 & 27 & 21 & 18 & 21 \\ 13 & 3 & 21 & 22 & 3 & 25 & 27 & 6 & 7 & 14 & 23 \\ 2 & 27 & 21 & 7 & 27 & 5 & 21 & 17 & 2 & 25 & 7 \end{pmatrix}$$

$$\text{ (b) } \mathbf{B}^T = \mathbf{M} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 37 & 38 & 61 & 56 & 51 & 33 & 51 & 50 & 30 & 57 & 51 \\ 24 & 35 & 40 & 34 & 48 & 8 & 24 & 44 & 23 & 43 & 28 \\ 11 & -24 & 0 & 15 & -24 & 20 & 6 & -11 & 5 & -11 & 16 \end{pmatrix}$$

$$(c) \mathbf{BA}^{-1} = \mathbf{B} \begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 1 & 0 & -1 \end{pmatrix} = \mathbf{M}$$

$$13. (a) \mathbf{B}' = \begin{pmatrix} 15 & 22 & 20 & 8 & 23 & 6 & 21 & 22 \\ 10 & 22 & 18 & 23 & 25 & 2 & 23 & 25 \\ 3 & 26 & 26 & 14 & 23 & 16 & 26 & 12 \end{pmatrix}$$

(b) Using correspondence (1) the encoded message is: OVTHWFUVJVRWYBWYCZZNWPZL.

$$(c) \mathbf{M}'\mathbf{A}^{-1}\mathbf{B}' = \begin{pmatrix} 1 & 4 & -3 \\ 2 & 3 & -2 \\ -2 & -4 & 3 \end{pmatrix} \mathbf{B}' = \begin{pmatrix} 46 & 32 & 14 & 58 & 54 & -34 & 35 & 86 \\ 54 & 58 & 42 & 57 & 75 & -14 & 59 & 95 \\ -61 & -54 & -34 & -66 & -77 & 28 & -56 & -108 \end{pmatrix}$$

$$\mathbf{M} = \mathbf{M}' \bmod 27 = \begin{pmatrix} 19 & 5 & 14 & 4 & 0 & 20 & 8 & 5 \\ 0 & 4 & 15 & 3 & 21 & 13 & 5 & 14 \\ 20 & 0 & 20 & 15 & 4 & 1 & 25 & 0 \end{pmatrix}.$$

Using correspondence (1) the encoded message is: SEND_THE_DOCUMENT_TODAY.

EXERCISES 8.14

An Error-Correcting Code

1. $(0 \ 1 \ 1 \ 0)$

2. $(1 \ 1 \ 1 \ 1)$

3. $(0 \ 0 \ 0 \ 1 \ 1)$

4. $(1 \ 0 \ 1 \ 0 \ 0)$

5. $(1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1)$

6. $(0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0)$

7. $(1 \ 0 \ 0)$

8. $(0 \ 0 \ 1)$

9. Parity error

10. $(1 \ 0 \ 1 \ 0)$

11. $(1 \ 0 \ 0 \ 1 \ 1)$

12. Parity error

In Problems 13-18, $\mathbf{D} = (c_1 \ c_2 \ c_3)$ and $\mathbf{P} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$.

13. $\mathbf{D}^T = \mathbf{P} (1 \ 1 \ 1 \ 0)^T = (0 \ 0 \ 0)^T$; $\mathbf{C} = (0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0)$

14. $\mathbf{D}^T = \mathbf{P} (0 \ 0 \ 1 \ 1)^T = (1 \ 0 \ 0)^T$; $\mathbf{C} = (1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1)$

15. $\mathbf{D}^T = \mathbf{P} (0 \ 1 \ 0 \ 1)^T = (0 \ 1 \ 0)^T$; $\mathbf{C} = (0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1)$

16. $\mathbf{D}^T = \mathbf{P} (0 \ 0 \ 0 \ 1)^T = (1 \ 1 \ 1)^T$; $\mathbf{C} = (1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1)$

17. $\mathbf{D}^T = \mathbf{P} (0 \ 1 \ 1 \ 0)^T = (1 \ 1 \ 0)^T$; $\mathbf{C} = (1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0)$

8.14 An Error-Correcting Code

18. $\mathbf{D}^T = \mathbf{P}(1 \ 1 \ 0 \ 0)^T = (0 \ 1 \ 1)^T$; $\mathbf{C} = (0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0)$

In Problems 19-28, $\mathbf{W}$ represents the correctly decoded message.

19. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0) = (0 \ 0 \ 0)^T$; a code word. $\mathbf{W} = (0 \ 0 \ 0 \ 0)$

20. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0) = (0 \ 1 \ 1)^T$; not a code word. The error is in the third bit.
 $\mathbf{W} = (1 \ 0 \ 0 \ 0)$

21. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1) = (1 \ 0 \ 1)^T$; not a code word. The error is in the fifth bit.
 $\mathbf{W} = (0 \ 0 \ 0 \ 1)$

22. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0) = (0 \ 0 \ 0)^T$; a code word. $\mathbf{W} = (0 \ 0 \ 1 \ 0)$

23. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1) = (0 \ 0 \ 0)^T$; a code word. $\mathbf{W} = (1 \ 1 \ 1 \ 1)$

24. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0) = (0 \ 0 \ 0)^T$; a code word. $\mathbf{W} = (0 \ 1 \ 1 \ 0)$

25. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(0 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1) = (0 \ 1 \ 0)^T$; not a code word. The error is in the second bit.
 $\mathbf{W} = (1 \ 0 \ 0 \ 1)$

26. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1) = (0 \ 1 \ 0)^T$; not a code word. The error is in the second bit.
 $\mathbf{W} = (0 \ 0 \ 0 \ 1)$

27. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 1) = (1 \ 1 \ 1)^T$; not a code word. The error is in the seventh bit.
 $\mathbf{W} = (1 \ 0 \ 1 \ 0)$

28. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1) = (0 \ 1 \ 0)^T$; not a code word. The error is in the second bit.
 $\mathbf{W} = (1 \ 0 \ 1 \ 1)$

29. (a) $2^7 = 128$

(b) $2^4 = 16$

(c) $(0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0)$, $(1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1)$, $(0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0)$, $(1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1)$,
 $(1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0)$, $(0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1)$, $(1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0)$, $(0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1)$,
 $(1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0)$, $(0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1)$, $(1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 0)$, $(0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1)$,
 $(0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0)$, $(1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1)$, $(0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0)$, $(1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1)$

30. (a) $c_4 = 0, c_3 = 1, c_2 = 1, c_1 = 0$; $(0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0)$

(b) $\mathbf{H} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$ (c) $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}(0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0)^T = (0 \ 0 \ 0 \ 0)^T$

EXERCISES 8.15

Method of Least Squares

1. We have $\mathbf{Y}^T = (1 \ 2 \ 3 \ 2)$ and $\mathbf{A}^T = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 54 & 14 \\ 14 & 4 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{20} \begin{pmatrix} 4 & -14 \\ -14 & 54 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} \frac{2}{5} \\ \frac{3}{5} \end{pmatrix}$ and the least squares line is $y = 0.4x + 0.6$.

2. We have $\mathbf{Y}^T = (-1 \ 3 \ 5 \ 7)$ and $\mathbf{A}^T = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 14 & 6 \\ 6 & 4 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{20} \begin{pmatrix} 4 & -6 \\ -6 & 14 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} \frac{13}{5} \\ -\frac{2}{5} \end{pmatrix}$ and the least squares line is $y = 2.6x - 0.4$.

3. We have $\mathbf{Y}^T = (1 \ 1.5 \ 3 \ 4.5 \ 5)$ and $\mathbf{A}^T = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 55 & 15 \\ 15 & 5 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{50} \begin{pmatrix} 5 & -15 \\ -15 & 55 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} 1.1 \\ -0.3 \end{pmatrix}$ and the least squares line is $y = 1.1x - 0.3$.

4. We have $\mathbf{Y}^T = (0 \ 1.5 \ 3 \ 4.5 \ 5)$ and $\mathbf{A}^T = \begin{pmatrix} 0 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 54 & 14 \\ 14 & 5 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{74} \begin{pmatrix} 5 & -14 \\ -14 & 54 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} 1.06757 \\ -0.189189 \end{pmatrix}$ and the least squares line is $y = 1.06757x - 0.189189$.

5. We have $\mathbf{Y}^T = (2 \ 3 \ 5 \ 5 \ 9 \ 8 \ 10)$ and $\mathbf{A}^T = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 91 & 21 \\ 21 & 7 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{196} \begin{pmatrix} 7 & -21 \\ -21 & 91 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} \frac{19}{14} \\ \frac{27}{14} \end{pmatrix}$ and the least squares line is $y = 1.35714x + 1.92857$.

6. We have $\mathbf{Y}^T = (2 \ 2.5 \ 1 \ 1.5 \ 2 \ 3.2 \ 5)$ and $\mathbf{A}^T = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 140 & 28 \\ 28 & 7 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{196} \begin{pmatrix} 7 & -28 \\ -28 & 140 \end{pmatrix}$

8.15 Method of Least Squares

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} 0.407143 \\ 0.828571 \end{pmatrix}$ and the least squares line is $y = 0.407143x + 0.828571$.

7. We have $\mathbf{Y}^T = (220 \ 200 \ 180 \ 170 \ 150 \ 135)$ and $\mathbf{A}^T = \begin{pmatrix} 20 & 40 & 60 & 80 & 100 & 120 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 36400 & 420 \\ 420 & 6 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{42000} \begin{pmatrix} 6 & -420 \\ -420 & 36400 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} -\frac{117}{140} \\ \frac{703}{3} \end{pmatrix}$ and the least squares line is $v = -0.835714T + 234.333$. At $T = 140$, $v \approx 117.333$ and at $T = 160$, $v \approx 100.619$.

8. We have $\mathbf{Y}^T = (0.47 \ 0.90 \ 2.0 \ 3.7 \ 7.5 \ 15)$ and $\mathbf{A}^T = \begin{pmatrix} 400 & 450 & 500 & 550 & 600 & 650 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 1697500 & 3150 \\ 3150 & 6 \end{pmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{262500} \begin{pmatrix} 6 & -3150 \\ -3150 & 1697500 \end{pmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} 0.0538 \\ -23.3167 \end{pmatrix}$ and the least squares line is $R = 0.0538T - 23.3167$. At $T = 700$, $R \approx 14.3433$.

EXERCISES 8.16

Discrete Compartmental Models

In Problems 1-5 we use the fact that the element τ_{ij} in the transfer matrix $\mathbf{T}$ is the rate of transfer from compartment j to compartment i , and the fact that the sum of each column in $\mathbf{T}$ is 1.

1. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{pmatrix} 90 \\ 60 \end{pmatrix} \quad \text{and} \quad \mathbf{T} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}.$$

(b) We have

$$\mathbf{X}_1 = \mathbf{T}\mathbf{X}_0 = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix} \begin{pmatrix} 90 \\ 60 \end{pmatrix} = \begin{pmatrix} 96 \\ 54 \end{pmatrix}$$

and

$$\mathbf{X}_2 = \mathbf{T}\mathbf{X}_1 = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix} \begin{pmatrix} 96 \\ 54 \end{pmatrix} = \begin{pmatrix} 98.4 \\ 51.6 \end{pmatrix}.$$

(c) From $\mathbf{T}\hat{\mathbf{X}} - \hat{\mathbf{X}} = (\mathbf{T} - \mathbf{I})\hat{\mathbf{X}} = \mathbf{0}$ and the fact that the system is closed we obtain

$$-0.2x_1 + 0.4x_2 = 0$$

$$x_1 + x_2 = 150.$$

The solution is $x_1 = 100$, $x_2 = 50$, so the equilibrium state is $\hat{\mathbf{X}} = \begin{pmatrix} 100 \\ 50 \end{pmatrix}$.

2. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{pmatrix} 100 \\ 200 \\ 150 \end{pmatrix} \quad \text{and} \quad \mathbf{T} = \begin{pmatrix} 0.7 & 0 & 0.5 \\ 0.3 & 0.8 & 0 \\ 0 & 0.2 & 0.5 \end{pmatrix}.$$

- (b) We have

$$\mathbf{X}_1 = \mathbf{T}\mathbf{X}_0 = \begin{pmatrix} 145 \\ 190 \\ 115 \end{pmatrix} \quad \text{and} \quad \mathbf{X}_2 = \mathbf{T}\mathbf{X}_1 = \begin{pmatrix} 159 \\ 195.5 \\ 95.5 \end{pmatrix}.$$

- (c) From $\mathbf{T}\hat{\mathbf{X}} - \hat{\mathbf{X}} = (\mathbf{T} - \mathbf{I})\hat{\mathbf{X}} = \mathbf{0}$ and the fact that the system is closed we obtain

$$\begin{aligned} -0.8x_1 + 0.5x_2 &= 0 \\ 0.3x_1 - 0.9x_2 &= 0 \\ x_1 + x_2 + x_3 &= 450. \end{aligned}$$

The solution is $x_1 = 145.161$, $x_2 = 217.742$, $x_3 = 87.0968$, so the equilibrium state is $\hat{\mathbf{X}} = \begin{pmatrix} 145.161 \\ 217.742 \\ 87.097 \end{pmatrix}$.

3. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{pmatrix} 100 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{T} = \begin{pmatrix} 0.2 & 0.5 & 0 \\ 0.3 & 0.1 & 0 \\ 0.5 & 0.4 & 1 \end{pmatrix}.$$

- (b) We have

$$\mathbf{X}_1 = \mathbf{T}\mathbf{X}_0 = \begin{pmatrix} 20 \\ 30 \\ 50 \end{pmatrix} \quad \text{and} \quad \mathbf{X}_2 = \mathbf{T}\mathbf{X}_1 = \begin{pmatrix} 19 \\ 9 \\ 72 \end{pmatrix}.$$

- (c) From $\mathbf{T}\hat{\mathbf{X}} - \hat{\mathbf{X}} = (\mathbf{T} - \mathbf{I})\hat{\mathbf{X}} = \mathbf{0}$ and the fact that the system is closed we obtain

$$\begin{aligned} -0.8x_1 + 0.5x_2 &= 0 \\ 0.3x_1 - 0.9x_2 &= 0 \\ x_1 + x_2 + x_3 &= 100. \end{aligned}$$

The solution is $x_1 = x_2 = 0$, $x_3 = 100$, so the equilibrium state is $\hat{\mathbf{X}} = \begin{pmatrix} 0 \\ 0 \\ 100 \end{pmatrix}$.

4. (a) The transfer matrix is

$$\mathbf{T} = \begin{pmatrix} 0.7 & 0.05 & 0.15 \\ 0.3 & 0.75 & 0 \\ 0 & 0.2 & 0.85 \end{pmatrix}.$$

8.16 Discrete Compartmental Models

(b)

Year	Bare Space	Grasses	Small Shrubs
0	10.00	0.00	0.00
1	7.00	3.00	0.00
2	5.05	4.35	0.60
3	3.84	4.78	1.38
4	3.14	4.74	2.13
5	2.75	4.49	2.76
6	2.56	4.19	3.24

5. From $\mathbf{T}\hat{\mathbf{X}} = 1\hat{\mathbf{X}}$ we see that the equilibrium state vector $\hat{\mathbf{X}}$ is the eigenvector of the transfer matrix $\mathbf{T}$ corresponding to the eigenvalue 1. It has the properties that its components add up to the sum of the components of the initial state vector.
6. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{pmatrix} 0 \\ 100 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{T} = \begin{pmatrix} 0.88 & 0.02 & 0 \\ 0.06 & 0.97 & 0.05 \\ 0.06 & 0.01 & 0.95 \end{pmatrix}.$$

(b)

Year	Phytoplankton	Water	Zooplankton
0	0.00	100.00	0.00
1	2.00	97.00	1.00
2	3.70	94.26	2.04
3	5.14	91.76	3.10
4	6.36	89.47	4.17
5	7.39	87.37	5.24
6	8.25	85.46	6.30
7	8.97	83.70	7.33
8	9.56	82.10	8.34
9	10.06	80.62	9.32
10	10.46	79.28	10.26
11	10.79	78.04	11.17
12	11.06	76.90	12.04

CHAPTER 8 REVIEW EXERCISES

1. $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{pmatrix}$

2. 4×3

3. $\mathbf{AB} = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}; \quad \mathbf{BA} = (11)$

4. $\mathbf{A}^{-1} = -\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$

5. False; consider $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ 6. True
7. $\det\left(\frac{1}{2}\mathbf{A}\right) = \left(\frac{1}{2}\right)^3(5) = \frac{5}{8}$; $\det(-\mathbf{A}^T) = (-1)^3(5) = -5$
8. $\det \mathbf{AB}^{-1} = \det \mathbf{A} / \det \mathbf{B} = 6/2 = 3$
9. 0 10. $\det \mathbf{C} = (-1)^3 / \det \mathbf{B} = -1/10^3(2) = -1/2000$
11. False; an eigenvalue can be 0. 12. True
13. True
14. True, since complex roots of real polynomials occur in conjugate pairs.
15. False; if the characteristic equation of an $n \times n$ matrix has repeated roots, there may not be n linearly independent eigenvectors.
16. True
17. True 18. True
19. False; $\mathbf{A}$ is singular and thus not orthogonal. 20. True
21. $\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^T)$ where $\frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$ is symmetric and $\frac{1}{2}(\mathbf{A} - \mathbf{A}^T)$ is skew-symmetric.
22. Since $\det \mathbf{A}^2 = (\det \mathbf{A})^2 \geq 0$ and $\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$, there is no $\mathbf{A}$ such that $\mathbf{A}^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
23. (a) $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ is nilpotent.
 (b) Since $\det \mathbf{A}^n = (\det \mathbf{A})^n = 0$ we see that $\det \mathbf{A} = 0$ and $\mathbf{A}$ is singular.
24. (a) $\sigma_x \sigma_y = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = -\sigma_y \sigma_x$; $\sigma_x \sigma_z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -\sigma_z \sigma_x$; $\sigma_y \sigma_z = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = -\sigma_z \sigma_y$
 (b) We first note that for anticommuting matrices $\mathbf{AB} = -\mathbf{BA}$, so $\mathbf{C} = 2\mathbf{AB}$. Then $\mathbf{C}_{xy} = \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix}$,
 $\mathbf{C}_{yz} = \begin{pmatrix} 0 & 2i \\ 2i & 0 \end{pmatrix}$, and $\mathbf{C}_{zx} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$.
25. $\left(\begin{array}{ccc|c} 5 & -1 & 1 & -9 \\ 2 & 4 & 0 & 27 \\ 1 & 1 & 5 & 9 \end{array} \right) \xrightarrow{R_{13}} \left(\begin{array}{ccc|c} 1 & 1 & 5 & 9 \\ 2 & 4 & 0 & 27 \\ 5 & -1 & 1 & -9 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 1 & 5 & 9 \\ 0 & 1 & -5 & \frac{9}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right).$
 The solution is $\mathbf{X} = \left(-\frac{1}{2} \quad 7 \quad \frac{1}{2}\right)^T$.
26. $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -2 & 3 & 2 \\ 2 & 0 & -3 & 3 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{2}{3} & \frac{4}{3} \\ 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right).$
 The solution is $x_1 = 3, x_2 = 2, x_3 = 1$.
27. Multiplying the second row by abc we obtain the third row. Thus the determinant is 0.
28. Expanding along the first row we see that the result is an expression of the form $ay + bx^2 + cx + d = 0$, which is a parabola since, in this case $a \neq 0$ and $b \neq 0$. Letting $x = 1$ and $y = 2$ we note that the first and second rows are the same. Similarly, when $x = 2$ and $y = 3$, the first and third rows are the same; and when $x = 3$

CHAPTER 8 REVIEW EXERCISES

and $y = 5$, the first and fourth rows are the same. In each case the determinant is 0 and the points lie on the parabola.

29. $4(-2)(3)(-1)(2)(5) = 240$

30. $(-3)(6)(9)(1) = -162$

31. Since $\begin{pmatrix} 1 & -1 & 1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{pmatrix} = 18 \neq 0$, the system has only the trivial solution.

32. Since $\begin{pmatrix} 1 & -1 & -1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{pmatrix} = 0$, the system has infinitely many solutions.

33. From $x_1\text{I}_2 + x_2\text{HNO}_3 \rightarrow x_3\text{HIO}_3 + x_4\text{NO}_2 + x_5\text{H}_2\text{O}$ we obtain the system $2x_1 = x_3$, $x_2 = x_3 + 2x_5$, $x_2 = x_4$, $3x_2 = 3x_3 + 2x_4 + x_5$. Letting $x_4 = x_2$ in the fourth equation we obtain $x_2 = 3x_3 + x_5$. Taking $x_1 = t$ we see that $x_3 = 2t$, $x_2 = 2t + 2x_5$, and $x_2 = 6t + x_5$. From the latter two equations we get $x_5 = 4t$. Taking $t = 1$ we have $x_1 = 1$, $x_2 = 10$, $x_3 = 2$, $x_4 = 10$, and $x_5 = 4$. The balanced equation is $\text{I}_2 + 10\text{HNO}_3 \rightarrow 2\text{HIO}_3 + 10\text{NO}_2 + 4\text{H}_2\text{O}$.

34. From $x_1\text{Ca} + x_2\text{H}_3\text{PO}_4 \rightarrow x_3\text{Ca}_3\text{P}_2\text{O}_8 + x_4\text{H}_2$ we obtain the system $x_1 = 3x_3$, $3x_2 = 2x_4$, $x_2 = 2x_3$, $4x_2 = 8x_3$. Letting $x_3 = t$ we see that $x_1 = 3t$, $x_2 = 2t$, and $x_4 = 3t$. Taking $t = 1$ we obtain the balanced equation $3\text{Ca} + 2\text{H}_3\text{PO}_4 \rightarrow \text{Ca}_3\text{P}_2\text{O}_8 + 3\text{H}_2$.

35. $\det \mathbf{A} = -84$, $\det \mathbf{A}_1 = 42$, $\det \mathbf{A}_2 = -21$, $\det \mathbf{A}_3 = -56$; $x_1 = \frac{42}{-84} = -\frac{1}{2}$, $x_2 = \frac{-21}{-84} = \frac{1}{4}$, $x_3 = \frac{-56}{-84} = \frac{2}{3}$

36. $\det = 4$, $\det \mathbf{A}_1 = 16$, $\det \mathbf{A}_2 = -4$, $\det \mathbf{A}_3 = 0$; $x_1 = \frac{16}{4} = 4$, $x_2 = \frac{-4}{4} = -1$, $x_3 = \frac{0}{4} = 0$

37. $\det \mathbf{A} = \cos^2 \theta + \sin^2 \theta$, $\det \mathbf{A}_1 = X \cos \theta - Y \sin \theta$, $\det \mathbf{A}_2 = Y \cos \theta + X \sin \theta$;
 $x_1 = X \cos \theta - Y \sin \theta$, $y = Y \cos \theta + X \sin \theta$

38. (a) $i_1 - i_2 - i_3 - i_4 = 0$, $i_2 R_1 = E$, $i_2 R_1 - i_3 R_2 = 0$, $i_3 R_2 - i_4 R_3 = 0$

(b) $\det \mathbf{A} = \begin{vmatrix} 1 & -1 & -1 & -1 \\ 0 & R_1 & 0 & 0 \\ 0 & R_1 & -R_2 & 0 \\ 0 & 0 & R_2 & -R_3 \end{vmatrix} = R_1 R_2 R_3$;

$$\det \mathbf{A}_1 = \begin{vmatrix} 0 & -1 & -1 & -1 \\ E & R_1 & 0 & 0 \\ 0 & R_1 & -R_2 & 0 \\ 0 & 0 & R_2 & -R_3 \end{vmatrix} = -E[-R_2 R_3 - R_1(R_3 + R_2)] = E(R_2 R_3 + R_1 R_3 + R_1 R_2);$$

$$i_1 = \frac{\det \mathbf{A}_1}{\det \mathbf{A}} = \frac{E(R_2 R_3 + R_1 R_3 + R_1 R_2)}{R_1 R_2 R_3} = E \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

39. $\mathbf{A}\mathbf{X} = \mathbf{B}$ is $\begin{pmatrix} 2 & 3 & -1 \\ 1 & -2 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix}$. Since $\mathbf{A}^{-1} = -\frac{1}{3} \begin{pmatrix} -2 & -3 & -2 \\ -1 & 0 & -1 \\ -4 & -6 & -7 \end{pmatrix}$, we have

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 7 \\ 5 \\ 23 \end{pmatrix}.$$

$$40. \text{ (a) } \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{4} & -\frac{9}{4} \\ -1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$\text{ (b) } \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{4} & -\frac{9}{4} \\ -1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -10 \\ 7 \\ -2 \end{pmatrix}$$

41. From the characteristic equation $\lambda^2 - 4\lambda - 5 = 0$ we see that the eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 5$. For $\lambda_1 = -1$ we have $2k_1 + 2k_2 = 0$, $4k_1 - 4k_2 = 0$ and $\mathbf{K}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. For $\lambda_2 = 5$ we have $-4k_1 + 2k_2 = 0$, $4k_1 - 2k_2 = 0$ and $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

42. From the characteristic equation $\lambda^2 = 0$ we see that the eigenvalues are $\lambda_1 = \lambda_2 = 0$. For $\lambda_1 = \lambda_2 = 0$ we have $4k_1 = 0$ and $\mathbf{K}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is a single eigenvector.

43. From the characteristic equation $-\lambda^3 + 6\lambda^2 + 15\lambda + 8 = -(\lambda + 1)^2(\lambda - 8) = 0$ we see that the eigenvalues are $\lambda_1 = \lambda_2 = -1$ and $\lambda_3 = 8$. For $\lambda_1 = \lambda_2 = -1$ we have

$$\begin{pmatrix} 4 & 2 & 4 & | & 0 \\ 2 & 1 & 2 & | & 0 \\ 4 & 2 & 4 & | & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}.$$

Thus $\mathbf{K}_1 = (1 \ -2 \ 0)^T$ and $\mathbf{K}_2 = (1 \ 0 \ -1)^T$. For $\lambda_3 = 8$ we have

$$\begin{pmatrix} -5 & 2 & 4 & | & 0 \\ 2 & -8 & 2 & | & 0 \\ 4 & 2 & -5 & | & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -\frac{2}{5} & -\frac{4}{5} & | & 0 \\ 0 & 1 & -\frac{1}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}.$$

Thus $\mathbf{K}_3 = (2 \ 1 \ 2)^T$.

44. From the characteristic equation $-\lambda^3 + 18\lambda^2 - 99\lambda + 162 = -(\lambda - 9)(\lambda - 6)(\lambda - 3) = 0$ we see that the eigenvalues are $\lambda_1 = 9$, $\lambda_2 = 6$, and $\lambda_3 = 3$. For $\lambda_1 = 9$ we have

$$\begin{pmatrix} -2 & -2 & 0 & | & 0 \\ -2 & -3 & 2 & | & 0 \\ 0 & 2 & -4 & | & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 1 & -2 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}.$$

Thus $\mathbf{K}_1 = (-2 \ 2 \ 1)^T$. For $\lambda_2 = 6$ we have

$$\begin{pmatrix} 1 & -2 & 0 & | & 0 \\ -2 & 0 & 2 & | & 0 \\ 0 & 2 & -1 & | & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -2 & 0 & | & 0 \\ 0 & 1 & -\frac{1}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}.$$

Thus $\mathbf{K}_2 = (2 \ 1 \ 2)^T$. For $\lambda_3 = 3$ we have

$$\begin{pmatrix} 4 & -2 & 0 & | & 0 \\ -2 & 3 & 2 & | & 0 \\ 0 & 2 & 2 & | & 0 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & -\frac{1}{2} & 0 & | & 0 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}.$$

Thus $\mathbf{K}_3 = (1 \ 2 \ -2)^T$.

CHAPTER 8 REVIEW EXERCISES

45. From the characteristic equation $-\lambda^3 - \lambda^2 + 21\lambda + 45 = -(\lambda + 3)^2(\lambda - 5) = 0$ we see that the eigenvalues are $\lambda_1 = \lambda_2 = -3$ and $\lambda_3 = 5$. For $\lambda_1 = \lambda_2 = -3$ we have

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 4 & -6 & 0 \\ -1 & -2 & 3 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Thus $\mathbf{K}_1 = (-2 \ 1 \ 0)^T$ and $\mathbf{K}_2 = (3 \ 0 \ 1)^T$. For $\lambda_3 = 5$ we have

$$\left(\begin{array}{ccc|c} -7 & 2 & -3 & 0 \\ 2 & -4 & -6 & 0 \\ -1 & -2 & -5 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & -\frac{2}{7} & \frac{3}{7} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Thus $\mathbf{K}_3 = (-1 \ -2 \ 1)^T$.

46. From the characteristic equation $-\lambda^3 + \lambda^2 + 2\lambda = -\lambda(\lambda + 1)(\lambda - 2) = 0$ we see that the eigenvalues are $\lambda_1 = 0$, $\lambda_2 = -1$, and $\lambda_3 = 2$. For $\lambda_1 = 0$ we have $k_3 = 0$, $2k_1 + 2k_2 + k_3 = 0$ and $\mathbf{K}_1 = (1 \ -1 \ 0)^T$. For $\lambda_2 = -1$ we have

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Thus $\mathbf{K}_2 = (0 \ 1 \ -1)^T$. For $\lambda_3 = 2$ we have

$$\left(\begin{array}{ccc|c} -2 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 2 & 2 & -1 & 0 \end{array} \right) \xrightarrow[\text{operations}]{\text{row}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Thus $\mathbf{K}_3 = (0 \ 1 \ 2)^T$.

47. Let $\mathbf{X}_1 = (a \ b \ c)^T$ be the first column of the matrix. Then $\mathbf{X}_1^T (-\frac{1}{\sqrt{2}} \ 0 \ \frac{1}{\sqrt{2}})^T = \frac{1}{\sqrt{2}}(c - a) = 0$ and $\mathbf{X}_1^T (\frac{1}{\sqrt{3}} \ \frac{1}{\sqrt{3}} \ \frac{1}{\sqrt{3}})^T = \frac{1}{\sqrt{3}}(a + b + c) = 0$. Also $\mathbf{X}_1^T \mathbf{X}_1 = a^2 + b^2 + c^2 = 1$. We see that $c = a$ and $b = -2a$ from the first two equations. Then $a^2 + 4a^2 + a^2 = 6a^2 = 1$ and $a = \frac{1}{\sqrt{6}}$. Thus $\mathbf{X}_1 = (\frac{1}{\sqrt{6}} \ -\frac{2}{\sqrt{6}} \ \frac{1}{\sqrt{6}})^T$.
48. (a) Eigenvalues are $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 = 5$ with corresponding eigenvectors $\mathbf{K}_1 = (0 \ 1 \ 0)^T$, $\mathbf{K}_2 = (2 \ 0 \ 1)^T$, and $\mathbf{K}_3 = (-1 \ 0 \ 2)^T$. Since $\|\mathbf{K}_1\| = 1$, $\|\mathbf{K}_2\| = \sqrt{5}$, and $\|\mathbf{K}_3\| = \sqrt{5}$, we have

$$\mathbf{P} = \begin{pmatrix} 0 & \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix} \quad \text{and} \quad \mathbf{P}^{-1} = \mathbf{P}^T = \begin{pmatrix} 0 & 1 & 0 \\ \frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} \end{pmatrix}.$$

$$(b) \ \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

49. We identify $\mathbf{A} = \begin{pmatrix} 1 & \frac{3}{2} \\ \frac{3}{2} & 1 \end{pmatrix}$. Eigenvalues are $\lambda_1 = -\frac{1}{2}$ and $\lambda_2 = \frac{5}{2}$ so $\mathbf{D} = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{5}{2} \end{pmatrix}$ and the equation becomes

$$(X \ Y) \mathbf{D} \begin{pmatrix} X \\ Y \end{pmatrix} = -\frac{1}{2}X^2 + \frac{5}{2}Y^2 = 1. \text{ The graph is a hyperbola.}$$

50. We measure years in units of 10, with 0 corresponding to 1890. Then $\mathbf{Y} = (63 \ 76 \ 92 \ 106 \ 123)^T$ and $\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}^T$, so $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 30 & 10 \\ 10 & 5 \end{pmatrix}$. Thus

$$\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \frac{1}{50} \begin{pmatrix} 5 & -10 \\ -10 & 30 \end{pmatrix} \mathbf{A}^T \mathbf{Y} = \begin{pmatrix} 15 \\ 62 \end{pmatrix},$$

and the least squares line is $y = 15t + 62$. At $t = 5$ (corresponding to 1940) we have $y = 137$. The error in the predicted population is 5 million or 3.7%.

51. The encoded message is

$$\begin{aligned} \mathbf{B} = \mathbf{A}\mathbf{M} &= \begin{pmatrix} 10 & 1 \\ 9 & 1 \end{pmatrix} \begin{pmatrix} 19 & 1 & 20 & 5 & 12 & 12 & 9 & 20 & 5 & 0 & 12 & 1 & 21 \\ 14 & 3 & 8 & 5 & 4 & 0 & 15 & 14 & 0 & 6 & 18 & 9 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 204 & 13 & 208 & 55 & 124 & 120 & 105 & 214 & 50 & 6 & 138 & 19 & 210 \\ 185 & 12 & 188 & 50 & 112 & 108 & 96 & 194 & 45 & 6 & 126 & 18 & 189 \end{pmatrix}. \end{aligned}$$

52. The encoded message is

$$\begin{aligned} \mathbf{B} = \mathbf{A}\mathbf{M} &= \begin{pmatrix} 10 & 1 \\ 9 & 1 \end{pmatrix} \begin{pmatrix} 19 & 5 & 3 & 0 & 1 & 7 & 14 & 20 & 0 & 1 & 18 \\ 18 & 22 & 19 & 0 & 20 & 21 & 5 & 19 & 0 & 1 & 13 \end{pmatrix} \\ &= \begin{pmatrix} 208 & 72 & 49 & 0 & 30 & 91 & 145 & 219 & 0 & 11 & 193 \\ 189 & 67 & 46 & 0 & 29 & 84 & 131 & 199 & 0 & 10 & 175 \end{pmatrix}. \end{aligned}$$

53. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} -3 & 2 & -1 \\ 1 & 0 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 19 & 0 & 15 & 14 & 0 & 20 \\ 35 & 10 & 27 & 53 & 1 & 54 \\ 5 & 15 & -3 & 48 & 2 & 39 \end{pmatrix} = \begin{pmatrix} 8 & 5 & 12 & 16 & 0 & 9 \\ 19 & 0 & 15 & 14 & 0 & 20 \\ 8 & 5 & 0 & 23 & 1 & 25 \end{pmatrix}.$$

From correspondence (1) we obtain: HELP_IS_ON_THE_WAY.

54. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} -3 & 2 & -1 \\ 1 & 0 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 2 & 21 \\ 27 & 17 & 40 \\ 21 & 13 & -2 \end{pmatrix} = \begin{pmatrix} 18 & 15 & 19 \\ 5 & 2 & 21 \\ 4 & 0 & 0 \end{pmatrix}.$$

From correspondence (1) we obtain: ROSEBUD__.

55. (a) The parity is even so the decoded message is (1 1 0 0 1)
 (b) The parity is odd; there is a parity error.

56. From $\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ we obtain the codeword (0 0 1 1 0 0 1).